

PHINMA UNIVERSITY OF ILOILO
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CAPSTONE PROJECT PROPOSAL

**Challenges in Digital Marketing and Client Communication for
Real Estate Properties**

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

1.1.1 The Global Digital Transformation in Real Estate

The real estate industry has undergone a profound and irreversible transformation over the last two decades, fundamentally reshaped by the exponential growth of digital technology and the increasing reliance of consumers on internet-based platforms for property-related needs. Across the globe, the traditional paradigms of property marketing, browsing, and transacting have shifted from face-to-face interactions and physical site visits to digital engagements conducted through sophisticated websites, mobile applications, social media platforms, and virtual reality technologies. This digital revolution has created unprecedented opportunities for real estate businesses while simultaneously presenting significant challenges, particularly for enterprises operating in developing economies where digital infrastructure and literacy levels vary considerably.

The proliferation of high-speed internet connectivity, the widespread adoption of smartphones, and the emergence of cloud computing have collectively dismantled the geographical barriers that once defined real estate markets. Property seekers in one continent can now seamlessly browse, evaluate, and initiate transactions for properties located thousands of miles away. This

globalization of real estate browsing has elevated consumer expectations, demanding that real estate businesses provide comprehensive digital experiences that rival those offered by leading e-commerce platforms and digital service providers. The modern property consumer expects not merely static listings with basic information, but immersive experiences featuring high-resolution imagery, virtual tours, interactive maps, real-time availability updates, and instant communication channels with property representatives.

Research conducted by the National Association of Realtors (NAR) in the United States indicates that over ninety-seven percent of homebuyers now begin their property search online, a figure that has remained consistently high and continues to underscore the critical importance of maintaining a robust, functional, and visually appealing digital presence. This statistic reflects a global trend that transcends developed markets and is increasingly relevant in emerging economies such as the Philippines, where internet penetration has grown exponentially across all demographic groups. The Philippine Statistics Authority reported that as of recent surveys, internet usage in the Philippines has reached unprecedented levels, with mobile internet access leading the charge due to the affordability of smartphones and data plans.

The digital transformation in real estate is not merely a matter of transferring existing information to online platforms; it represents a fundamental reimagining of how properties are presented, discovered, evaluated, and reserved. Traditional real estate operations relied heavily on personal networks, physical signage, printed brochures, and word-of-mouth referrals. While these methods retain some value, they are increasingly insufficient in a market where consumers conduct extensive independent research online before ever contacting a real estate agent. The digital-first consumer expects

to access comprehensive property information, compare multiple options, read reviews, and communicate with agents through their preferred digital channels, all without the constraints of business hours or physical office locations.

Furthermore, the COVID-19 pandemic served as a powerful catalyst for digital adoption across all industries, with real estate experiencing particularly dramatic shifts. During periods of lockdown and restricted physical movement, businesses that lacked robust digital infrastructure struggled to maintain operations, while those with established online platforms were able to continue serving their clients remotely through virtual tours, digital document signing, and online consultations. The pandemic demonstrated conclusively that digital systems are not merely advantageous but essential for business continuity and resilience. Even as physical restrictions have eased globally, consumer behavior has permanently shifted, with the preference for online property browsing, virtual property tours, and digital communication remaining firmly entrenched.

The implications of this digital transformation extend beyond consumer convenience to fundamentally alter the competitive dynamics of the real estate industry. Businesses that successfully leverage digital technologies gain significant advantages in market reach, operational efficiency, data analytics, and customer relationship management. Conversely, businesses that fail to adapt find themselves increasingly marginalized, unable to compete effectively for the attention and loyalty of digitally empowered consumers. This digital divide is particularly pronounced in regional and secondary markets, where the adoption of property technology (PropTech) solutions often lags behind major urban centers.

1.1.2 The Philippine Real Estate Landscape

The Philippine real estate market represents one of the most dynamic and rapidly growing sectors in Southeast Asia, characterized by robust demand driven by a young and growing population, sustained economic growth, significant infrastructure investments, and the substantial purchasing power of Overseas Filipino Workers (OFWs). The real estate industry contributes significantly to the Philippine economy, encompassing residential, commercial, industrial, and agricultural property segments, with residential real estate representing the largest and most active market segment.

The Philippine real estate boom of the past decade has been fueled by several converging factors. The business process outsourcing (BPO) industry has created a large class of young professionals with disposable income and housing needs. Infrastructure development programs under successive government administrations have improved connectivity and accessibility, opening new areas for real estate development. The sustained growth of the Philippine economy, despite global uncertainties, has maintained consumer confidence and investment appetite. Additionally, the remittances from millions of OFWs, which consistently rank among the highest in the world, provide a steady stream of capital flowing into real estate investments.

However, the Philippine real estate market also exhibits significant structural challenges that differentiate it from more mature markets. The industry remains highly fragmented, with numerous small to medium-sized real estate agencies and brokerages operating alongside large property developers. Many of these smaller operations lack the capital, technical expertise, and strategic

vision to invest in sophisticated digital infrastructure. The regulatory environment, while improving, still presents complexities in property transactions, titling, and documentation that can slow down digital transformation efforts. Furthermore, the digital literacy gap between urban and rural areas, and between different socioeconomic segments, means that digital real estate solutions must be designed with accessibility and ease of use as paramount considerations.

The Philippine real estate consumer has evolved considerably in recent years. The typical property seeker is now more informed, more connected, and more demanding than ever before. Empowered by access to vast amounts of information online, consumers arrive at initial agent consultations with detailed knowledge of market prices, comparable properties, and neighborhood characteristics. They expect transparency, responsiveness, and professionalism from real estate service providers. This evolution in consumer sophistication has placed additional pressure on real estate businesses to upgrade their service delivery mechanisms, with digital platforms serving as the primary interface between businesses and their prospective clients.

The residential real estate market in particular has seen shifting preferences, with increasing demand for properties in suburban and peri-urban areas that offer more space, better value, and improved quality of life compared to congested urban centers. Municipalities such as Pavia in Iloilo Province have emerged as attractive destinations for residential development, benefiting from their proximity to major urban centers while offering more affordable property options and a less congested environment. This trend toward decentralized urbanization has created new opportunities for real estate businesses willing to establish a strong presence in these emerging markets.

1.1.3 The Rise of PropTech in Developing Economies

Property Technology, commonly referred to as PropTech, encompasses the application of information technology and platform economics to the real estate industry. PropTech solutions range from simple property listing websites to sophisticated artificial intelligence-driven valuation models, blockchain-based transaction platforms, and Internet of Things (IoT) integrated smart buildings. The global PropTech sector has attracted billions of dollars in investment over the past decade, reflecting the enormous potential for technology to disrupt and improve traditional real estate processes.

In developing economies such as the Philippines, the adoption of PropTech solutions presents both unique challenges and distinctive opportunities. Unlike in mature markets where PropTech often focuses on incremental improvements to already digitized processes, in developing economies, PropTech can serve as a leapfrogging mechanism, allowing markets to bypass intermediate stages of technological development and move directly to modern, mobile-first digital platforms. The widespread adoption of smartphones in the Philippines, even among lower-income segments, creates a foundation for mobile-optimized real estate platforms that can reach previously underserved market segments.

However, the PropTech landscape in the Philippines remains uneven. Major property developers and large real estate agencies in Metro Manila and other primary cities have invested in sophisticated digital platforms, virtual reality showrooms, and comprehensive customer relationship management systems.

In contrast, smaller agencies and businesses operating in secondary cities and municipalities often continue to rely on traditional methods, creating a significant technology gap within the industry itself. This intra-industry digital divide means that the benefits of PropTech are not evenly distributed, with smaller, regional players at risk of being left behind.

The development of localized PropTech solutions tailored to the specific needs, regulatory environments, and consumer behaviors of Philippine regional markets represents a significant opportunity. Generic international platforms often fail to account for local nuances such as the importance of family networks in property decisions, the specific documentation requirements of Philippine real estate transactions, the prevalence of OFW buyers, and the varying levels of digital literacy across different demographics. Locally developed solutions, such as the proposed ILOILO TOP PROPERTY HOMES (ITPH) system, can address these specific contextual factors more effectively than imported platforms.

Moreover, the Philippine government's increasing emphasis on digital transformation, as evidenced by initiatives such as the Philippine Digital Transformation Strategy and various e-government programs, creates a supportive policy environment for the adoption of digital solutions in sectors such as real estate. The recognition that digitalization is essential for economic competitiveness and inclusive growth provides a macro-level rationale for investments in real estate technology at the local and regional levels.

1.1.4 Digital Marketing Evolution in Property Sales

Digital marketing has fundamentally transformed how real estate properties are promoted, discovered, and evaluated by potential buyers. The evolution from traditional marketing methods, print advertisements, billboards, flyers, and personal networks, to comprehensive digital marketing strategies represents one of the most significant shifts in real estate business operations. In the digital marketing paradigm, properties are promoted through multiple online channels including search engines, social media platforms, email campaigns, content marketing, and dedicated property portals, each offering distinct advantages in terms of reach, targeting, engagement, and measurability.

The effectiveness of digital marketing in real estate stems from its ability to precisely target specific demographic segments, geographic areas, and interest profiles. Unlike traditional advertising, which broadcasts messages to broad audiences with limited ability to measure impact, digital marketing enables real estate businesses to identify and reach potential buyers based on their online behavior, search patterns, demographic characteristics, and expressed interests. This precision targeting is particularly valuable in real estate, where the target market for any given property may be highly specific in terms of income level, family composition, lifestyle preferences, and investment objectives.

Search Engine Optimization (SEO) has become a critical component of real estate digital marketing, as property seekers increasingly rely on search engines such as Google to initiate their property search process. Real estate businesses that optimize their online presence for relevant search terms can attract significant organic traffic from motivated buyers actively seeking

properties in specific locations. The technical and content requirements of effective SEO, including mobile optimization, fast loading times, quality content, and local search optimization, demand that real estate businesses maintain technically sophisticated web platforms.

Social media marketing has emerged as another powerful channel for real estate promotion, with platforms such as Facebook, Instagram, and increasingly TikTok, serving as important venues for property showcasing and client engagement. The visual nature of real estate makes it particularly well-suited to image and video-focused social media platforms. Successful real estate social media marketing requires not merely posting property listings, but creating engaging content that tells stories about properties, neighborhoods, and lifestyles, thereby building emotional connections with potential buyers.

Despite the clear advantages of digital marketing, many real estate businesses, particularly smaller operations in regional markets, struggle to implement effective digital marketing strategies. The complexity of digital marketing, encompassing SEO, content creation, social media management, paid advertising, email marketing, and analytics, requires specialized skills and sustained effort that many traditional real estate businesses do not possess. The absence of integrated digital marketing tools within property management systems further complicates efforts to maintain consistent, professional, and effective online marketing presence.

The proposed ILOILO TOP PROPERTY HOMES (ITPH) system addresses this challenge by providing an integrated web-based platform that serves as both a property management system and a digital marketing tool. By ensuring that property listings are professionally presented, consistently formatted,

mobile-optimized, and easily discoverable, the system enhances the digital marketing capabilities of real estate businesses without requiring them to master the full complexity of digital marketing independently.

1.1.5 Client Communication in the Digital Age

Effective communication lies at the heart of successful real estate transactions, and the digital age has fundamentally altered both the expectations and the mechanisms of client-agent communication in the property sector. Modern consumers, shaped by their experiences with instant messaging, social media, and on-demand digital services, expect rapid, responsive, and personalized communication from service providers across all industries, including real estate. The days when clients would accept delayed responses, limited availability, or communication restricted to business hours are rapidly fading, replaced by expectations of near-instantaneous engagement across multiple digital channels.

The nature of real estate inquiries makes responsive communication particularly critical. Property decisions involve significant financial commitments and emotional investment, and prospective buyers often operate under time constraints, whether due to lease expirations, investment timelines, or competitive market conditions. When a potential buyer expresses interest in a property, the speed and quality of the agent's response can significantly influence the likelihood of conversion. Research in customer relationship

management consistently demonstrates that response time is a critical factor in lead conversion, with each hour of delay substantially reducing the probability of successful transaction completion.

Traditional real estate communication methods, phone calls, text messages, emails managed through personal accounts, and face-to-face meetings, while retaining value, present significant scalability challenges. An individual agent managing multiple client relationships simultaneously faces inherent limitations in their capacity to respond promptly to all inquiries. During peak periods, when multiple clients may be expressing interest simultaneously, the manual nature of traditional communication creates inevitable bottlenecks, delays, and oversights.

The fragmentation of communication across multiple platforms and channels further complicates effective client communication management. An agent might receive inquiries through Facebook Messenger, Viber, WhatsApp, SMS, email, and phone calls, each requiring separate monitoring and response. This fragmentation not only increases the cognitive load on agents but also creates risks of missed messages, inconsistent responses, and incomplete records of client interactions. Without a centralized communication management system, maintaining comprehensive awareness of all client interactions becomes increasingly difficult as inquiry volumes grow.

The importance of structured communication records extends beyond immediate transaction management to encompass long-term relationship building and business intelligence. Comprehensive records of client interactions enable agents to personalize future communications, remember client preferences, track the progression of inquiries from initial contact to

transaction completion, and identify patterns in client behavior that can inform marketing and service strategies. Manual communication methods, by their nature, produce fragmented and incomplete records that undermine these long-term business objectives.

Automated communication tools, including chatbots, automated email responses, inquiry routing systems, and integrated messaging platforms, offer significant potential to address these challenges. By handling routine inquiries automatically, providing instant acknowledgment of client messages, routing complex inquiries to appropriate agents, and maintaining comprehensive interaction records, these tools can dramatically improve both the efficiency and effectiveness of real estate client communication. The integration of such tools within a comprehensive web-based real estate management system represents a strategic approach to modernizing communication practices while maintaining the personal touch that remains important in real estate transactions.

1.1.6 The Overseas Filipino Worker Market Segment

Overseas Filipino Workers represent one of the most economically significant and demographically important segments of the Philippine real estate market. With millions of Filipinos employed abroad across diverse industries and geographies, OFWs contribute billions of dollars in remittances annually, a substantial portion of which flows into real estate investments. The Bangko Sentral ng Pilipinas consistently reports remittances as one of the largest sources of foreign exchange for the country, with real estate ranking among the top investment destinations for these funds.

The unique circumstances of OFW investors create specific requirements and challenges for real estate service provision. Physically located outside the Philippines, often in time zones that differ significantly from Philippine time, OFWs cannot easily visit properties, conduct in-person inspections, or meet with agents during standard business hours. Their entire property search and evaluation process must be conducted remotely, relying entirely on digital information, online communication, and virtual interactions. This complete dependence on digital channels makes the quality, comprehensiveness, and reliability of online real estate platforms critically important for serving this market segment.

OFW property investors typically exhibit distinct behavioral patterns compared to domestic buyers. Their investment decisions are often influenced by long-term financial planning objectives, including retirement preparation, family housing provision, and capital preservation. They may conduct extensive research over extended periods, comparing multiple properties and locations before making decisions. The physical distance from the investment location increases their risk perception and amplifies their need for trustworthy, detailed, and transparent information. Any indication of unprofessionalism, outdated information, or communication delays can quickly erode their confidence and drive them to alternative providers.

The time zone challenges inherent in serving OFW clients cannot be overstated. An OFW working in the Middle East may be available to communicate during hours that correspond to late night or early morning in the Philippines. An OFW in North America faces a twelve to sixteen-hour time difference. Traditional real estate office hours and communication methods are poorly suited to accommodate these temporal disparities.

Without systems that enable asynchronous communication, comprehensive information access independent of agent availability, and automated responses to routine inquiries, real estate businesses struggle to provide satisfactory service to this valuable client segment.

The economic significance of OFW investments in Philippine real estate makes this market segment impossible to ignore for serious real estate businesses. Properties marketed specifically to OFW investors, including affordable housing developments, condominium units, and suburban residential lots, represent a substantial portion of new property sales in many Philippine markets. Real estate businesses that successfully develop systems and processes for effectively reaching, engaging, and serving OFW clients can access a large, motivated, and financially capable market segment that competitors may struggle to serve effectively.

The proposed ILOILO TOP PROPERTY HOMES (ITPH) system, with its emphasis on comprehensive online property browsing, automated communication capabilities, and twenty-four-seven accessibility, directly addresses the specific needs of OFW property investors. By providing a platform where OFWs can independently research properties, submit inquiries at any time, receive automated responses to common questions, and track their reservation status without requiring synchronous agent communication, the system enables real estate businesses to serve this critical market segment far more effectively than traditional methods allow.

1.1.7 Operational Efficiency and Workload Management in Real Estate

The operational structure of traditional real estate businesses places significant burdens on agents and administrative staff, burdens that increase

proportionally with business volume in the absence of systematic automation. Real estate agents function as the primary interface between the business and its clients, responsible for property marketing, client acquisition, inquiry management, property showings, negotiation, documentation, and transaction coordination. In small to medium-sized agencies, agents often additionally assume administrative responsibilities including listing management, record keeping, and basic marketing tasks.

The manual processing of client inquiries represents one of the most time-consuming and repetitive aspects of real estate operations. Each inquiry must be received, acknowledged, evaluated for seriousness and fit, responded to with appropriate information, tracked for follow-up purposes, and integrated into the agent's overall pipeline management. When inquiry volumes are low, this manual processing may be manageable. However, as marketing effectiveness improves and inquiry volumes increase, the linear relationship between inquiry volume and agent workload creates a scalability ceiling beyond which service quality inevitably degrades.

The cognitive burden of managing multiple simultaneous client relationships through manual methods should not be underestimated. Human working memory has inherent limitations, and agents managing numerous clients across different stages of the decision-making process face significant challenges in maintaining comprehensive awareness of each client's status, preferences, and history. Important details may be forgotten, follow-up actions delayed, and clients inadvertently neglected as attention is diverted to more immediate demands. These human limitations are not indicative of poor performance but rather reflect the inherent constraints of manual information management.

The administrative overhead associated with traditional real estate operations extends beyond client communication to encompass property listing management, appointment scheduling, document preparation, and reporting. Each of these activities, when performed manually, consumes time that could otherwise be directed toward higher-value activities such as client relationship development, strategic market analysis, and transaction negotiation. The cumulative effect of these administrative burdens is reduced agent productivity, increased operational costs, and limited business growth capacity.

Management oversight and business intelligence represent additional areas where manual operations prove inadequate. Without systematic data collection and reporting, business owners and managers lack visibility into key performance indicators such as inquiry volumes, response times, conversion rates, agent performance, and property-level interest metrics. Decision-making becomes based on intuition and anecdote rather than empirical evidence, reducing the reliability and effectiveness of strategic planning. The inability to identify trends, measure marketing effectiveness, or allocate resources based on data represents a significant competitive disadvantage in an increasingly data-driven business environment.

Automation and systematization offer clear pathways to addressing these operational challenges. By automating routine tasks, centralizing information management, standardizing processes, and providing real-time visibility into business operations, technology-enabled real estate businesses can achieve substantially higher levels of operational efficiency. Agents can handle larger client volumes without proportional increases in workload. Management can make informed decisions based on comprehensive data. The business as a

whole can scale its operations more effectively while maintaining or improving service quality.

1.1.8 The Real Estate Market in Pavia, Iloilo City

Pavia, a municipality located in the province of Iloilo in the Western Visayas region of the Philippines, represents a compelling case study in the dynamics of emerging real estate markets in Philippine secondary cities. Situated immediately adjacent to Iloilo City, the regional capital and economic hub of Western Visayas, Pavia has experienced significant development momentum driven by infrastructure investments, population growth, and the spillover effects of Iloilo City's economic expansion. The municipality's strategic location, combined with relatively more affordable land prices compared to Iloilo City proper, has positioned it as an increasingly attractive destination for residential, commercial, and mixed-use development.

The development trajectory of Pavia reflects broader patterns of peri-urbanization occurring around major Philippine cities. As primary urban centers become increasingly congested and expensive, surrounding municipalities absorb population and economic growth, transforming from primarily agricultural or rural areas into dynamic suburban communities. Pavia has benefited from this pattern, with numerous residential subdivisions, commercial establishments, and institutional facilities establishing operations within its boundaries. The Iloilo Business Park, the Iloilo Convention Center, and various infrastructure projects have enhanced the connectivity and attractiveness of the broader Iloilo area, with Pavia positioned to capture significant benefits from these developments.

The real estate market in Pavia encompasses a diverse range of property types, including residential lots, house and lot packages, townhouses, and commercial spaces. The target market for these properties includes young professionals working in Iloilo City, local business owners, families seeking more spacious and affordable housing options than central city locations provide, and OFW investors attracted by the area's growth potential and relative affordability. This diverse market composition creates both opportunities and challenges for real estate businesses, requiring marketing and service approaches that can effectively reach and engage multiple distinct customer segments.

Despite the evident growth potential of Pavia's real estate market, many local real estate businesses continue to operate using traditional methods that are increasingly misaligned with market realities and consumer expectations. Property marketing relies heavily on physical signage, printed materials, and personal networks rather than comprehensive digital presence. Client communication occurs through informal messaging applications and phone calls without systematic tracking or management. Property listings lack standardization, professional presentation, and real-time updating. These operational limitations constrain the ability of local real estate businesses to fully capitalize on the market opportunities presented by Pavia's growth trajectory.

The competitive landscape in Pavia's real estate market is evolving, with increased entry of property developers and real estate agencies recognizing the area's potential. Businesses that successfully modernize their operations through digital transformation will be positioned to capture disproportionate market share, while those that persist with traditional methods risk gradual

marginalization. The window for proactive digital adoption is finite, as competitors and consumer expectations continue to advance.

1.1.9 Research Gap and Opportunity

The convergence of the factors discussed above, global digital transformation in real estate, the specific characteristics of the Philippine market, the importance of the OFW segment, operational efficiency imperatives, and the emerging opportunities in Pavia, Iloilo City, creates a compelling case for the development of localized, integrated web-based real estate management systems. While numerous property technology solutions exist in the Philippine market, significant gaps remain that the proposed ILOILO TOP PROPERTY HOMES (ITPH) system seeks to address.

Existing real estate platforms in the Philippines tend to fall into two categories: large-scale property portals that aggregate listings from multiple sources but offer limited tools for individual business management, and generic content management systems that lack the specific features required for effective real estate operations. The former category includes platforms such as Lamudi, Property24, and DotProperty, which provide broad exposure but do not address the internal operational challenges of inquiry management, agent coordination, and administrative oversight. The latter category includes various website builders and CRM systems that can be adapted for real estate use but require significant customization and lack integrated property-specific functionality.

What is notably absent from the current market, particularly at the regional and municipal levels, are integrated systems that combine client-facing property browsing and reservation capabilities with back-end administrative

management, automated communication features, and digital marketing support, all tailored to the specific operational context of small to medium-sized real estate businesses in Philippine secondary cities. The proposed ILOILO TOP PROPERTY HOMES (ITPH) system addresses this gap by providing a unified platform that serves the needs of clients, agents, and administrators within a single integrated system designed specifically for the Pavia, Iloilo City real estate market.

The opportunity presented by this research gap extends beyond the immediate functional benefits of the system to encompass broader contributions to digital transformation in Philippine regional real estate markets. By demonstrating that effective, affordable, and locally appropriate PropTech solutions can be developed and implemented in contexts such as Pavia, the study provides a model and reference point for similar initiatives in other secondary cities and municipalities. The successful development of the ILOILO TOP PROPERTY HOMES (ITPH) system would validate the proposition that digital transformation in real estate is not exclusively the domain of large enterprises in major urban centers but is achievable and valuable for smaller businesses in regional markets.

1.2 Theoretical Framework

The development and implementation of the ILOILO TOP PROPERTY HOMES (ITPH) web-based real estate management system is grounded in established theoretical frameworks that provide conceptual foundations for understanding technology adoption, innovation diffusion, and customer relationship management in the context of real estate operations. These theories inform the design decisions, feature prioritization, and

implementation strategies of the proposed system, ensuring that the system is not merely a collection of technical features but a theoretically grounded solution to identified operational challenges.

1.2.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model, originally proposed by Fred Davis in 1989, provides a foundational theoretical framework for understanding how users come to accept and use new information technologies. TAM posits that two primary factors determine an individual's intention to use a technology: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). Perceived Usefulness refers to the degree to which a person believes that using a particular system would enhance their job performance or achieve their objectives. Perceived Ease of Use refers to the degree to which a person believes that using a particular system would be free of effort.

In the context of the ILOILO TOP PROPERTY HOMES (ITPH) system, TAM provides valuable guidance for system design and implementation. For real estate agents, the system's perceived usefulness depends on its ability to demonstrably improve their efficiency in managing inquiries, tracking client interactions, and coordinating appointments. If agents perceive that the system adds administrative burden without corresponding benefits, adoption resistance is likely. Conversely, if the system clearly reduces manual workload, prevents oversight of important inquiries, and provides tools that enhance their professional effectiveness, perceived usefulness will be high, facilitating adoption.

For clients, perceived usefulness relates to the system's ability to provide convenient access to comprehensive property information, enable efficient

reservation processes, and facilitate responsive communication with agents. The system's value proposition for clients must be immediately apparent, with intuitive navigation, comprehensive property details, and clear pathways to inquiry and reservation. Perceived ease of use is equally critical, particularly given the diverse technical proficiencies of potential users including OFWs, young professionals, and older investors. The system interface must be designed to minimize cognitive load, provide clear guidance, and accommodate users with varying levels of digital literacy.

TAM has been extensively validated across diverse technology contexts and cultures, and its applicability to the Philippine context is well-established in information systems research. The model suggests that successful technology implementation requires attention not merely to technical functionality but to the subjective perceptions and experiences of users. This theoretical insight informs the ILOILO TOP PROPERTY HOMES (ITPH) system's emphasis on user-friendly design, comprehensive training and support, and clear demonstration of value to all user groups.

1.2.2 Diffusion of Innovations Theory

Everett Rogers' Diffusion of Innovations theory, first published in 1962 and subsequently refined through multiple editions, provides a comprehensive framework for understanding how new ideas, practices, and technologies spread through societies and organizations. The theory identifies five categories of adopters, innovators, early adopters, early majority, late majority, and laggards, and examines the factors that influence the rate and pattern of innovation adoption. Key factors affecting diffusion include relative advantage, compatibility, complexity, trialability, and observability.

Relative advantage refers to the degree to which an innovation is perceived as better than the idea it supersedes. For the ILOILO TOP PROPERTY HOMES (ITPH) system, relative advantage must be clearly established in comparison to traditional real estate operational methods. The system's advantages in terms of inquiry management efficiency, communication responsiveness, listing accuracy, and administrative oversight must be demonstrable and significant enough to motivate behavioral change among users accustomed to established work patterns.

Compatibility refers to the degree to which an innovation is perceived as consistent with existing values, past experiences, and needs of potential adopters. The ILOILO TOP PROPERTY HOMES (ITPH) system must align with the operational realities, business objectives, and cultural context of real estate businesses in Pavia. This compatibility consideration informs decisions regarding system features, terminology, workflow design, and integration with existing business practices. A system that requires radical restructuring of established business processes faces greater adoption barriers than one that enhances and systematizes existing practices.

Complexity refers to the degree to which an innovation is perceived as difficult to understand and use. Consistent with TAM's emphasis on ease of use, Diffusion of Innovations theory suggests that innovations perceived as complex diffuse more slowly than those perceived as simple and accessible. The ILOILO TOP PROPERTY HOMES (ITPH) system's design prioritizes simplicity and intuitive interaction, recognizing that user resistance increases with perceived complexity. This consideration is particularly important given the varying levels of technical expertise among potential users in the target market.

Trialability refers to the degree to which an innovation may be experimented with on a limited basis. The ability to trial the ILOILO TOP PROPERTY HOMES (ITPH) system with limited commitment reduces perceived risk and facilitates adoption. Observability refers to the degree to which the results of an innovation are visible to others. The visible improvements in operational efficiency, client responsiveness, and business organization resulting from ILOILO TOP PROPERTY HOMES (ITPH) implementation can serve as powerful motivators for adoption by other businesses observing the system's benefits.

1.2.3 Customer Relationship Management Theory

Customer Relationship Management (CRM) theory provides the conceptual foundation for understanding and optimizing the interactions between businesses and their customers throughout the customer lifecycle. CRM encompasses strategies, processes, and technologies designed to acquire, retain, and grow customer relationships by understanding and responding to customer needs, preferences, and behaviors. In the digital era, CRM has evolved from primarily database-driven approaches to encompass comprehensive platforms that integrate marketing, sales, and service functions.

The core principles of CRM directly inform the design of the ILOILO TOP PROPERTY HOMES (ITPH) system's communication and inquiry management features. The system implements CRM principles by maintaining comprehensive records of all client interactions, enabling personalized communication based on client history and preferences, supporting systematic follow-up processes, and providing visibility into the

status of each client relationship. These capabilities transform client communication from ad-hoc, memory-dependent interactions into systematically managed relationships.

CRM theory emphasizes the importance of customer data as a strategic asset. Every client interaction generates data that, when properly collected and analyzed, can yield insights into customer preferences, behavior patterns, and satisfaction levels. The ILOILO TOP PROPERTY HOMES (ITPH) system's centralized database architecture ensures that client interaction data is systematically captured and made available for analysis, supporting data-driven decision-making in marketing, service improvement, and business strategy.

The lifetime value concept in CRM theory highlights the importance of viewing customer relationships as long-term assets rather than transactional encounters. Real estate transactions, while individually significant, represent points in ongoing relationships that may include future transactions, referrals, and advocacy. The ILOILO TOP PROPERTY HOMES (ITPH) system's comprehensive client records and communication history support long-term relationship management by ensuring that client context is preserved and accessible regardless of staff changes or the passage of time.

1.3 Conceptual Framework

1.3.1 Input-Process-Output Model

The conceptual framework of this study is structured using the Input-Process-Output (IPO) model, a widely recognized systems analysis framework that provides a structured approach to understanding and designing information

systems. The IPO model conceptualizes any system as a set of inputs that are transformed through defined processes to produce specific outputs, with feedback mechanisms enabling system adjustment and improvement.

Inputs represent the resources, data, and requirements that feed into the system development process. For the ILOILO TOP PROPERTY HOMES (ITPH) system, inputs include: (1) the operational requirements of real estate businesses in Pavia, Iloilo City, derived from analysis of current practices and identified challenges; (2) the functional requirements specified by stakeholders including business owners, agents, and prospective clients; (3) the technological infrastructure and development tools available for system implementation; (4) the regulatory and business environment constraints within which the system must operate; and (5) the theoretical foundations and best practices from related literature and existing systems that inform design decisions.

Processes represent the activities and transformations applied to inputs to produce desired outputs. The ILOILO TOP PROPERTY HOMES (ITPH) system development process encompasses: (1) systems analysis and requirements gathering through interviews, observation, and document analysis; (2) system design including database architecture, user interface design, and security framework development; (3) system implementation involving coding, database creation, and feature integration; (4) system testing including unit testing, integration testing, system testing, and user acceptance testing; and (5) system deployment and evaluation against defined performance criteria.

Outputs represent the deliverables and results produced by the system development process. The primary output is the ILOILO TOP PROPERTY

HOMES (ITPH) web-based real estate management system itself, comprising the integrated platform with its property browsing, reservation, communication, and administrative management functionalities. Secondary outputs include system documentation, user manuals, testing reports, evaluation results, and the knowledge generated through the development process that contributes to the broader understanding of PropTech implementation in Philippine regional markets.

Feedback represents the mechanisms through which system performance information is used to drive improvement. Feedback in the ILOILO TOP PROPERTY HOMES (ITPH) system includes user feedback collected during testing and evaluation, performance metrics generated by system analytics, and ongoing stakeholder input that informs system refinement and future development priorities.

1.3.2 Paradigm of the Study

The research paradigm illustrates the relationships among the key variables and components of the study. The paradigm is structured around the central problem of operational inefficiency in traditional real estate operations and the proposed technological solution.

The Independent Variables consist of the components and features of the ILOILO TOP PROPERTY HOMES (ITPH) system: (1) the property browsing and listing management module; (2) the online reservation system; (3) the automated communication and chat functionality; (4) the administrative dashboard and management tools; and (5) the security framework and access control mechanisms.

The Dependent Variables represent the outcomes and improvements that the system aims to achieve: (1) operational efficiency in real estate business processes; (2) client engagement and satisfaction levels; (3) inquiry response times and communication effectiveness; (4) agent productivity and workload management; and (5) overall business performance and competitiveness.

The Intervening Variables include factors that may influence the relationship between the system implementation and the desired outcomes: (1) user technology acceptance and adoption rates; (2) the quality of internet connectivity and device availability; (3) the training and support provided to system users; (4) the organizational culture and change management practices of implementing businesses; and (5) external market conditions affecting real estate demand and competition.

The paradigm posits that the implementation of the ILOILO TOP PROPERTY HOMES (ITPH) system (independent variables) will produce measurable improvements in operational and business outcomes (dependent variables), with the magnitude of these improvements moderated by various intervening factors that must be considered in system design, implementation, and evaluation.

1.4 Statement of the Problem

The real estate business plays a vital role in providing housing opportunities, investment options, and property services for individuals, families, and investors. However, despite its importance in the economy and society, many real estate businesses are currently experiencing significant operational challenges. These challenges are mainly caused by the continued use of outdated marketing practices, manual processing of transactions, and

inefficient handling of client inquiries. As the industry continues to evolve in a digital-first environment, businesses that fail to adopt modern technologies face difficulties in remaining competitive and relevant in the market.

In today's digital age, clients expect real estate services to be fast, accessible, and technology-driven. Most property seekers now rely on online platforms to search for listings, compare options, and communicate with agents. They expect real-time responses, updated property information, and visually appealing presentations of available properties. However, many real estate businesses are still dependent on traditional methods that are no longer aligned with modern consumer behavior. This gap between customer expectations and business capabilities creates inefficiencies that negatively affect customer satisfaction and business performance.

The general problem addressed by this study is: How can a web-based real estate management system with integrated digital marketing and automated communication features be developed and implemented to improve the operational efficiency, client engagement, and overall business performance of real estate businesses in Pavia, Iloilo City?

This general problem encompasses the fundamental challenge of bridging the gap between traditional real estate operational practices and the digital capabilities required to compete effectively in the modern market. It recognizes that technological solutions must be not merely technically functional but effectively integrated into business operations to produce meaningful improvements in performance outcomes.

1.4.1 Specific Problems

To address the general problem comprehensively, this study identifies and seeks to resolve the following specific problems:

1. The business is not reaching enough new customers, especially its intended target audience.

Despite having available properties that meet market needs, real estate businesses in Pavia are constrained by marketing approaches that lack the reach and targeting capabilities of modern digital strategies. The absence of a comprehensive digital presence effectively excludes these businesses from the consideration set of a large and growing segment of potential buyers who conduct their property searches predominantly through online channels. This problem is particularly acute in relation to younger, digitally native consumers and Overseas Filipino Workers who rely almost exclusively on internet-based platforms for property discovery and evaluation.

Traditional marketing channels, word-of-mouth referrals, printed flyers, physical signage, and informal social media posting, while retaining some local effectiveness, are inherently limited in their reach, targeting precision, and measurability. A flyer distributed in Pavia reaches only those who physically encounter it, with no ability to target specific demographics, track engagement, or measure return on investment. Word-of-mouth marketing, while valuable, depends on existing client networks and cannot be scaled systematically. The result is a persistent underutilization of market potential, with qualified buyers remaining unaware of available properties that might meet their needs.

The digital marketing deficiency creates a compounding effect: limited online visibility reduces inquiry volumes, which reduces transaction volumes, which limits business growth and investment capacity, which further constrains marketing capabilities. Breaking this cycle requires the implementation of digital marketing infrastructure that enables sustained, measurable, and scalable market reach. Without such infrastructure, real estate businesses in Pavia risk gradual marginalization as competitors and consumer expectations continue to advance digitally.

2. Marketing materials and property listings appear outdated and less appealing to clients.

The quality and presentation of property listings significantly influence client perceptions of business professionalism, property desirability, and overall trustworthiness. In the digital environment, where consumers form initial impressions within seconds of encountering online content, the visual and informational quality of property listings can determine whether a prospective buyer engages further or moves on to alternative providers.

Many property listings maintained by traditional real estate operations suffer from multiple quality deficiencies. Photographs may be of low resolution, poorly composed, or fail to accurately represent property features. Property descriptions may be incomplete, inconsistently formatted, or lacking in the detail that modern consumers expect. Critical information such as exact location, lot area, floor area, property condition, and availability status may be ambiguous or outdated. Some listings may continue to display properties that have already been sold or reserved, creating confusion and frustration among prospective buyers and eroding trust in the business's reliability.

The absence of a systematic content management process for property listings means that quality control is inconsistent and often absent. Without standardized templates, image requirements, and review processes, listing quality depends entirely on individual agent practices, which vary considerably. The result is a collection of listings that fails to present the business and its properties in the most favorable light, reducing engagement rates and undermining competitive positioning compared to businesses that maintain professionally presented, consistently updated digital listings.

3. Responses to client inquiries are slow because most inquiries are handled manually.

Client communication in traditional real estate operations relies heavily on manual processes including personal messaging applications, phone calls, and face-to-face interactions. While these channels enable personal connection, they are inherently limited in scalability and responsiveness. When agents are handling multiple clients simultaneously, attending property showings, traveling between locations, or outside of business hours, inquiry responses are inevitably delayed.

The speed of initial response to client inquiries is a critical success factor in real estate lead conversion. Research in sales and customer service consistently demonstrates that response time significantly influences conversion probability, with leads contacted within minutes of inquiry showing dramatically higher conversion rates than those experiencing hours or days of delay. In the competitive real estate environment, where multiple agencies may be contacted by the same prospective buyer, the agency that responds first and most effectively often secures the client relationship.

Manual communication processes also introduce variability in response quality. The information provided to different clients inquiring about the same property may vary depending on which agent responds, their current workload, their knowledge of the specific property, and their communication skills. This inconsistency undermines the professionalism of the business and may result in clients receiving incomplete or inaccurate information. The absence of standardized response templates, automated information delivery, and systematic inquiry routing means that communication quality depends on individual agent performance rather than organizational systems.

4. Agents are overloaded with follow-ups and inquiry management tasks.

Real estate agents in traditional operations bear the primary responsibility for managing the entire client communication lifecycle, from initial inquiry through follow-up, appointment scheduling, property showing coordination, negotiation, and transaction completion. When inquiry volumes increase, whether due to seasonal factors, marketing campaigns, or market conditions, the manual nature of these tasks creates significant workload burdens.

The repetitive and administrative aspects of inquiry management consume substantial portions of agent time that could otherwise be directed toward higher-value activities. Every inquiry requires acknowledgment, evaluation, response, and tracking. Follow-up communications must be scheduled, prepared, and sent. Appointment details must be coordinated and confirmed. These tasks, while necessary, do not directly generate revenue and represent opportunity costs in terms of agent time that could be spent on client relationship development, market analysis, or transaction negotiation.

The cognitive demands of manually tracking multiple client relationships simultaneously should not be underestimated. Agents must remember the

status of each inquiry, the content of previous communications, client preferences and constraints, and scheduled follow-up actions. As the number of active client relationships increases, the probability of oversight, delay, or error increases correspondingly. Important inquiries may be forgotten amidst the volume of communications. Follow-up actions may be delayed or missed entirely. Clients may perceive neglect or unprofessionalism when their expectations for responsiveness are not met.

This agent overload creates a paradox: successful marketing that generates increased inquiry volumes can actually degrade service quality and business performance if the operational infrastructure cannot scale to handle the increased demand. Without systematic automation and workflow management, business growth becomes self-limiting as service quality degrades beyond acceptable thresholds.

5. There is no automated system for tracking and updating client inquiries.

The absence of a centralized, automated inquiry tracking system creates multiple operational deficiencies that affect both immediate service delivery and long-term business intelligence. Without systematic tracking, the status of individual inquiries becomes opaque, with agents and management unable to readily determine which inquiries require attention, which are progressing toward transactions, and which have become inactive.

The fragmentation of inquiry records across multiple communication channels, different messaging applications, email accounts, phone call logs, handwritten notes, compounds the tracking challenge. No single source of truth exists for client interaction history, making it difficult for multiple agents to collaborate on client service, for management to monitor overall inquiry pipeline status, or for the business to maintain continuity when agent

assignments change. When an agent is unavailable due to illness, vacation, or departure from the business, their client relationships and inquiry histories may be effectively lost, damaging client trust and business continuity.

Beyond immediate operational concerns, the lack of systematic inquiry tracking deprives the business of valuable data that could inform strategic decision-making. Without data on inquiry volumes, sources, response times, conversion rates, and property-specific interest levels, management cannot evaluate marketing effectiveness, identify high-potential properties, allocate resources optimally, or set meaningful performance targets. Decision-making becomes based on intuition and anecdote rather than empirical evidence, reducing strategic reliability and competitive responsiveness.

The absence of automated tracking also prevents the implementation of systematic follow-up processes that are essential for maximizing conversion rates. Many real estate inquiries do not result in immediate transactions but represent potential future business if properly nurtured through systematic follow-up. Without automated tracking and reminder systems, these potential future transactions are frequently lost due to inadequate follow-up discipline.

6. There is a need to determine how the business can effectively attract more customers through improved marketing strategies.

The current marketing approaches employed by many real estate businesses in Pavia do not fully exploit the capabilities of modern digital marketing tools and strategies. The absence of automated notification systems, integrated online engagement platforms, search engine optimized property listings, and data-driven marketing analytics limits the effectiveness and efficiency of marketing investments. Without these capabilities, marketing activities tend

to be reactive rather than proactive, generic rather than targeted, and immeasurable rather than optimized.

The competitive landscape for real estate marketing is intensifying, with property seekers exposed to increasingly sophisticated digital marketing from national property portals, large developers, and digitally advanced competitors. To compete effectively for client attention and engagement, real estate businesses must elevate their marketing capabilities to comparable levels of professionalism, targeting, and responsiveness. The development and implementation of a web-based real estate management system with integrated digital marketing features represents an opportunity to fundamentally transform marketing effectiveness, shifting from reliance on traditional, limited-reach methods to leveraged, data-informed digital strategies.

1.5 Objectives of the Study

The rapid advancement of digital technology has fundamentally altered the landscape of business operations across all industries, with real estate experiencing particularly significant impacts due to the information-intensive and relationship-dependent nature of property transactions. Clients now demand services that are faster, more accessible, more transparent, and more responsive than traditional real estate operational models can consistently deliver. The expectations shaped by digital experiences in e-commerce, social media, and on-demand services have transferred to real estate, creating both challenges for businesses that fail to adapt and opportunities for those that successfully embrace digital transformation.

In response to these evolving market realities, this study undertakes the development of a comprehensive web-based real estate management system specifically designed to address the operational challenges identified in the Pavia, Iloilo City real estate market. The system's design is informed by thorough analysis of current operational practices, examination of best practices in property technology from both local and international contexts, and direct engagement with stakeholders including real estate business owners, agents, and prospective clients. The objectives are formulated to be specific, measurable, achievable, relevant, and time-bound (SMART), ensuring that the study's outcomes can be clearly evaluated against defined success criteria.

1.5.1 General Objective

The general objective of this study is to develop and implement a web-based real estate management system called ILOILO TOP PROPERTY HOMES (ITPH) that enables clients to conveniently browse and reserve real estate properties online while providing automated communication features such as chat functionality. The system also aims to assist administrators in efficiently managing property listings, monitoring appointments, and handling client inquiries in a more organized, systematic, and data-driven manner.

Through the strategic integration of digital tools, automated processes, and user-centered design principles, the proposed system seeks to reduce manual workloads, improve response times, enhance the quality and consistency of client service, and modernize traditional real estate operations. The system aims to shift real estate businesses from fragmented, manual processes to a centralized, technology-driven platform that supports improved client

engagement, operational efficiency, and business competitiveness in the Pavia, Iloilo City real estate market.

1.5.2 Specific Objectives

Objective 1: Develop a user-friendly interface for property browsing.

This objective focuses on the design and implementation of an intuitive, responsive, and visually appealing user interface that allows clients to easily browse available real estate properties in Pavia, Iloilo City. The interface will present comprehensive property information including high-resolution images, detailed descriptions, location information, pricing, availability status, and property specifications in a clear and organized manner.

The interface design will adhere to established user experience (UX) principles including clear information hierarchy, consistent visual design language, intuitive navigation structures, and accessibility features that accommodate users with varying levels of technical proficiency. Special attention will be given to mobile responsiveness, recognizing the high prevalence of mobile internet access in the Philippines and the importance of enabling property browsing across all device types. The interface will support multiple search and filtering options, allowing clients to narrow down property listings based on criteria such as property type, price range, location, and specific features.

The achievement of this objective will be evaluated through usability testing with representative users, measuring metrics such as task completion rates, time to complete common browsing tasks, user satisfaction ratings, and error rates. The interface must enable users to locate relevant properties and access

detailed information without requiring extensive technical knowledge or training.

Objective 2: Implement an online reservation system.

This objective involves the creation of a secure, efficient, and reliable reservation system that allows clients to submit reservation requests for selected properties directly through the web-based platform. The system will automatically record reservation details in the central database, ensuring accuracy, eliminating manual data entry errors, and providing immediate confirmation to clients.

The reservation system will incorporate essential features including real-time availability checking to prevent double-booking, automatic conflict detection and resolution, confirmation notifications sent to both clients and administrators, and comprehensive reservation tracking capabilities. Clients will be able to view the status of their reservations through their account interface, while administrators will have access to complete reservation management tools including status updates, scheduling coordination, and historical reporting.

The system will maintain a comprehensive audit trail of all reservation transactions, recording timestamps, user actions, and status changes to support accountability, transparency, and dispute resolution. Security measures will ensure that reservation data is protected from unauthorized access or modification. The achievement of this objective will be evaluated through functional testing, security testing, and user acceptance testing.

Objective 3: Provide automated communication through chat functionality.

This objective addresses the critical need for improved client communication by integrating an automated communication feature that enables clients to send inquiries and receive timely, consistent responses regarding property details, availability, pricing, and reservation status. The chat functionality will be accessible directly within the web platform, providing a seamless communication channel that does not require clients to use external messaging applications or email.

The communication feature will include multiple components: a real-time chat interface for direct client-agent messaging, automated acknowledgment messages that confirm receipt of client inquiries, notification systems that alert agents to new messages ensuring prompt attention, and message history storage that maintains comprehensive records of all communications for future reference and quality assurance purposes.

By automating basic communication processes and providing structured inquiry management tools, this feature aims to reduce agent workload while maintaining effective client interaction. The system will support both automated responses to common inquiries and seamless escalation to human agents for complex or personalized questions.

Objective 4: Develop an administrative dashboard for system management.

This objective focuses on the creation of a comprehensive administrative dashboard that provides administrators with centralized control over all system functions and real-time visibility into business operations. The dashboard will enable efficient management of property listings including adding new properties, updating existing listings, removing unavailable properties, and ensuring information accuracy and currency.

Beyond listing management, the dashboard will provide tools for monitoring client appointments and reservations, managing user accounts and access permissions, tracking inquiry volumes and response metrics, and generating reports on system usage, property interest levels, and agent performance. Real-time analytics will present key performance indicators through visual dashboards, enabling data-driven decision-making and rapid identification of operational issues or opportunities.

The administrative interface will be designed for efficiency, enabling administrators to complete common management tasks with minimal steps and providing clear, organized access to all system functions.

Objective 5: Ensure secure data handling and system protection.

This objective addresses the critical importance of security in any system handling sensitive personal and business information. The study will implement comprehensive security measures including robust login authentication with password encryption, role-based access control that restricts system functionality based on user roles, input validation to prevent injection attacks and data corruption, and defensive programming techniques to protect against common web application vulnerabilities.

The security framework will implement three primary user roles with differentiated access privileges: Administrators with full system access, Agents with access to inquiry management and property update functions relevant to their responsibilities, and Clients with access to property browsing, reservation, and communication features. Data encryption will protect sensitive information such as user credentials and personal contact details both in transit and at rest.

Regular security auditing processes will be established to identify and address potential vulnerabilities. The achievement of this objective will be evaluated through security testing including penetration testing, access control verification, input validation testing, and review of security documentation and procedures.

1.6 Scope and Delimitation

A clearly defined scope is essential for ensuring that the development of the ILOILO TOP PROPERTY HOMES (ITPH) system remains focused, achievable, and aligned with the study's objectives within the constraints of available time, resources, and technical capabilities. The scope delineates the boundaries of the system, what it will accomplish, for whom, and under what conditions, while the delimitation explicitly identifies what falls outside the current development phase, managing expectations and directing development efforts toward the highest-impact features.

1.6.1 Scope of the Study

This project encompasses the complete development of a web-based real estate management system designed specifically for real estate properties located in Pavia, Iloilo City. The system provides a centralized digital platform accessible through standard web browsers, enabling multiple categories of users to perform their respective functions within a unified, integrated environment.

Geographic Scope: The system focuses exclusively on real estate properties situated within the municipality of Pavia, Iloilo City. This geographic focus ensures that property data, location information, market context, and system

features are specifically relevant and accurate for this defined area. The system will support property listings across all barangays and development areas within Pavia, encompassing residential, commercial, and mixed-use properties available in the local market.

Functional Scope, Client-Facing Features: The system provides comprehensive property browsing capabilities allowing clients to view available properties, access detailed property information including images, descriptions, specifications, pricing, and availability status, and utilize search and filtering functions to identify properties matching their criteria. Clients can create personal accounts to manage their browsing history, saved properties, and reservation requests.

The reservation management feature enables authenticated clients to submit reservation requests for properties of interest. These reservations are automatically recorded in the system database with timestamp and client information, generating notifications to administrators and initiating the reservation management workflow. Clients can track their reservation status through their account dashboard and receive updates as their reservation progresses.

The integrated communication feature provides clients with the ability to send inquiries directly through the system interface, initiating structured conversations with assigned agents. All communications are stored within the system, creating comprehensive interaction records accessible to both clients and administrators.

Functional Scope, Administrative Features: The administrative module provides comprehensive property listing management capabilities, enabling authorized administrators and agents to add new properties, upload images,

enter detailed specifications, set pricing and availability status, and update or remove listings as needed. The system ensures that all client-facing information reflects current, accurate data through centralized content management.

The inquiry and communication management features provide administrators with tools to view, organize, assign, and respond to client inquiries through the integrated chat interface. Comprehensive communication histories enable quality monitoring, agent performance assessment, and service continuity across staff changes.

The reservation monitoring and management features enable administrators to view all pending and confirmed reservations, update reservation statuses, coordinate appointment scheduling, and generate reports on reservation activity. The dashboard provides real-time visibility into key business metrics including inquiry volumes, response times, reservation rates, and property interest levels.

User Scope: The system serves three primary categories of users. Clients are individuals seeking to browse, inquire about, or reserve real estate properties. Agents are authorized real estate professionals assigned by administrators to manage client inquiries, update property information, and coordinate appointments. Administrators are senior authorized users responsible for overall system management, user administration, property oversight, and business intelligence.

Technical Scope: The system is developed using established web technologies including HTML, CSS, and JavaScript for frontend user interface development, PHP for server-side processing and business logic implementation, and MySQL for relational database management. This

technology stack was selected based on widespread availability, extensive documentation, proven reliability, and compatibility with standard web hosting infrastructure commonly used by small to medium-sized businesses in the Philippines.

The system implements responsive web design principles to ensure optimal user experience across desktop computers, tablets, and smartphones. Security measures include password-based authentication, session management, input validation, SQL injection prevention, and role-based access control.

1.6.2 Delimitation of the Study

While the ILOILO TOP PROPERTY HOMES (ITPH) system provides comprehensive functionality for its intended scope, certain features and capabilities are explicitly excluded from the current development phase. These delimitations are not permanent restrictions but rather boundaries that define the current capstone project scope, with the understanding that future development iterations may expand system capabilities beyond these boundaries.

Financial Transaction Processing: The system does not include online payment processing, bank integration, e-wallet transactions, or any form of electronic financial transaction handling. All financial dealings, including reservation fees, down payments, and full property payments, are assumed to be handled externally between the client and the real estate office through traditional payment methods. This delimitation reflects the significant technical complexity, regulatory requirements, and security considerations involved in implementing secure online payment systems, which extend beyond the scope and resources of the current capstone project. The system

does, however, record reservation intentions and facilitate the scheduling of property viewings and consultations that may lead to offline financial transactions.

Advanced Digital Marketing Automation: While the system improves digital marketing effectiveness through professional property presentation and web presence, it does not include advanced digital marketing automation features such as automated social media posting, search engine optimization tools, paid advertising management, email marketing campaigns, or customer segmentation for targeted marketing. The system provides the foundational digital platform upon which such marketing capabilities could be built in future development phases, but the current implementation focuses on core operational functionality rather than sophisticated marketing automation.

Legal Documentation Processing: The system does not handle legal documentation such as contract generation, deed preparation, ownership transfer processing, title verification, or any other legally binding document management. These processes remain outside the system and are handled manually by authorized legal and real estate professionals. The inclusion of legal document processing would require specialized legal expertise, compliance with Philippine real estate law and regulations, and integration with government registry systems, all of which are beyond the current project scope.

Property Valuation and Automated Pricing: The system does not provide automated property valuation, pricing recommendations, or market analysis tools. Property pricing decisions remain the exclusive responsibility of the real estate business based on their market knowledge, comparative analysis, and internal policies. Automated valuation models require extensive historical

transaction data, sophisticated analytical algorithms, and continuous market data feeds that are not available within the scope of this project.

External Communication Integration: The system does not integrate with external real-time communication channels such as direct phone systems, SMS messaging services, email servers, or third-party messaging platforms. All client-agent communication is contained within the system's integrated chat interface to maintain security, ensure comprehensive record-keeping, and provide a consistent user experience. While this design choice ensures data integrity and security, it means that clients must access the web platform to initiate and review communications rather than using their preferred external messaging applications.

Mobile Application Development: The system is implemented as a responsive web application accessible through standard web browsers rather than as a native mobile application for iOS or Android platforms. While the responsive design ensures usability across mobile devices, the system does not leverage native mobile capabilities such as push notifications, offline storage, camera integration, or GPS functionality that would be available in a dedicated mobile application.

Offline Functionality: The system requires active internet connectivity for all functions and does not include offline capabilities or data synchronization features for environments with intermittent or unreliable internet access. Users must have access to a stable internet connection and a compatible device to utilize the platform fully.

Multi-Location Expansion: While the system architecture could theoretically support expansion to additional locations, the current implementation is configured exclusively for Pavia, Iloilo City. The system does not include

features for managing properties across multiple municipalities, cities, or regions, nor does it support franchise or multi-office operational models.

1.7 Significance of the Study

The proposed ILOILO TOP PROPERTY HOMES (ITPH) web-based real estate management system represents more than a technical development project; it constitutes a practical intervention aimed at addressing real operational challenges and creating tangible value for multiple stakeholder groups. The significance of the study extends across immediate operational benefits for direct users, broader contributions to industry development, academic value, and foundational support for future technological innovation in Philippine regional real estate markets.

1.7.1 For Clients and Property Seekers

The system provides transformative benefits for clients and property seekers by fundamentally improving the accessibility, convenience, and quality of real estate service delivery. The most immediate benefit is the ability to browse comprehensive property listings at any time and from any location with internet access. Unlike traditional real estate operations constrained by office hours and physical location, the web-based platform provides twenty-four-seven availability, enabling clients to conduct property research at their convenience, whether during lunch breaks, late evenings, or weekends. This flexibility is particularly valuable for working professionals, busy parents, and Overseas Filipino Workers in distant time zones who cannot easily visit physical offices during standard business hours.

The comprehensiveness of property information available through the system addresses a significant pain point in traditional real estate browsing. Clients can access high-resolution photographs, detailed property descriptions, accurate specifications, location information, pricing, and availability status without requiring direct agent involvement for basic information gathering. This self-service capability empowers clients to conduct preliminary screening and comparison independently, making their subsequent interactions with agents more focused and productive. The reduction in information asymmetry between clients and agents supports more informed decision-making and builds client confidence in the transaction process.

The automated communication features significantly improve the client experience by reducing response delays and ensuring consistent information quality. When clients submit inquiries through the system, they receive immediate acknowledgment, and agents are automatically notified, enabling faster response times than traditional communication methods typically achieve. The structured nature of system-based communication ensures that inquiries are not lost, overlooked, or forgotten, addressing a common source of frustration in traditional real estate interactions.

For Overseas Filipino Workers, the system addresses specific challenges that traditional real estate services struggle to accommodate. The complete online accessibility eliminates the need for physical presence, enabling OFWs to research properties, submit inquiries, and initiate reservations entirely remotely. The comprehensive property information and visual presentation partially compensate for the inability to conduct in-person property inspections. The asynchronous communication capabilities accommodate time zone differences, allowing OFWs to interact with the system and leave

messages that agents can respond to during Philippine business hours. These features collectively make the Pavia real estate market more accessible to this important investor segment.

1.7.2 For Real Estate Agents and Administrators

Real estate agents and administrators represent the primary operational beneficiaries of the ILOILO TOP PROPERTY HOMES (ITPH) system, as it directly addresses the workload, efficiency, and management challenges that characterize traditional real estate operations. The centralized platform eliminates the fragmentation of information and communication that agents typically struggle with when managing multiple clients across diverse channels. All property information, client inquiries, communication histories, and reservation records are stored in a single accessible location, dramatically reducing the cognitive and administrative burden of information management.

The automation of routine tasks represents one of the most significant benefits for agents. Inquiry acknowledgment, basic information provision, appointment scheduling, and status updates can be handled through system automation, freeing agent time for higher-value activities such as personalized client consultation, property showings, negotiation, and relationship building. The reduction in repetitive administrative tasks not only improves productivity but also enhances job satisfaction by allowing agents to focus on the interpersonal and strategic aspects of real estate service that drew them to the profession.

The systematic inquiry tracking and management capabilities ensure that no client inquiry is overlooked or forgotten. Automated notifications alert agents to new inquiries and pending follow-up actions, while comprehensive

interaction histories provide immediate context for every client communication. These capabilities improve service consistency, reduce the risk of missed opportunities, and enable agents to manage larger client volumes without proportional increases in workload or stress.

For administrators, the dashboard provides unprecedented visibility into business operations. Real-time data on inquiry volumes, response times, reservation rates, property interest levels, and agent performance enables evidence-based management decisions. Rather than relying on intuition or periodic reports, administrators can monitor business health continuously, identify trends as they emerge, and make timely adjustments to marketing strategies, resource allocation, and operational procedures. This data-driven management approach represents a significant advancement over traditional real estate business oversight.

1.7.3 For Real Estate Business Owners

Real estate business owners stand to gain strategic benefits from the ILOILO TOP PROPERTY HOMES (ITPH) system that extend beyond operational efficiency to encompass competitive positioning, market expansion, and business scalability. The professional digital presence provided by the system enhances the business's credibility and market image, positioning it as a modern, technologically capable service provider. In a market where consumer expectations are increasingly shaped by digital experiences, a professional web platform signals competence, reliability, and commitment to client service quality.

The system's digital marketing capabilities expand the business's market reach beyond the limitations of traditional marketing methods. Properties

listed on the web platform are discoverable by any internet user, including potential buyers who may be located in other cities, provinces, or countries. This expanded reach is particularly valuable for attracting OFW investors and clients from outside the immediate Pavia area who might not encounter traditional local marketing. The ability to reach a broader audience without proportional increases in marketing expenditure improves marketing efficiency and return on investment.

The operational scalability enabled by the system supports business growth without the linear cost increases associated with traditional expansion. As inquiry volumes increase, the automated features of the system absorb much of the additional workload, enabling the business to serve more clients with the same or modestly increased staffing levels. This scalability is essential for small to medium-sized real estate businesses seeking to grow their market share without requiring capital-intensive infrastructure investments.

1.7.4 For the Real Estate Industry

At the industry level, the ILOILO TOP PROPERTY HOMES (ITPH) system contributes to the broader digital transformation of Philippine real estate, particularly in secondary and regional markets that have lagged behind major urban centers in technology adoption. By demonstrating that effective, affordable, and locally appropriate PropTech solutions can be developed and implemented in contexts such as Pavia, the study provides a replicable model for similar initiatives in other municipalities and cities across the Philippines.

The system highlights the feasibility and value of digital transformation for small to medium-sized real estate businesses, challenging the perception that advanced property technology is accessible only to large developers and

national agencies with substantial capital and technical resources. By documenting the development process, challenges encountered, solutions implemented, and outcomes achieved, the study provides practical guidance for other real estate businesses considering similar digital initiatives.

1.7.5 For Future Researchers and Developers

The ILOILO TOP PROPERTY HOMES (ITPH) system and the documentation of its development process serve as a valuable reference for future researchers and developers undertaking similar projects. The study provides a comprehensive example of how property management systems can be designed, developed, and evaluated within the constraints and requirements of a capstone project, offering realistic insights into achievable scope, necessary trade-offs, and effective methodologies.

Future researchers can build upon the foundations established by this study, exploring extensions such as mobile application development, artificial intelligence integration for property recommendations or chatbot enhancement, advanced analytics and business intelligence features, integration with external payment and communication systems, or adaptation to additional geographic markets. The evaluation framework developed for this study can be adapted and applied to the evaluation of similar systems in other contexts.

1.8 Definition of Terms

To ensure clarity, precision, and shared understanding throughout this study, the following terms are defined both conceptually and operationally as they are used within the context of the ILOILO TOP PROPERTY HOMES (ITPH)

web-based real estate management system and the broader research on digital real estate platforms in the Philippine context. Terms are presented in alphabetical order for ease of reference.

Administrator. Conceptually, an administrator refers to an individual with the highest level of authority and responsibility within an organizational system. Operationally, in the context of the ILOILO TOP PROPERTY HOMES (ITPH) system, an Administrator is the highest-level authorized user who possesses full control over all system features, functions, and configurations. Administrators are responsible for managing the overall operation of the real estate platform, including creating and managing user accounts for agents and other staff, overseeing property listing management, monitoring all reservation activities, supervising agent performance, configuring system settings, and ensuring data security and integrity.

Agent. Conceptually, an agent is an individual authorized to act on behalf of another party in business transactions. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, an Agent is an authorized user assigned by the Administrator to perform specific operational responsibilities including handling client inquiries, updating property information, following up on appointments and reservations, and communicating with clients through the integrated chat interface. Agents have access to a defined subset of system features relevant to their designated responsibilities but do not possess full administrative control.

Appointment/Reservation. Conceptually, a reservation refers to the act of reserving or holding something for future use, while an appointment refers to a scheduled meeting or arrangement. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, Appointment and Reservation are used

interchangeably to refer to a confirmed arrangement recorded in the system where a client expresses intent to view, inspect, or secure a property within a specified period.

Automated Communication. Conceptually, automated communication refers to the use of technology to generate and deliver messages without requiring continuous human intervention. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, automated communication encompasses system-generated responses and notifications that assist in handling client inquiries and managing operational workflows.

Client. Conceptually, a client refers to an individual or organization that uses the professional services of another. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, a Client is an individual or potential property buyer who accesses the system to view available real estate listings, submit inquiries, make reservations, and communicate with agents.

Digital Marketing. Conceptually, digital marketing refers to the use of digital channels, platforms, and technologies to promote products and services to consumers. Operationally, in this study, digital marketing encompasses the strategies and tools used to promote real estate properties through the ILOILO TOP PROPERTY HOMES (ITPH) web platform and associated online channels.

MySQL. Conceptually, MySQL is an open-source relational database management system. Operationally, MySQL serves as the database management system for the ILOILO TOP PROPERTY HOMES (ITPH) system, providing reliable, scalable, and efficient storage and retrieval of all system data.

Overseas Filipino Workers (OFWs). Conceptually, OFWs are Filipino nationals who are employed in foreign countries. Operationally, in this study, OFWs represent a significant target market segment for the ILOILO TOP PROPERTY HOMES (ITPH) system, characterized by their reliance on digital platforms for property research and transactions due to physical absence from the Philippines.

PHP (Hypertext Preprocessor). Conceptually, PHP is a widely used open-source server-side scripting language designed for web development. Operationally, PHP serves as the primary programming language for the server-side components of the ILOILO TOP PROPERTY HOMES (ITPH) system.

Property Listing. Conceptually, a property listing is a published advertisement or record of a real estate property available for sale, lease, or reservation. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, a Property Listing is a digital record within the database that represents an individual real estate property.

PropTech (Property Technology). Conceptually, PropTech refers to the application of technology to the real estate industry to improve efficiency, transparency, and user experience. Operationally, PropTech encompasses the digital tools, platforms, and systems used to modernize real estate operations.

Role-Based Access Control (RBAC). Conceptually, RBAC is a security approach that restricts system access based on the roles of individual users. Operationally, the ILOILO TOP PROPERTY HOMES (ITPH) system implements RBAC by defining three primary user roles, Administrator, Agent, and Client, each with specific permissions and access rights.

User Acceptance Testing (UAT). Conceptually, UAT is the final phase of software testing in which intended end users test the system. Operationally, UAT for the ILOILO TOP PROPERTY HOMES (ITPH) system involves representative users from each user category evaluating the system's functionality, usability, and overall performance.

Web-Based System. Conceptually, a web-based system is a software application that operates through a web browser and can be accessed via the internet. Operationally, the web-based system refers to the ILOILO TOP PROPERTY HOMES (ITPH) platform, which serves as the main interface where clients can browse properties, make reservations, and communicate with agents.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive, critical, and integrative review of related literature and studies that establish the theoretical, empirical, and technological foundations for the development of the web-based real estate management system with integrated digital marketing and automated communication features for Pavia, Iloilo City. The review is systematically organized into four major sections: Foreign Studies, Local Studies, Related Literature, and Synthesis of the Reviewed Literature and Studies. Each section is designed to build upon the preceding one, creating a coherent argumentative arc that moves from empirical observation to theoretical abstraction, and finally to integrated justification for the current research.

The Foreign Studies section examines empirical research and system development projects conducted outside the Philippines, focusing on web-

based real estate platforms, digital marketing automation, artificial intelligence in property management, client relationship management systems, and reservation management technologies. These international perspectives provide benchmarks for best practices, technological innovations, and proven methodologies that can be adapted to the Philippine context. The selection of foreign studies prioritizes recent publications (2023–2026) to ensure relevance to contemporary technological standards, while also including seminal works that established foundational concepts in real estate informatics.

The Local Studies section investigates research, market analyses, and platform developments within the Philippines, with particular attention to the real estate digital ecosystem, the unique needs of Overseas Filipino Workers (OFWs) as property investors, government initiatives supporting digital transformation, and the specific challenges facing local real estate markets. These studies ground the proposed system in the realities of the Philippine market and ensure cultural, economic, and regulatory relevance. The local studies section deliberately emphasizes the OFW demographic because this population segment represents both a significant economic force in Philippine real estate and a user group with distinct technological requirements, including cross-timezone accessibility, mobile-first interfaces, and remote decision-making capabilities.

The Related Literature section provides the theoretical scaffolding for the study, drawing from established frameworks in information systems, digital marketing, web development, customer relationship management, security engineering, and software development methodologies. This section transforms the proposed system from a mere technical project into a

theoretically grounded scholarly work. Each theoretical framework is not merely presented but actively interpreted in terms of its implications for real estate system design, user acceptance, and operational effectiveness in the specific context of a mid-sized Philippine municipality.

Finally, the Synthesis section integrates all reviewed materials to identify convergent themes, critical research gaps, and the specific contributions that the current study makes to existing knowledge and practice. Through this synthesis, the justification for the proposed system emerges as both empirically supported and theoretically sound. The synthesis also addresses potential counterarguments and limitations of existing research, positioning the current study as a necessary corrective to identified deficiencies in the literature.

2.1 Foreign Studies

This section presents a detailed examination of studies conducted in international contexts that investigate the development, implementation, and impact of technologies relevant to the proposed web-based real estate management system. The studies are organized thematically to cover web-based real estate platforms, digital marketing strategies, automated communication and CRM systems, reservation and booking technologies, and security frameworks in property management systems. The thematic organization allows for progressive accumulation of evidence, beginning with core platform functionality and extending toward advanced automation and security considerations.

2.1.1 Web-Based Real Estate Management Systems and Property Platforms

The development of web-based systems for real estate management has been a significant area of international research, driven by the global need to modernize property transactions and improve accessibility for buyers, sellers, and agents. These systems represent a fundamental shift from traditional, paper-based, and face-to-face real estate operations to digitized, centralized, and automated platforms that transcend geographic limitations. The transition from analog to digital processes in real estate is not merely a matter of convenience but a structural transformation that reconfigures power dynamics between market participants, democratizes access to information, and introduces new forms of accountability and transparency that were previously impossible in fragmented, informal markets.

Al Ghawail (2026), in a study conducted at Alasmarya Islamic University in Libya, undertook comprehensive research on designing a web-based real estate services management system intended to replace traditional methods and align with modern technological advances. The study employed a mixed-methods approach combining literature review, surveys, interviews, and prototype development. The literature review provided critical insights into existing property management systems and industry best practices, while surveys and interviews collected detailed user requirements, preferences, and identified pain points in traditional real estate operations. The applied analytical approach utilized Unified Modeling Language (UML) to systematically analyze traditional real estate marketing processes and identify precise areas where digital transformation could yield the greatest benefits. The use of UML is particularly noteworthy because it demonstrates that the

development of real estate systems benefits from formal software engineering methodologies rather than ad hoc coding approaches. By modeling use cases, class diagrams, sequence diagrams, and activity diagrams, Al Ghawail ensured that the system's architecture reflected actual business processes rather than imposing alien workflows upon users.

The technological architecture of the developed system utilized PHP as the server-side scripting language, combined with HTML, CSS, and JavaScript for frontend development, all implemented through Visual Studio Code. The database layer employed MySQL, with Google Maps integrated as a map server and reference tool to provide geographic visualization of property locations. Notably, the Agile methodology was chosen as the system development life cycle model, emphasizing repeating progress, continuous stakeholder feedback, and adaptability to changing requirements throughout the development process. The selection of Agile methodology for an academic project is significant because it challenges the traditional assumption that capstone projects must follow rigid waterfall models. Instead, Al Ghawail demonstrated that repeating development is not only feasible but preferable when requirements are emergent and stakeholder needs are multifaceted.

The results of Al Ghawail's study demonstrated that the developed real estate services website successfully addressed the needs of both industry professionals and customers seeking property-related services such as buying, selling, and renting buildings or land. The integration of GPS technology to display properties on interactive maps emerged as a particularly significant feature, as it enabled customers to visualize property locations within their broader geographic context, assess neighborhood accessibility, and make more informed decisions without requiring physical site visits. The researcher

concluded that web-based real estate systems function as comprehensive software solutions designed to streamline and automate diverse property management tasks, offering user-friendly interfaces, efficient data management capabilities, and enhanced communication channels between property owners, tenants, and managers.

This study is significantly relevant to the current research for several reasons. First, the use of PHP and MySQL as core technologies validates the proposed technology stack for the current system, demonstrating that this combination is sufficiently robust for real estate applications in contexts with resource constraints similar to those anticipated in Pavia, Iloilo City. Second, the emphasis on geographic visualization through Google Maps integration supports the current study's planned inclusion of location-based property display, which is essential for helping OFW investors understand neighborhood contexts without physical presence. Third, the Agile methodology employed aligns with the current study's chosen development approach, providing example for repeating development in academic real estate system projects. Fourth, the study's focus on replacing traditional methods with digital solutions mirrors the current study's primary motivation of addressing inefficiencies in manual real estate operations in Pavia, Iloilo City, where many agencies still rely on printed brochures, physical signage, and informal word-of-mouth networks.

Complementing this research, an international study on an Online Platform for Property Listing and Real Estate Operation (International Research Journal of Modernization in Engineering Technology and Science, 2025) developed a centralized web-based system utilizing the MERN stack architecture (MongoDB, Express.js, React.js, and Node.js). This research

addressed the fragmentation and inefficiency of traditional property handling methods by creating a unified digital environment for property owners, buyers, and administrators. The platform featured comprehensive user authentication, property listing creation with multiple image uploads, interactive map-based property visualization using Google Maps API, and direct enquiry management between buyers and property owners.

The methodology adopted for this platform followed a structured and systematic approach encompassing requirement analysis, system design, implementation, and testing. Functional requirements included user registration and authentication for property owners and buyers, property listing with detailed information such as location, price, images, and descriptions, enquiry submission and contact integration for buyers, map-based property visualization, and dashboard management for owners to update or delete listings. Non-functional requirements emphasized data security through JWT authentication, scalable and responsive interfaces for multiple devices, fast data retrieval using MongoDB indexing, and cloud-based image storage for optimized performance.

Testing results demonstrated that the application met all expected objectives with high accuracy and stability. The platform effectively displayed property details, handled user inquiries, and ensured secure user interaction. The integration of Google Maps provided a visual and user-friendly way to explore properties based on geographic locations, while cloud-based image management allowed faster image retrieval and display during property searches. Performance testing confirmed that the application maintained stability under simultaneous user operations, proving its scalability for larger deployments.

The study concluded that compared to conventional manual property handling methods, the developed system significantly reduced time, paperwork, and communication delays. Administrators benefited from simplified monitoring, as all activities could be tracked in real-time through the connected database. The research acknowledged minor limitations, including the lack of advanced analytics, push notifications, and multi-language support, suggesting that future versions could incorporate AI-based recommendation features to further enhance user experience by providing personalized property suggestions.

While the current study differs in its choice of PHP and MySQL rather than the MERN stack, this research provides valuable insights into essential features for real estate platforms, including the importance of map integration, image management, enquiry systems, and scalability testing. The study's emphasis on modular development and comprehensive testing procedures informs the current study's methodology. Furthermore, the explicit identification of missing features (analytics, notifications, AI recommendations) in the MERN platform study provides the current research with a clear roadmap of capabilities that should be prioritized or at least architecturally anticipated in the proposed system's design, even if full implementation exceeds capstone scope.

Further supporting the development of property management systems, research conducted in Siargao Island, Philippines, and published in an international academic journal (Dergipark, 2025) presented the creation and evaluation of an advanced Property Management System (PMS) specifically designed for tourist destination properties. Although geographically situated in the Philippines, the study's publication and methodological rigor position it

within international academic discourse on property management technology. The PMS featured a robust database for managing property, tenant, and invoice information, with essential functionalities including property search, online payment processing, maintenance request management, lease management, and data analytics.

The development process utilized the Unified Process for Object-Oriented Analysis and Design, with particular attention to technical architecture, security, and privacy protocols given the system's handling of sensitive property and payment information. The system leveraged Python Flask as its web application framework, with GraphQL facilitating communication between frontend and backend components. Third-party service integration included payment processing and Google Maps API for geolocation features, while cloud services provided scalability and secure data storage.

Security measures in this system were notably comprehensive, addressing multiple layers of protection. Authentication security employed bcrypt algorithm for password hashing, enforced strong password complexity policies, required password updates every 90 days, and implemented multi-factor authentication using Google Authenticator with time-based one-time passwords. OAuth 2.0 supported single-sign-on and secure API authentication, while an account lockout policy restricted access after five failed login attempts. Authorization mechanisms utilized Role-Based Access Control (RBAC), enforcing the principle of least privilege by restricting data and functionality access based on user roles. Tenants could access only their lease information, submit maintenance requests, and make payments; property managers could oversee tenants, property listings, financial reports,

and maintenance requests; administrators maintained complete system control.

Evaluation through ISO/IEC 25010 Software Quality Standards yielded an overall mean rating of 4.52, indicating high user satisfaction and compliance with international software quality benchmarks. Specific dimension ratings included 4.48 for functional suitability, 4.54 for performance efficiency, 4.49 for compatibility, 4.62 for usability, 4.45 for reliability, and 4.53 for security. These results demonstrated that the developed system met user requirements across all critical quality dimensions.

This study provides the current research with essential guidance on security implementation, quality evaluation standards, and system architecture design. The emphasis on RBAC, multi-factor authentication, and ISO standards informs the current study's approach to system security and testing procedures. The use of ISO/IEC 25010 as an evaluation framework is particularly valuable because it provides the current study with a validated instrument for assessing system quality beyond mere functionality. The current research can adapt this evaluation approach to measure the proposed system's performance across the same quality dimensions, ensuring that the final product meets international standards rather than merely functional requirements. Additionally, the study's handling of sensitive payment information, while not directly replicated in the current study (which excludes financial transaction processing), establishes security protocols that should be preemptively incorporated should future iterations of the system expand to include payment features.

2.1.2 Digital Marketing Strategies and Automation in Real Estate

The transformation of real estate marketing from traditional to digital channels has been extensively studied in international contexts, with researchers examining the effectiveness of various digital strategies, the role of social media, search engine optimization, and the integration of comprehensive marketing automation tools. The shift from print media and physical signage to digital platforms represents more than a change in medium; it constitutes a reconceptualization of how properties are discovered, evaluated, and transacted. Digital marketing enables persistent, searchable, and shareable property presentations that extend the temporal and geographic reach of listings far beyond the limitations of traditional methods.

Research on digital marketing strategy for property developers in Indonesia (European Journal of Business and Management Research, 2025) examined the case of PT Widyaka Propertindo to analyze how digital marketing strategies could improve property sales and market reach. The study grounded its analysis in established marketing theories, specifically the 7Ps Marketing Mix Model originally developed by Booms and Bitner (1981) and further elaborated by marketing scholars including Kotler and Armstrong (2014). The research also drew from digital marketing frameworks by Chaffey and Chadwick (2019), who defined digital marketing as achieving marketing objectives through applying digital technologies and media.

The study's findings revealed that property developers implementing comprehensive digital marketing strategies experienced significantly improved market reach and customer engagement compared to those relying solely on traditional methods such as printed brochures, billboards, and personal networking. The integration of social media marketing, search engine optimization, and online property listings created multiple touchpoints

for potential buyers, particularly younger demographics and overseas investors who primarily rely on digital channels for property research and decision-making. The research emphasized that digital marketing in real estate involves strategic content creation, targeted advertising based on demographic and behavioral data, and data-driven decision making to optimize marketing effectiveness and return on investment.

The study concluded that the transition from traditional promotional methods to digital marketing through web-based systems is essential for competitive advantage in modern real estate markets. This finding directly supports the current study's objective of integrating digital marketing features into the real estate management platform, ensuring that properties listed in Pavia, Iloilo City can reach broader audiences including OFW investors and younger local buyers. The Indonesian context is particularly analogous to the Philippine context because both markets feature rapidly growing middle classes, significant overseas worker populations investing in domestic real estate, and high mobile device penetration rates that make digital channels the most efficient means of market access.

Expanding on digital marketing automation, Birdeye (2026) conducted extensive research on real estate marketing automation solutions, analyzing how automated workflows transform property marketing and client communication across diverse market contexts. The study examined various automation tools including CRM integrations, social media automation platforms, online review management systems, and lead nurturing workflows that operate across multiple digital channels simultaneously.

The research defined real estate marketing automation as the use of software and technology to streamline and automate various marketing tasks,

including lead generation, email campaigns, social media management, online reviews, and client follow-ups. The study found that automation enables agents to create scalable, repeatable, and measurable marketing strategies that attract high-quality leads, nurture relationships throughout the sales cycle, and ultimately shorten the time from initial inquiry to closed transaction. From drip email campaigns and automated workflows to predictive analytics and AI-driven follow-ups, marketing automation empowers agents to work smarter and grow faster.

A particularly compelling case study within this research examined Moss & Company, a property management firm managing over 300 properties. After implementing comprehensive automation tools including automated review responses, AI-generated social media posts, and omnichannel messaging, the company achieved remarkable results: a 761% increase in map views, 218% rise in search views, and 290% boost in web traffic. These quantitative improvements translated directly into increased lead generation and client engagement, demonstrating the tangible business impact of marketing automation.

The study emphasized that automation does not replace personal relationships but enhances them by allowing agents to maintain timely and relevant communication across every stage of the client journey. This insight is crucial for the current study, as it validates the approach of automating routine marketing and communication tasks while preserving the human elements of real estate transactions that require empathy, negotiation skills, and personal judgment. The Moss & Company case study provides particularly strong evidence because it demonstrates results across multiple metrics simultaneously, suggesting that automation creates synergistic effects

where improvements in one channel (map visibility) reinforce gains in others (web traffic and search visibility).

Further supporting the role of automation, NetSuite (2025) conducted comprehensive research on automating real estate processes, examining how automation technology encompasses property listings across multiple platforms, chatbots for customer inquiries, simplified contract management, and AI-powered lead generation tools. The study found that by automating repetitive and time-consuming tasks, real estate professionals can redirect their efforts from administrative paperwork to high-value activities such as building client relationships, conducting property showings, and closing transactions.

Citing the National Association of Realtors Research Group's 2023 Technology Survey, the study noted that 92% of real estate businesses now use at least one social media network to promote listings, maintain relationships with existing clients, generate leads, and track industry trends. Automated CRM systems collect contact information and preferences from customer interactions across social media posts, property listing websites, and other digital touchpoints, organizing this data into centralized databases. Agents can then use these unified profiles to more easily track and follow up with potential clients, identifying those most likely to proceed to the next stage of the transaction process.

The research concluded that automation helps businesses reduce operational costs and scale their operations without losing their ability to deliver personalized attention to clients. As technology continues to evolve, automation will likely bring even more benefits, but only for businesses that strategically implement, apply, and continuously update these tools. This

finding underscores the importance of building automation capabilities into the proposed system from the outset, rather than treating them as optional add-ons. For the Pavia, Iloilo City context, where local real estate agencies may lack dedicated marketing staff, built-in automation features level the competitive playing field by enabling small agencies to maintain consistent digital presence comparable to larger firms with dedicated marketing departments.

2.1.3 Artificial Intelligence and Automated Communication in Real Estate

The integration of artificial intelligence and automated communication systems into real estate operations represents one of the most significant technological advances in the industry, with international research demonstrating substantial improvements in efficiency, client satisfaction, and conversion rates. AI in real estate is not limited to futuristic applications but encompasses practical, deployable tools that address specific operational bottlenecks, particularly in client communication and lead management.

Research on AI-Enhanced CRM with Chatbots for Real Estate Lead Optimization (2026) investigated the integration of artificial intelligence and conversational agent technology into customer relationship management systems specifically designed for real estate applications. The study focused on three critical automation functions: inquiry handling through natural language processing, lead scoring based on prospect behavior and stated preferences, and automatic tour scheduling integrated with calendar systems.

The chatbot integration process followed a structured workflow: users send queries regarding property details, pricing, or location; the chatbot processes these queries using NLP techniques to understand intent and extract key parameters; the system communicates with the CRM database through

APIs to retrieve relevant information; and the chatbot provides instant, contextual responses to the user. This architecture ensures that routine inquiries receive immediate attention regardless of time zones or agent availability, while complex or high-value inquiries can be seamlessly escalated to human agents.

The research employed case studies from five real estate companies to conduct rigorous before-and-after comparisons of system performance. Key performance indicators included containment rate (percentage of queries handled without human intervention), conversion rate (percentage of leads converted to customers), and return on investment (calculated from cost savings and revenue increases). The comparative results demonstrated dramatic improvements across all metrics: response time decreased from high to low (specifically, from hours or days to seconds), conversion rates doubled from 20% to 40%, query handling capacity improved from 40% to 85%, and operational costs associated with initial inquiry management were significantly reduced.

The study concluded that AI chatbot integration transforms CRM systems from passive databases into active engagement tools that operate continuously. This research is particularly relevant to the current study as it provides empirical validation for the proposed chat functionality, establishes performance benchmarks that the current system can aim to achieve, and demonstrates that automated communication does not degrade service quality but rather enhances it by ensuring immediate responsiveness. The doubling of conversion rates is especially significant for the current study because it suggests that the inclusion of automated chat functionality in the proposed

system could directly improve business outcomes for the target agency, not merely provide technological novelty.

Complementing this research, Whippy (2025) investigated AI automation applications in real estate, focusing on specific use cases including automated follow-ups, appointment scheduling, and predictive lead prioritization. The study found that timely follow-ups build rapport and significantly boost referral rates, with AI assistants automatically sending thank-you messages after property showings, sharing property recaps, gathering client feedback, and delivering personalized new listings based on stated preferences and browsing history.

The research demonstrated that AI-driven scheduling tools can consider multiple variables simultaneously, including travel time between showings, client preferences for specific times or days, and priority levels assigned to different leads. This intelligent scheduling helps agents plan more efficient days with fewer conflicts and less unproductive travel time. Predictive lead scoring analyzed prospect behavior across multiple touchpoints including email opens, website property views, virtual tour engagement, and responses to SMS or chatbot prompts, creating composite scores that indicate likelihood to convert.

Real-time alerts notified agents immediately when leads clicked on listings, viewed 3D tours, liked or commented on social media posts, or submitted website forms. This immediacy enabled agents to follow up during moments of peak interest, dramatically improving conversion rates compared to delayed responses. The study concluded that AI tools help schedule outreach at ideal post-sale moments, maximizing response rates and ensuring

satisfied clients share their experiences online, thereby generating valuable social proof and referral business.

This body of research on AI and automated communication provides robust empirical support for the current study's planned chat functionality, automated inquiry responses, and appointment management features. The consistent finding across studies is that automation enhances rather than replaces human agent capabilities, allowing professionals to focus on high-value relationship-building activities while technology handles routine interactions. For the proposed system serving Pavia, Iloilo City, these findings suggest that even a basic automated inquiry acknowledgment and follow-up system can yield measurable improvements in client satisfaction and conversion efficiency.

Nylas (2024) examined how integrated communication platforms enhance real estate software experiences, with a compelling case study of LionDesk, which became the #1 Real Estate CRM in 2022. By integrating email and calendar functionalities through APIs rather than building custom solutions, LionDesk saved 28,242 development hours and \$1.8 million while efficiently serving over 165,000 real estate professionals with centralized communication tools. The study found that streamlined agent-client interactions through integrated email and messaging, automated virtual property tour scheduling, and AI-driven personalization significantly improve productivity and client satisfaction.

The LionDesk case study is instructive for the current research because it demonstrates the economic and operational value of communication integration. While the proposed system may not initially integrate external email and calendar APIs at the same scale, the principle of centralized

communication—where all client interactions are logged, searchable, and actionable within a single interface—directly informs the proposed system's messaging and inquiry management modules. The quantified savings (development hours and costs) provide a business case for prioritizing communication integration over peripheral features.

2.1.4 Reservation Systems and Online Booking Technologies

While much international research on reservation systems focuses on hospitality and tourism applications, the underlying principles and technologies are directly applicable to real estate appointment and reservation management. These studies provide frameworks for designing systems that manage scheduling, prevent conflicts, and enhance user experience through digital booking capabilities. The hospitality industry, with its mature reservation infrastructure and high consumer expectations for instant booking confirmation, offers a model that real estate systems can adapt, albeit with modifications for longer transaction cycles and higher financial stakes.

Research on the design and development of website-based reservation systems for hospitality applications (Jurnal Galaksi, 2025) examined how digital booking systems overcome the limitations of manual reservation processes, including confirmation delays, data duplication, and operational inefficiencies. The study developed a comprehensive web-based booking information system that provided key features including facility information display, real-time availability checking, online booking submission, customer and booking data management, and structured reporting for administrators.

The methodology employed black box testing across 28 test scenarios (24 on the admin side and 4 on the visitor side), demonstrating 100% functional validity. These results confirmed that all designed features operated

properly and met expected functional requirements, ensuring the system was ready for production use. The study concluded that web-based reservation systems successfully overcome manual process limitations, significantly improving operational efficiency and service quality.

The black box testing methodology described in this study is directly applicable to the current research. The proposed system's reservation module can adopt similar testing protocols, ensuring that both administrative and client-side functions operate correctly without requiring users to understand internal system logic. The 100% functional validity result sets a benchmark that the current study should aspire to in its own testing phase, though the smaller scale of the current project may warrant adjusted testing scope.

Similarly, research on web-based reservation systems for tourism education (International Journal of Engineering and Information Systems, 2023) developed an online reservation system for teaching students how to carry out hotel room, airline ticket, and food ordering reservations. The system was designed to provide hands-on experience in online reservation processes, emphasizing the importance of practical digital skills in modern service industries. The study reviewed existing literature on online tourism demand, digital pedagogy in hospitality education, and agile software development methodologies to establish its theoretical and methodological foundations.

These reservation system studies inform the current research regarding essential features for the proposed real estate appointment management module: real-time availability display, immediate confirmation upon submission, automated reminder notifications, structured data management for administrators, and comprehensive testing to ensure functional validity.

While real estate reservations differ from hospitality bookings in their longer cycles and higher financial stakes, the core technological requirements for scheduling, conflict prevention, and user experience remain consistent. The current study adapts these hospitality-derived principles to the specific context of property viewings and reservation requests, where the resource being reserved is not a hotel room but an agent's time and a property's availability status.

2.2 Local Studies

This section examines research, market analyses, and system development projects conducted within the Philippine context. These studies provide essential localized insights into market conditions, user behaviors, regulatory environments, and technological infrastructures that directly influence the design, functionality, and potential success of the proposed web-based real estate management system for Pavia, Iloilo City. The Philippine context presents unique characteristics that distinguish it from international markets: high mobile-first internet usage, significant OFW investment flows, fragmented local property markets, and government initiatives specifically supporting digital transformation for overseas workers.

2.2.1 Philippine Real Estate Digital Platforms Market Analysis

Ken Research (2025) conducted a comprehensive market analysis of the Philippines Real Estate Digital Platforms market, providing detailed segmentation and identifying major players in the expanding digital property ecosystem. The report segmented the market by type (residential platforms, commercial platforms, rental platforms, investment platforms, auction platforms, property management platforms, industrial platforms, agricultural land platforms, vacation rental platforms, foreclosed property platforms), end-

user (first-time home buyers, real estate agents, property developers, investors, renters, corporates), application (property listing, virtual tours, market analysis tools, customer relationship management, online auctions, digital payment integration), and distribution mode (mobile applications, web platforms, social media channels).

The research identified major players in the Philippine digital real estate market, including dedicated property platforms such as Lamudi Philippines, Property24 Philippines, MyProperty.ph, Dot Property Philippines, and Hoppler; general marketplace platforms including Carousell Philippines; and major property developers with digital presences such as Ayala Land, Megaworld Corporation, DMCI Homes, Century Properties, Vista Land, SM Development Corporation, Rockwell Land, and Filinvest Land. The presence of both dedicated real estate platforms and general marketplace sites indicates a fragmented but rapidly expanding digital ecosystem with varying levels of specialization and service quality.

The study found that the Philippine real estate digital platform market is experiencing rapid growth driven by several factors: increasing internet penetration across the archipelago, widespread smartphone adoption, changing consumer preferences toward online property research, and the growing importance of digital channels for reaching overseas Filipino investors. Web platforms and mobile applications serve as primary distribution channels, with property listing and virtual tours being the most utilized applications by consumers.

This market analysis establishes the commercial viability and competitive context for the proposed system. While major platforms dominate national-level property searches, the study reveals opportunities for localized,

specialized platforms serving specific geographic markets such as Pavia, Iloilo City. The current study addresses this market gap by providing a dedicated platform for a defined local area, potentially offering more relevant content, better community context, and more personalized service than national platforms that treat all locations uniformly. The identification of specific market segments (residential, commercial, rental, investment) also suggests that the proposed system should architecturally support multiple property types and user roles from the outset, even if initial implementation focuses on residential sales.

2.2.2 Local Real Estate Platform Development: The Tokkatok Case Study

Codica (2025) documented the development of Tokkatok, a real estate marketplace specifically designed to address the unique challenges of the Philippine property market. This case study is particularly valuable because it emerged directly from the founder's personal experience of attempting to find rental properties in the Philippines through Facebook groups, where listings were frequently outdated, unverifiable, poorly structured, and lacking accountability mechanisms.

The study identified several structural challenges in the Philippine real estate market that Tokkatok was designed to address:

Unstructured market environment: The absence of reliable, dedicated digital platforms specifically built for property rentals and sales in the Philippines forced property seekers to rely on informal channels such as Facebook groups, word-of-mouth referrals, and physical signage. These channels lack verification mechanisms, standardized listing formats, and reliable communication tools.

Technical barriers for non-technical founders: Many local real estate entrepreneurs lack technical backgrounds but recognize the need for digital platforms. They require development partners who can guide product decisions, technology choices, and implementation strategies without overwhelming them with technical complexity.

Regional integration challenges: Connecting Philippine platforms with local services such as GCash for payments and Facebook Business for marketing requires navigating local regulatory constraints, international compliance requirements, and technical compatibility issues.

Accessibility requirements: Given the diversity of digital literacy levels in the Philippines, platforms must be simple and usable for people unfamiliar with sophisticated digital tools, including older users and those in rural areas with limited technology exposure.

Distributed team collaboration: Technology development for Philippine markets often involves distributed teams working remotely across different time zones, requiring strong communication protocols and comprehensive documentation.

Tokkatok was architected around three foundational pillars: security, usability, and performance. The secure and flexible user management system implemented email verification, social login options, multi-factor authentication via email, JWT-based session handling with refresh token rotation, secure time-limited password recovery, step-by-step profile setup, and profile photo uploads stored in AWS S3 with automatic optimization.

The streamlined listing management system featured a multi-step wizard to guide users through property creation, context-aware fields that

adjusted based on property category (residential, commercial, rental), batch drag-and-drop image uploads, map-based address selection with validation, draft mode and publishing controls, admin moderation queue with approve/reject logic, and optional paid features to highlight premium listings.

Advanced search and discovery capabilities were powered by Elasticsearch, providing faceted filters, price range selectors, category-specific filters for precision searching, proximity-based search by location, recommendations for similar listings on property detail pages, and emphasized featured listings in search results.

The study placed particular emphasis on mobile responsiveness and performance optimization, recognizing that smartphones serve as the primary internet access device for most Filipinos. The interface prioritized mobile fluidity through responsive layouts for all screen sizes, optimized image loading for faster rendering, stable performance across older devices and slower networks, and intuitive mobile UX designed for non-technical users.

This case study directly informs multiple aspects of the current research. The emphasis on mobile-first design validates the proposed system's responsive design approach. The identified market challenges (fragmented listings, lack of verification, informal communication channels) mirror the problems identified in the current study's problem statement. The technology choices and feature priorities provide practical benchmarks for the proposed system's development. The Tokkatok experience also serves as a cautionary illustration of the consequences of inadequate digital infrastructure: without structured platforms, markets devolve into inefficient, trust-deficient environments where buyers and sellers waste resources on verification and coordination tasks that automated systems could handle seamlessly.

2.2.3 Government Support for Digital Real Estate Platforms Serving OFWs

The Philippine News Agency (2025) reported on significant government backing for Bahay Deals, an AI-powered online platform specifically designed to help Overseas Filipino Workers invest in real estate in the Philippines. This government endorsement represents a notable development in national policy, recognizing the critical economic role of OFW remittances and investments in the domestic property market while acknowledging the need for reliable digital tools to facilitate remote investment decisions.

The report detailed the specific challenges that OFWs face when investing in Philippine real estate from abroad: difficulty verifying property legitimacy and developer credibility, lack of transparent and standardized pricing information, limited communication channels with developers and agents across time zones, inability to physically inspect properties before making purchase decisions, and vulnerability to fraudulent schemes targeting overseas investors.

Government support for digital platforms addresses these challenges by providing implicit credibility and trust markers for OFW investors. When government agencies endorse or support specific platforms, overseas investors gain confidence that these platforms meet certain standards of transparency, security, and reliability. This endorsement also signals national recognition of the importance of digital transformation in real estate to serve overseas investors effectively.

This study is directly relevant to the current research as it validates the importance of web-based real estate systems in serving OFW clients, a primary target demographic identified in the proposal. The government's

active support for such platforms indicates that the current study aligns with national digitalization priorities and economic development strategies. Furthermore, the documented challenges faced by OFWs (verification, transparency, communication, remote inspection) directly inform the feature requirements for the proposed system, suggesting that comprehensive property information, high-quality imagery, automated communication, and transparent availability status are essential rather than optional features. The current study can position itself as contributing to national development goals by providing municipal-level digital infrastructure that complements national initiatives like Bahay Deals.

2.2.4 OFW Investment Behavior and Digital Platform Preferences

Unicapital Inc. (2024) conducted research on how OFWs build wealth and invest their money, revealing that real estate remains the most favored investment option among overseas workers due to its tangible nature, perceived stability, and potential for generating rental income. The study found that OFWs are increasingly taking proactive approaches to financial growth, moving beyond simply sending remittances to actively building diversified investment portfolios.

The research identified traditional investment preferences among OFWs: real estate for tangible stability and rental income potential; small business ventures or franchises for creating sustainable income streams upon return to the Philippines; and savings accounts or mutual funds for lower-risk, steady returns. However, the study also identified an emerging trend toward stock market investment, driven by higher growth potential, better liquidity compared to real estate, and easy diversification across industries and markets.

Crucially, the research found that modern digital platforms have made investing significantly easier for Filipinos working abroad. Platforms providing 24/7 access to investment opportunities using only smartphones or laptops have democratized access to financial markets. OFWs prefer platforms that offer convenience (access anytime, anywhere), low-cost entry (allowing small initial investments), and educational resources (guides, tips, and real-time data to support confident decision-making).

The study concluded that the best wealth-building strategy for OFWs involves diversification, balancing stable traditional assets like real estate with higher-growth, liquid investments such as stocks. This research supports the current study's focus on creating an accessible, user-friendly web-based platform for property browsing and reservation. The findings emphasize that OFW clients require platforms that operate across time zones, provide comprehensive information to compensate for inability to visit properties physically, and offer immediate, reliable communication channels to address questions and concerns promptly. The identification of convenience, low-cost entry, and educational resources as preferred platform characteristics directly informs the proposed system's design priorities: 24/7 availability, minimal barriers to browsing and inquiry submission, and potentially informative content about the Pavia market and property acquisition processes.

2.2.5 Home Financing and Real Estate Platforms for Non-Traditional Income Earners

Reelist8 (2023) explored home financing solutions specifically for OFWs and freelancers in the Philippine property market, highlighting how digital platforms can serve as intermediaries connecting buyers with appropriate financing options. The study found that navigating home

financing in the Philippines is particularly complex for those with non-traditional income sources, as standard employment documentation may not be available or may not adequately represent their earning capacity.

The research cataloged various financing options available to OFWs and freelancers: home loans from local banks requiring proof of income and employment documentation; government-sponsored housing loans through Pag-IBIG Fund and the Social Security System offering low-interest rates and extended repayment terms; in-house financing from property developers with less stringent documentation requirements and faster approval times; private lending institutions and non-bank financial companies providing alternative options with more flexible requirements; and rent-to-own programs allowing lease-to-purchase arrangements ideal for those without immediate full purchase capacity.

The study emphasized that digital platforms serving OFWs must extend beyond simple property listings to include comprehensive educational resources about financing options, requirements, and application processes. By providing this guidance within the platform, OFWs can explore financing options concurrently with property browsing, making more informed decisions about affordability and purchase feasibility.

This research informs the current study by suggesting that the proposed system could enhance its value to OFW clients by including property financing information, connecting buyers with financing resources, or at minimum providing educational content about the various paths to property acquisition. While the current study's scope excludes direct financial transaction processing, the inclusion of financing guidance aligns with the broader goal of serving OFW clients comprehensively. Even basic

informational content about Pag-IBIG requirements, typical down payment percentages, or developer financing options could significantly reduce the cognitive load on OFW buyers who otherwise must research financing separately from property searching.

2.2.6 Philippine Real Estate Industry Challenges and Digital Imperatives

Multiple local studies collectively establish the challenging context of the Philippine real estate industry and the imperative for digital transformation. Research by the Philippine Institute for Development Studies indicates that housing affordability remains a critical national issue, as property prices continue to rise faster than household incomes, limiting homeownership capacity for many Filipino families, particularly in urban areas where prices are significantly higher than rural regions.

Studies on Philippine real estate industry practices identify persistent problems including unethical practices by some developers and agents, unlicensed real estate practitioners operating without proper accreditation, documentation issues in property transactions, and insufficient regulatory enforcement. These challenges affect buyer trust and decision-making confidence when investing in real estate properties, creating an environment where transparency and accountability mechanisms are desperately needed.

Despite these challenges, the real estate sector contributes significantly to national economic development by providing employment across construction, sales, property management, and ancillary services; generating housing solutions for growing populations; and contributing government revenue through taxes, transfer fees, and registration costs. This economic importance underscores the national interest in improving industry practices and transaction efficiency.

The Philippine Statistics Authority (2021) documented the increasing use of internet and mobile devices across the Philippines, fundamentally changing how businesses promote services and how consumers research purchases. This digital shift affects all sectors, including real estate, where traditional methods of property promotion through printed flyers, newspaper advertisements, and physical signage are rapidly losing effectiveness compared to digital channels.

The Department of Trade and Industry (2022) formally emphasized that small and medium enterprises, including local real estate agencies, must shift from traditional marketing methods to digital platforms to remain competitive and reach broader customer bases. This government guidance establishes digital transformation not merely as an option but as a necessity for business survival and growth.

University of the Philippines studies demonstrate that online property listings and web-based systems measurably improve accessibility for property seekers, enhance customer engagement through interactive features and immediate feedback, and support better-informed decision-making through comprehensive, easily comparable property information. These findings support the transition from traditional estate promotion to digital marketing strategies using web-based systems in the local real estate industry.

The convergence of these local studies creates a compelling narrative of necessity: the Philippine real estate industry faces structural challenges (affordability, unethical practices, documentation issues) that digital systems can partially mitigate through transparency, standardization, and automated record-keeping. Simultaneously, national digital adoption trends and government policy create favorable conditions for system implementation.

The proposed system for Pavia, Iloilo City enters this environment as a timely intervention that addresses documented market failures while aligning with national digitalization trajectories.

2.2.7 Digital Investment Education and OFW Empowerment

Bria Homes (2024) researched smart investment options for OFWs, emphasizing the critical role of online trading platforms and digital educational tools in facilitating property investment decisions. The study found that numerous online platforms now assist OFW investors in learning fundamentals of real estate investment, property valuation techniques, market analysis methodologies, and legal considerations in property acquisition.

The research highlighted that digital platforms serving OFW investors must prioritize three core principles: transparency in pricing and property details, comprehensive property information including virtual tours and neighborhood data, and responsive customer support capable of addressing inquiries across international time zones. The availability of online resources and virtual property tours has become essential for OFWs who cannot easily visit properties before making purchase commitments.

The study concluded that empowering OFWs with digital tools and educational resources enables them to make more confident, informed investment decisions while reducing their vulnerability to misinformation, fraudulent schemes, and high-pressure sales tactics. This research reinforces the current study's emphasis on providing detailed property information, high-quality imagery, location details, and automated communication features as essential components for serving OFW clients effectively. The three principles identified—transparency, comprehensiveness, and

responsiveness—can serve as direct design requirements for the proposed system's user interface and feature set.

2.3 Related Literature

This section presents the theoretical foundations, conceptual frameworks, and technological principles that underpin the proposed web-based real estate management system. The literature is organized into thematic areas covering information systems theory, digital marketing concepts, web development architectures, customer relationship management, reservation system design, security engineering, specific technology platforms, software development methodologies, user experience design, database principles, and geographic information systems. Each thematic area is developed with sufficient depth to support scholarly analysis while maintaining clear relevance to the practical requirements of real estate system development.

2.3.1 Information Systems Theory and Technology Acceptance Frameworks

Information Systems (IS) theory provides the fundamental conceptual lens through which the proposed real estate management system can be understood as more than a technical artifact, but as a socio-technical intervention designed to transform organizational processes and improve stakeholder outcomes. An information system is formally defined as an organized combination of people, hardware, software, communication networks, data resources, and policies that stores, retrieves, transforms, and disseminates information in an organization (Laudon & Laudon, 2020). In the context of real estate, information systems serve as the digital backbone for managing property portfolios, client relationships, marketing campaigns, and transaction workflows.

The Socio-Technical Systems theory, originally developed by Trist and Bamforth (1951) in their seminal study of coal mining operations and subsequently expanded by numerous organizational theorists, posits that effective organizational systems must jointly optimize both social and technical subsystems. The theory argues that optimizing technical components alone while ignoring social factors such as user behaviors, organizational culture, power dynamics, and workflow patterns leads to suboptimal outcomes and system rejection. Conversely, focusing solely on social aspects without adequate technical capability results in inefficient, manual processes that cannot scale.

This theory is particularly relevant to real estate management systems because successful implementation requires careful attention to how agents currently handle client relationships, how administrators manage property listings, and how clients prefer to search for and inquire about properties. The proposed system must not merely digitize existing processes but reimagine them in ways that respect established professional relationships while introducing efficiencies that stakeholders find genuinely valuable. For example, automating inquiry responses must be implemented in ways that do not make clients feel ignored or depersonalized, and administrative dashboards must align with how managers actually oversee operations rather than imposing alien workflows. In the Pavia context, where real estate agencies may be family-owned businesses with long-established personal networks, the socio-technical balance is especially delicate; the system must enhance rather than disrupt the personal trust relationships that drive local real estate transactions.

The Technology Acceptance Model (TAM), developed by Fred Davis (1989) at the MIT Sloan School of Management, provides the most widely cited theoretical framework for understanding how users come to accept and use new information technologies. TAM posits that two primary beliefs determine user acceptance: perceived usefulness and perceived ease of use. Perceived usefulness is defined as the degree to which a person believes that using a particular system would enhance his or her job performance or effectiveness. Perceived ease of use refers to the degree to which a person believes that using a system would be free of physical and mental effort.

TAM further proposes that perceived usefulness and perceived ease of use are influenced by external variables such as system characteristics, training, documentation, and user support. These beliefs then shape behavioral intentions to use the system, which in turn determine actual system usage. Finally, actual usage influences job performance, productivity gains, and user satisfaction, creating feedback loops that reinforce or diminish continued use.

In the context of the proposed real estate system, TAM provides specific guidance for design decisions. To maximize perceived usefulness, the system must clearly demonstrate benefits in terms of time savings (faster inquiry handling, automated listing updates), improved client management (organized contact histories, automated follow-ups), and increased sales effectiveness (better lead tracking, faster response times). To maximize perceived ease of use, the interface must be intuitive enough that agents and administrators can perform core tasks with minimal training, navigation must follow familiar patterns, and help resources must be readily accessible. For the Pavia market, where agents may have varying levels of digital literacy,

perceived ease of use is not merely a convenience factor but a potential adoption barrier that must be aggressively addressed through simple interfaces, clear labeling, and minimal required steps for common tasks.

The Unified Theory of Acceptance and Use of Technology (UTAUT), proposed by Venkatesh, Morris, Davis, and Davis (2003), extends TAM by incorporating additional determinants of technology acceptance and identifying moderating variables that influence the strength of relationships in the model. UTAUT identifies four key constructs that directly influence behavioral intention and use behavior:

Performance expectancy refers to the degree to which users believe the system will help them attain gains in job performance. For real estate agents, this translates to beliefs about whether the system will help them close more deals, manage more clients efficiently, or provide better service quality.

Effort expectancy is the degree of ease associated with the use of the system. This encompasses not just interface simplicity but also the learning curve required to become proficient, the availability of training resources, and the clarity of system documentation.

Social influence is the extent to which users perceive that important others (managers, colleagues, clients) believe they should use the system. In real estate agencies, if management actively encourages system use and colleagues demonstrate successful outcomes, individual agents are more likely to adopt the platform enthusiastically.

Facilitating conditions refer to the degree to which users believe organizational and technical infrastructure exists to support system use. This includes reliable internet connectivity, adequate hardware, technical support

availability, and organizational policies that accommodate the time needed to learn and use the system effectively.

UTAUT also identifies four key moderating variables: gender, age, experience, and voluntariness of use. For example, research suggests that performance expectancy may be more influential for men, while effort expectancy may be more important for women; that social influence may be stronger for older users; and that facilitating conditions matter more when system use is mandatory rather than voluntary.

For the proposed real estate system, UTAUT suggests that successful adoption depends on a holistic approach addressing all four direct determinants. The system must demonstrably improve agent and administrator performance, minimize learning effort, garner visible support from agency leadership, and be supported by adequate technical infrastructure and organizational policies. The moderating variables are particularly relevant in the Philippine context, where real estate agencies often employ agents across wide age ranges and varying experience levels, necessitating flexible training approaches and interface designs that accommodate diverse user profiles.

The Information Systems Success Model, originally developed by DeLone and McLean (1992) and significantly updated in 2003, provides a comprehensive framework for evaluating IS effectiveness across six interrelated dimensions:

System quality refers to technical performance aspects including response time, system reliability, usability, navigability, and visual appeal. For web-based real estate systems, high system quality means pages load quickly even with multiple high-resolution property images, the system

remains available during peak usage periods, and the interface is visually professional and easy to navigate.

Information quality encompasses the accuracy, completeness, timeliness, relevance, and consistency of information provided by the system. In real estate applications, information quality requires that property listings accurately reflect current availability and pricing, that property descriptions are comprehensive and honest, that images faithfully represent actual conditions, and that status updates (available, reserved, sold) are applied immediately.

Service quality addresses the quality of support services provided to system users, including technical support responsiveness, training quality, and documentation completeness. Even well-designed systems require ongoing support to address edge cases, user questions, and technical issues.

Use encompasses the actual consumption of system features by intended users, including frequency of use, duration of sessions, and variety of features utilized. High use indicates that users find the system valuable enough to integrate into their regular workflows.

User satisfaction represents users' overall positive or negative feelings about the system, measured through surveys, interviews, or behavioral indicators such as continued use and recommendation to others.

Net benefits capture the ultimate positive impacts of the system on individual productivity, organizational effectiveness, and broader stakeholder outcomes. For real estate systems, net benefits include increased sales, reduced administrative costs, improved client satisfaction, and enhanced competitive positioning.

These theoretical frameworks collectively establish that the success of the proposed web-based real estate system depends not merely on technical functionality but on how users perceive, adopt, and derive value from the system in their professional contexts. The frameworks also provide validated instruments and constructs that the current study can use to evaluate system success beyond simple technical checklists, enabling a more nuanced and academically rigorous assessment of project outcomes.

2.3.2 Digital Marketing Theory and Real Estate Promotion

Digital marketing represents the strategic application of internet-based technologies, platforms, and channels to achieve marketing objectives and create value for both businesses and consumers. Chaffey and Chadwick (2019), in their comprehensive text on digital marketing strategy, define digital marketing as "achieving marketing objectives through applying digital technologies and media." This definition emphasizes that digital marketing is not about technology for its own sake but about leveraging technology to accomplish concrete business goals such as lead generation, brand awareness, customer engagement, and transaction facilitation.

In real estate contexts, digital marketing encompasses a broad spectrum of activities: property listings on dedicated websites and general marketplace platforms; social media promotion through organic posts and paid advertising; search engine optimization to improve visibility in organic search results; email marketing campaigns targeting specific buyer segments; content marketing through blog posts, market reports, and neighborhood guides; and online advertising through pay-per-click campaigns, display ads, and retargeting strategies.

The 7Ps Marketing Mix Model, originally developed by Booms and Bitner (1981) as a services marketing extension of McCarthy's traditional 4Ps (Product, Price, Place, Promotion), adds three additional elements critical for service contexts: People, Process, and Physical Evidence. This expanded framework provides comprehensive guidance for real estate digital marketing:

Product in real estate digital marketing refers to the properties being marketed, encompassing not just physical structures but also locations, neighborhoods, amenities, and lifestyle associations. Effective digital presentation requires comprehensive property details, accurate specifications, and compelling descriptions that help potential buyers envision living or investing in the property.

Price encompasses listing prices, payment terms, financing options, and value propositions. Digital platforms enable dynamic pricing displays, comparison tools, and transparent cost breakdowns that help buyers assess affordability and value. The ability to filter and sort properties by price range is one of the most frequently used features in real estate websites.

Place in digital contexts refers to the channels through which properties are marketed and discovered. This includes the real estate agency's own website, national property portals (Lamudi, Property24), social media platforms (Facebook, Instagram), search engines (Google), and email inboxes. Effective digital marketing requires strategic presence across multiple channels where target buyers are likely to search.

Promotion includes all digital communications designed to attract attention, generate interest, and drive action. This encompasses social media posts showcasing properties, email newsletters featuring new listings, blog

content addressing buyer concerns, paid advertising targeting specific demographics, and search engine optimization to improve organic visibility.

People refers to the human element of real estate service delivery, including agents, administrators, and support staff who interact with clients through digital channels. In digital marketing, people are represented through agent profiles, biography pages, video introductions, and responsive communication that projects professionalism, expertise, and trustworthiness.

Process involves the workflows and procedures for handling inquiries, managing listings, processing reservations, and completing transactions. Digital marketing must accurately represent these processes, setting clear expectations about response times, viewing procedures, documentation requirements, and reservation protocols.

Physical Evidence in digital real estate marketing translates to the tangible proof of service quality that users can observe through screens. This includes website design quality, property photograph resolution and composition, virtual tour production values, the professionalism of listing descriptions, and the overall user experience of interacting with the digital platform. High-quality physical evidence builds trust and credibility, while poor-quality presentation undermines confidence regardless of actual property quality.

Kotler and Armstrong (2014), in their foundational marketing text, emphasize that the marketing mix must be adapted to specific target markets, competitive environments, and service characteristics. For real estate in Pavia, Iloilo City, this means tailoring digital marketing strategies to local market conditions, the specific demographics of likely buyers (local families,

investors, OFWs), and the competitive landscape of existing property marketing channels.

Search Engine Optimization (SEO) theory provides systematic frameworks for improving the visibility of web-based real estate platforms in organic search engine results. SEO encompasses technical optimization (site structure, page speed, mobile responsiveness, secure connections), on-page optimization (keyword-rich content, meta tags, header structure, image alt text), and off-page optimization (backlink building, social signals, local citations). For real estate systems, effective SEO ensures that potential buyers searching for location-specific terms such as "house for sale Pavia Iloilo," "property investment Iloilo City," or "affordable homes Pavia" can easily discover the platform.

Local SEO is particularly important for real estate, as property searches are inherently location-based. Local SEO strategies include claiming and optimizing Google Business Profile listings, ensuring consistent Name-Address-Phone (NAP) information across directories, generating local reviews, and creating location-specific content that signals geographic relevance to search engines. For the proposed system, local SEO strategies should target not just "Iloilo City" but specific barangays within Pavia, neighborhood landmarks, and proximity to key infrastructure (schools, markets, churches) that local buyers search for.

Content marketing theory, articulated by Pulizzi (2013) and others, emphasizes the creation and distribution of valuable, relevant, and consistent content to attract and retain a clearly defined audience. Unlike traditional advertising that interrupts consumers with promotional messages, content marketing attracts consumers by providing genuinely useful information. In

real estate, effective content marketing includes neighborhood guides that describe local amenities, schools, and transportation; market trend reports that analyze pricing patterns and investment potential; home buying guides that explain financing options and legal procedures; and property investment case studies that demonstrate successful outcomes.

The theory posits that providing valuable content builds trust and credibility with potential clients over time, establishing the real estate business as a knowledgeable authority in the local market. This trust translates into preferential consideration when content consumers are ready to buy or sell property, effectively shortening sales cycles and improving conversion rates. For the Pavia context, content marketing might include guides to barangay characteristics, flood risk assessments, transportation access to Iloilo City proper, and development projections that help buyers understand long-term property value trajectories.

Social media marketing theory leverages platforms such as Facebook, Instagram, TikTok, and LinkedIn to engage with potential clients, showcase properties, build brand awareness, and foster community relationships. The theory emphasizes the visual nature of real estate marketing, as property photographs and videos generate significantly higher engagement rates than text-only posts. Research consistently shows that social media posts with images receive substantially more interactions (likes, comments, shares) than text-only content, and video content achieves even higher engagement levels.

Social media algorithms favor content that generates meaningful interactions, making it essential for real estate platforms to create engaging, shareable content that encourages user participation. This includes asking questions in posts, creating polls about neighborhood preferences, sharing

behind-the-scenes content about property preparation, and encouraging satisfied clients to share their experiences and tag the agency. For Philippine markets, Facebook remains the dominant social platform, making Facebook integration and sharing capabilities particularly important features for the proposed system.

2.3.3 Web-Based System Architectures and E-Commerce Principles

Web-based systems, also known as web applications, represent a fundamental category of software applications that run on remote web servers and are accessed by users through web browsers over internet connections. Unlike traditional desktop applications that require installation on local computers, web-based systems offer the critical advantage of platform independence, enabling access from any device with a browser and internet connectivity. This characteristic makes web-based systems particularly suitable for real estate applications where clients may search for properties from home computers, work laptops, or mobile devices; where agents may need to update listings while away from the office; and where administrators may require access during travel or after hours.

The three-tier architecture represents the most widely adopted design pattern for robust web-based systems, separating concerns across three distinct logical and often physical layers:

The Presentation Tier, also known as the Client Layer or User Interface Layer, handles all user interaction functions including interface rendering, input capture, and client-side validation. Technologies employed at this tier typically include HTML for document structure, CSS for visual styling and layout, and JavaScript for interactive behavior and dynamic content updates. In modern web development, this tier may also incorporate frontend

frameworks such as React, Vue, or Angular, though the proposed system will utilize foundational technologies to maintain simplicity and compatibility.

The Application Tier, variously called the Business Logic Layer, Logic Tier, or Middle Tier, processes business rules, manages workflows, handles data processing, and coordinates communication between the presentation and data tiers. This tier implements the core functionality of the system, enforcing rules such as "only administrators can approve property listings," "clients cannot reserve already-sold properties," and "inquiry responses must be logged for accountability." The application tier is typically implemented using server-side languages such as PHP, Python, Java, or Node.js.

The Data Tier, also referred to as the Database Layer or Persistence Layer, stores and retrieves data, manages transactions, ensures data integrity, and handles backup and recovery operations. Relational database management systems such as MySQL, PostgreSQL, or Microsoft SQL Server typically serve this tier, though NoSQL databases such as MongoDB are increasingly used for specific application requirements.

This three-tier architecture provides several significant advantages for web-based real estate systems. Separation of concerns enables independent development, testing, and maintenance of each tier, allowing frontend designers to modify user interfaces without affecting business logic, and database administrators to optimize queries without disrupting application functionality. Scalability is enhanced because each tier can be scaled independently based on demand; for example, additional web servers can be added to the presentation tier during high-traffic periods without modifying the database tier. Maintainability improves because changes to one tier's

technology or implementation can often be made with minimal impact on other tiers, provided that inter-tier interfaces remain consistent.

Client-server architecture forms the foundational communication model for web-based systems. In this model, clients (typically web browsers running on user devices) request services and resources from centralized servers. Clients initiate communication by sending HTTP requests containing specific methods (GET for retrieving data, POST for submitting data, PUT for updating resources, DELETE for removing resources) along with relevant parameters, headers, and body content. Servers process these requests, interact with databases or other backend services as needed, and return HTTP responses containing status codes (200 for success, 404 for not found, 500 for server errors), response headers, and response bodies containing requested data or confirmation messages.

In real estate systems, this client-server interaction occurs continuously: when a client searches for properties, the browser sends a GET request with search parameters; the server queries the database for matching properties and returns formatted results. When a client submits an inquiry, the browser sends a POST request with form data; the server validates the data, stores it in the database, triggers any automated responses, and returns a confirmation. When an administrator updates a property status, the browser sends a PUT request; the server updates the database record and may broadcast the change to other users viewing that property.

E-commerce principles, while developed primarily for online retail transactions, provide valuable frameworks for designing real estate web-based workflows. E-commerce research has extensively studied product catalog management, which directly parallels property listing management in real

estate. Principles such as comprehensive product descriptions, high-quality imagery, customer reviews, and related product recommendations all have direct analogues in real estate systems.

The e-commerce concept of shopping cart functionality parallels property shortlisting or favorites features in real estate platforms, where clients can save interesting properties for later comparison and review. The checkout process in e-commerce, involving review of selected items, confirmation of details, and final submission, parallels the reservation submission process in real estate, where clients review property details, confirm their contact information, and submit reservation requests.

Trust theory in e-commerce, developed by researchers such as McKnight, Choudhury, and Kacmar (2002), identifies factors that build consumer confidence in online platforms. These factors include website quality and professional design, security indicators such as SSL certificates and trust badges, privacy policies explaining how personal data is used, clear return and refund policies, and social proof through customer reviews and testimonials. Applied to real estate systems, these trust factors translate to professional website design, visible security measures protecting client data, clear privacy policies, transparent terms for reservations and appointments, and testimonials or success stories from satisfied clients.

Responsive web design theory, articulated by Ethan Marcotte (2010) in his seminal article, advocates for creating websites that provide optimal viewing and interaction experiences across a wide range of devices, from desktop computer monitors to mobile phone screens. Responsive design employs three core technical components: fluid grids that use relative units (percentages) rather than fixed units (pixels) for layout dimensions; flexible

images that scale within their containing elements to prevent overflow on smaller screens; and CSS media queries that apply different styling rules based on device characteristics such as screen width, height, orientation, and resolution.

Given the high prevalence of mobile device usage in the Philippines, where smartphones often serve as the primary or sole internet access device for many users, responsive design is not merely a nice-to-have feature but an essential requirement. OFW clients accessing the platform from abroad may predominantly use mobile devices due to convenience or limited access to desktop computers. Local clients in Pavia may similarly rely on smartphones for property browsing. A mobile-first design approach, which prioritizes the mobile experience during design and progressively enhances features for larger screens, ensures that the broadest possible audience can effectively use the platform.

2.3.4 Customer Relationship Management and Communication Automation

Customer Relationship Management (CRM) theory encompasses the strategic approaches, processes, and technologies that organizations use to manage and analyze customer interactions and data throughout the customer lifecycle. The fundamental goal of CRM is to improve business relationships with customers, assist in customer retention, and drive sales growth. CRM systems consolidate customer information from various communication channels and touchpoints to provide a comprehensive, unified view of each customer's history, preferences, interactions, and potential value to the organization.

Buttle (2009) proposed the CRM value chain model, which identifies five primary stages of CRM implementation that create value for both the organization and its customers:

Customer portfolio analysis involves examining the customer base to identify high-value prospects, segmenting customers by characteristics and behaviors, and understanding the distribution of customer value across the portfolio. In real estate applications, this translates to analyzing which types of properties attract the most inquiries, which client segments (local buyers, OFWs, investors) generate the highest conversion rates, and which geographic areas or price points show the strongest demand.

Customer intimacy refers to building deep understanding of individual customer needs, preferences, timelines, and constraints. For real estate agents, this means maintaining detailed profiles of client preferences including desired locations, property types, budget ranges, financing needs, and decision timelines. Digital CRM systems enable systematic capture and retrieval of this information, ensuring that agents can personalize interactions even when managing large numbers of prospects.

Network development involves creating and maintaining relationships with partners, suppliers, and complementary service providers who can enhance value delivery to customers. In real estate, networks include property developers, financial institutions offering mortgages, home inspectors, contractors, and legal professionals. A comprehensive CRM system may track these relationships and facilitate referrals or coordinated service delivery.

Value proposition development entails creating compelling offers and experiences tailored to specific customer segments. For real estate platforms, value propositions might include exclusive early access to new listings for

registered users, personalized property recommendations based on search history, or bundled services such as financing consultation and legal referral.

Managing the customer lifecycle addresses the progression of customer relationships from initial awareness through consideration, purchase, ownership, and eventual resale or upgrade. In real estate, this lifecycle is typically longer than in many other industries, with clients potentially returning years later for additional investments or referring family members. CRM systems must support long-term relationship maintenance, including periodic check-ins, market updates, and anniversary acknowledgments.

Automated communication theory examines how technology can streamline, standardize, and enhance communication processes while maintaining sufficient personalization to avoid impersonal or robotic interactions. Automation in client communication involves using software to send triggered messages based on specific client actions, predefined timelines, or system events. The goal is not to eliminate human communication but to ensure that routine communications occur consistently and immediately, freeing human agents to focus on complex, emotionally nuanced, or high-value interactions.

In real estate applications, automated communication workflows might include: immediate acknowledgment messages when inquiries are submitted, confirming receipt and setting expectations for response time; scheduled follow-up messages if no agent response occurs within a specified timeframe (e.g., 24 hours); appointment reminder messages sent 24 hours before scheduled property viewings; status update notifications when reserved properties change status; and post-transaction follow-up sequences requesting

feedback, offering assistance, and maintaining relationships for future referrals.

Chatbot technology and conversational agents represent increasingly sophisticated components of automated communication systems. Rule-based chatbots operate using predefined scripts, decision trees, and pattern matching to respond to common inquiries with structured answers. For example, a rule-based real estate chatbot might recognize questions about property availability, pricing, or location and provide standardized responses drawn from the property database. While limited in their ability to handle unanticipated queries, rule-based chatbots are relatively simple to implement, maintain, and control.

AI-powered chatbots leverage natural language processing (NLP), machine learning, and sometimes large language models to understand and respond to more complex, conversational queries. These systems can interpret intent even when queries are phrased in varied ways, maintain context across multi-turn conversations, and generate human-like responses. In real estate applications, AI chatbots can handle nuanced inquiries about neighborhood characteristics, financing options, or comparison between properties. However, AI chatbots require more extensive training data, ongoing refinement, and careful monitoring to ensure accurate, appropriate responses.

The theory of service quality in digital communication, derived from the SERVQUAL model originally developed by Parasuraman, Zeithaml, and Berry (1988) and subsequently adapted for online contexts, identifies five key dimensions of service quality that apply to digital real estate communication:

Responsiveness refers to the willingness and speed with which service providers respond to customer requests and inquiries. In digital contexts, this

translates to response time for chat messages, email replies, and inquiry submissions. Research consistently shows that response time significantly impacts customer satisfaction and conversion rates in real estate, with faster responses correlating strongly with higher engagement and transaction completion.

Reliability concerns the ability to perform promised services dependably and accurately. For real estate digital communication, reliability means that automated responses contain accurate information, that promised follow-ups occur as scheduled, and that system-generated messages reflect current property status rather than outdated data.

Assurance involves the knowledge, courtesy, and ability of service providers to convey trust and confidence. In automated systems, assurance is communicated through professional language, clear identification of the sender (system vs. human agent), transparent disclosure of automation, and easy escalation paths to human agents when needed.

Empathy refers to the caring, individualized attention provided to customers. While automation inherently risks depersonalization, empathetic automated communication can be achieved through personalized greetings, reference to specific properties or previous interactions, acknowledgment of individual circumstances, and tone that conveys warmth and understanding.

Tangibles encompass the physical facilities, equipment, and appearance of service providers. In digital contexts, tangibles translate to the visual design of communication interfaces, the professionalism of email templates, the clarity of chat window layouts, and the overall aesthetic quality of digital touchpoints.

2.3.5 Reservation and Appointment Management Systems

Reservation systems theory provides structured frameworks for designing information systems that manage the booking, scheduling, and allocation of resources, services, or appointments. While extensively developed in hospitality, healthcare, and transportation sectors, these principles apply directly to real estate where clients schedule property viewings, reserve properties for purchase consideration, and book consultation appointments with agents.

The theory of online appointment scheduling emphasizes several critical success factors: real-time availability display that shows current open slots and prevents double-booking; instant confirmation that provides users with immediate feedback that their reservation has been successfully recorded; automated reminder notifications sent via email or SMS to reduce no-show rates; easy rescheduling and cancellation options that respect user autonomy while capturing freed slots for reallocation; and waitlist functionality that allows interested clients to register for fully booked time slots or properties.

Research in healthcare appointment systems, while serving a different industry, provides robust evidence for the benefits of online scheduling. Studies consistently demonstrate that online appointment systems reduce administrative workload for staff, decrease no-show rates by 20-40% through automated reminders, improve patient satisfaction by offering scheduling convenience outside business hours, and optimize resource utilization by filling gaps in schedules more efficiently. These benefits translate directly to real estate contexts, where agent time is a valuable resource, missed viewings

represent lost opportunities, and client convenience directly influences engagement levels.

Database design theory for reservation systems emphasizes the importance of maintaining data integrity through proper entity relationships, transaction management, and concurrency control. In relational database design, reservation systems typically involve several interconnected entities: resources being reserved (properties, time slots, agent availability); customers making reservations (clients, buyers, investors); reservation records linking customers to resources with timestamps and status indicators; and potentially payment or deposit records associated with reservations.

Critical database design principles for reservation systems include:

Normalization to eliminate data redundancy and prevent update anomalies. For example, customer contact information should be stored in a dedicated customers table and referenced through foreign keys in reservation records, rather than duplicated in each reservation entry.

Foreign key constraints to enforce referential integrity, ensuring that reservation records cannot reference non-existent properties or customers, and that deletion of referenced records is handled appropriately (either prevented or cascaded based on business rules).

Atomic transactions to ensure that multi-step reservation processes complete entirely or not at all. If a client submits a reservation request, the system must record the reservation, update property status, notify the agent, and send confirmation to the client as a single atomic unit. If any step fails, the entire transaction should roll back to prevent inconsistent states.

Concurrency control to prevent race conditions when multiple clients attempt to reserve the same property or time slot simultaneously. Database locking mechanisms or optimistic concurrency control techniques ensure that only one client successfully claims a resource, while others receive clear notification that the resource is no longer available.

Calendar integration theory examines how reservation systems can synchronize with external calendar applications such as Google Calendar, Microsoft Outlook, or Apple Calendar. While full calendar integration may exceed the scope of the proposed capstone system, the underlying principles inform the design of internal scheduling displays: clear date and time formatting, timezone awareness for international clients (particularly important for OFWs), visual indicators of availability and booked slots, and conflict detection algorithms that prevent overlapping appointments.

2.3.6 Property Management and Real Estate Operations Theory

Property management theory encompasses the comprehensive processes, systems, and strategies used to oversee residential, commercial, and industrial real estate properties on behalf of owners or investors. Effective property management spans the entire property lifecycle, from initial acquisition and marketing through leasing or sale, ongoing occupancy or ownership, maintenance and improvement, and eventual disposition or resale.

The property lifecycle model identifies distinct stages that digital systems must support:

Acquisition involves identifying, evaluating, and purchasing properties for the management portfolio. While the proposed system focuses primarily on marketing and transaction management, understanding acquisition helps

contextualize how properties enter the system and what information must be captured initially.

Marketing and leasing/sale encompasses all activities required to attract and secure tenants or buyers. This stage includes property preparation, photography and description creation, listing distribution across channels, inquiry management, viewing scheduling, application processing, and lease or sale agreement execution. The proposed system concentrates heavily on this stage, providing tools for listing creation, inquiry handling, appointment management, and reservation tracking.

Occupancy and ownership management involves rent collection, lease enforcement, tenant relations, and ongoing communication during the tenancy period. While not the primary focus of the proposed system, understanding occupancy management informs the design of post-reservation workflows and client relationship maintenance features.

Maintenance and improvement addresses the physical upkeep and enhancement of properties. Digital systems may track maintenance requests, schedule repairs, and document property condition over time.

Disposition involves the eventual sale or transfer of properties from the portfolio. Accurate records of property history, improvements, and financial performance support disposition decisions and valuation.

Listing management theory addresses how properties are presented in digital platforms to maximize visibility, engagement, and conversion. Key principles derived from e-commerce and real estate research include:

Comprehensive information provision ensures that listings answer the most common buyer questions without requiring additional inquiry. This

includes not just basic specifications (bedrooms, bathrooms, floor area, lot area) but also detailed descriptions of features, neighborhood characteristics, nearby amenities, utility arrangements, and any restrictions or special conditions.

High-quality visual content significantly influences buyer interest and engagement. Research consistently demonstrates that listings with professional photographs receive substantially more views and inquiries than those with poor-quality, limited, or no images. The number of images also matters; studies suggest that listings with 10-20 high-quality photos perform better than those with fewer images, while excessive images (50+) may overwhelm viewers. Virtual tours, drone footage, and video walkthroughs further enhance engagement, particularly for remote buyers such as OFWs.

Search optimization requires structuring listing data to support filtering and sorting by the criteria buyers actually use. Common search parameters include location (city, neighborhood, barangay), price range, property type (house, lot, condominium, commercial), number of bedrooms and bathrooms, lot size, floor area, and specific features (garage, garden, security). The database schema and search interface must accommodate these parameters efficiently.

Status accuracy is critical for maintaining trust and operational efficiency. Listings must immediately reflect current availability status. Properties that are reserved or sold should be promptly updated to prevent wasted inquiries on unavailable properties, which frustrate buyers and create administrative burden for agents. Automated status updates triggered by reservation system events help maintain accuracy without requiring manual intervention for every change.

Categorization and tagging organize properties into logical groups that match how clients think about real estate. Beyond basic property types, useful categorizations might include investment properties, family homes, retirement properties, vacation homes, or development opportunities. Tags can identify specific features such as "near school," "flood-free," "gated community," or "ready for occupancy" that attract specific buyer segments.

Lead management theory in real estate contexts examines how potential clients (leads) are captured, qualified, nurtured, and converted into transactions. The lead management process typically includes:

Lead capture involves collecting contact information and inquiry details through web forms, chat interactions, phone calls, or walk-in visits. Digital systems should minimize friction in capture processes while ensuring data quality through validation rules. Progressive profiling, which collects additional information over multiple interactions rather than requiring extensive forms upfront, can improve conversion rates.

Lead qualification assesses lead quality and readiness based on factors such as stated budget, desired location, timeline for purchase, financing pre-approval status, and level of engagement with the platform. Qualified leads can be prioritized for agent attention, while unqualified leads may be nurtured through automated content until they become ready to proceed.

Lead distribution assigns qualified leads to appropriate agents based on expertise, workload, geographic specialization, or performance metrics. Automated distribution rules ensure equitable allocation while matching leads with agents best positioned to serve their specific needs.

Lead nurturing maintains regular, valuable communication with leads who are not yet ready to transact. Automated nurture sequences might include market updates, new listings matching their criteria, educational content about buying processes, and periodic check-ins. Effective nurturing keeps the agency top-of-mind until the lead's circumstances or readiness changes.

Conversion tracking monitors which leads progress through successive stages of the sales funnel: initial inquiry, property viewing, formal offer, reservation, and final transaction. Conversion metrics enable performance evaluation, process optimization, and accurate forecasting.

2.3.7 Security Frameworks and Defensive Programming for Web Applications

Web application security theory addresses the systematic protection of web-based systems from threats, vulnerabilities, and malicious attacks. The field has matured significantly, with standardized frameworks identifying common risks and established methodologies for mitigating them.

The Open Web Application Security Project (OWASP), an open community dedicated to enabling organizations to develop, purchase, and maintain trustworthy applications, maintains regularly updated lists of the most critical web application security risks. The OWASP Top 10 identifies common vulnerabilities including:

Injection attacks, where malicious data is sent as part of commands or queries to trick the interpreter into executing unintended commands or accessing unauthorized data. SQL injection, where attackers insert malicious SQL code into input fields to manipulate database queries, is particularly

dangerous for systems like the proposed real estate platform that rely heavily on database interactions.

Broken authentication occurs when authentication mechanisms are implemented incorrectly, allowing attackers to compromise passwords, keys, session tokens, or exploit implementation flaws to assume other users' identities. In real estate systems, broken authentication could allow unauthorized access to client data, property financial details, or administrative functions.

Sensitive data exposure results from inadequate protection of data such as financial information, personal identifiers, or authentication credentials. Real estate systems handle sensitive client contact information, property pricing, and potentially financial documentation that requires strong encryption and access controls.

Broken access control happens when restrictions on authenticated users are not properly enforced, allowing attackers to access unauthorized functionality or data. For example, a client might be able to access other clients' reservation records or an agent might access administrative functions beyond their role.

Security misconfiguration represents the most commonly seen vulnerability, resulting from insecure default configurations, incomplete configurations, open cloud storage, misconfigured HTTP headers, or verbose error messages containing sensitive information.

Cross-site scripting (XSS) allows attackers to execute scripts in victims' browsers to hijack user sessions, deface websites, or redirect users to

malicious sites. XSS vulnerabilities in real estate systems could be used to steal client session cookies or present fraudulent property listings.

Using components with known vulnerabilities occurs when software components such as libraries, frameworks, or other software modules run with privileges that attackers can exploit. Keeping all components updated with security patches is essential.

Insufficient logging and monitoring prevents timely detection and response to security breaches. Effective systems must log security-relevant events and establish monitoring processes to detect suspicious patterns.

Authentication theory examines methods for verifying user identity before granting system access. Password-based authentication, while common, requires careful implementation to be secure. Best practices include requiring minimum password length and complexity, implementing password hashing using strong algorithms such as bcrypt or Argon2 (never storing plaintext passwords or using weak hashing like MD5 or SHA1), enforcing periodic password changes, and preventing password reuse.

Session management maintains authenticated state between HTTP requests, which are inherently stateless. Secure session management requires generating cryptographically strong session identifiers, storing session data server-side rather than in client cookies, implementing session expiration and idle timeout, providing secure logout functionality that invalidates server-side session data, and protecting against session fixation attacks.

Input validation ensures that all user-provided data meets expected formats, types, lengths, and ranges before processing. Validation should occur at multiple levels: client-side validation for immediate user feedback, server-

side validation as the authoritative security control (since client-side validation can be bypassed), and database-level constraints to enforce data integrity. For real estate systems, input validation prevents injection attacks, ensures that property prices are numeric, verifies that email addresses follow proper formats, and restricts file uploads to expected image types and sizes.

Output encoding transforms data before rendering it in web pages to prevent XSS attacks. When displaying user-generated content such as client names, inquiry messages, or property descriptions, the system must encode special characters so that browsers interpret them as text rather than executable code.

Authorization theory, distinct from authentication, governs what authenticated users are permitted to do within a system. Role-Based Access Control (RBAC) is the most widely adopted authorization model for business applications. In RBAC, permissions are associated with roles rather than individual users, and users are assigned to roles based on their job functions. This simplifies administration because permissions can be managed at the role level, and users automatically inherit appropriate permissions when assigned to roles.

For the proposed real estate system, RBAC would typically define roles such as:

Administrator: Full system control including user management, property listing approval, system configuration, and access to all reports and data.

Agent: Ability to manage assigned properties, respond to client inquiries, view their own appointment schedules, and update client interaction

records. Agents cannot access other agents' private client data or administrative functions.

Client: Ability to browse properties, submit inquiries, make reservation requests, and view their own inquiry and reservation history. Clients cannot access other clients' data, administrative functions, or unapproved property drafts.

Guest/Unauthenticated User: Ability to browse public property listings and access general information, but inability to submit inquiries or reservations without registration.

The principle of least privilege, a foundational security concept, dictates that users should be granted only the minimum permissions necessary to perform their legitimate functions. This principle minimizes the potential damage from compromised accounts, insider threats, or accidental misuse. RBAC implementations should regularly review role assignments and remove unnecessary permissions.

Data protection theory emphasizes the confidentiality, integrity, and availability (CIA triad) of information:

Confidentiality ensures that data is accessible only to authorized individuals. In real estate systems, this means encrypting sensitive data in transit (using HTTPS/TLS) and at rest (encrypting database files or specific sensitive columns), implementing strict access controls, and preventing unauthorized disclosure.

Integrity guarantees that data is accurate, complete, and unaltered by unauthorized parties. Mechanisms ensuring integrity include database

transaction controls, input validation, checksums or hashes for critical data, and audit logs tracking changes to sensitive records.

Availability ensures that systems and data are accessible when needed by authorized users. Availability is protected through redundant infrastructure, backup and recovery procedures, denial-of-service protection, and disaster planning.

Defensive programming theory advocates for writing code that continues to function correctly even under unexpected conditions, invalid inputs, or malicious attacks. Defensive programming techniques include:

Fail-safe defaults: Systems should default to the most restrictive, secure state when errors occur. For example, if authorization checking fails due to a database error, the system should deny access rather than grant it.

Complete mediation: Every access to every resource must be checked for authorization, not just during initial authentication. Each page request, API call, and database query should verify that the requesting user has appropriate permissions.

Economy of mechanism: Simple designs with fewer components and interactions are easier to secure and verify. Unnecessary complexity increases vulnerability surface area.

Separation of privilege: Critical operations should require multiple independent conditions or approvals. For example, deleting a property listing might require both administrative role and confirmation of intent.

Least common mechanism: Shared resources and communication channels should be minimized to prevent information leakage between users or processes.

Psychological acceptability: Security mechanisms must be user-friendly enough that legitimate users will actually use them correctly. Overly burdensome security measures encourage users to circumvent them, creating greater vulnerability.

2.3.8 Technology Stack Analysis: PHP, MySQL, HTML, CSS, and JavaScript

The selection of specific technologies for system development carries significant implications for development efficiency, system performance, security, maintainability, and long-term viability. The proposed system's technology stack—PHP for server-side logic, MySQL for data persistence, HTML for document structure, CSS for presentation, and JavaScript for client-side interactivity—represents a time-tested, widely supported, and academically validated combination for web application development.

PHP (Hypertext Preprocessor) is a server-side scripting language specifically designed for web development, created by Rasmus Lerdorf in 1994 and continuously evolved through active open-source community contribution. PHP powers a substantial portion of the World Wide Web, with estimates suggesting that PHP runs on over 75% of all websites and serves as the foundation for major platforms including WordPress (which powers approximately 40% of all websites), Facebook, Wikipedia, and numerous e-commerce and content management systems.

PHP's advantages for web-based real estate systems are numerous and well-documented. As a server-side language, PHP executes on the web server before delivering HTML to client browsers, enabling dynamic content generation based on database queries, user authentication status, and business logic. This server-side execution model protects sensitive code and database credentials from exposure to end users. PHP offers native, robust support for

MySQL and other database systems through extensions such as PDO (PHP Data Objects) and MySQLi, providing both object-oriented and procedural interfaces for database operations. PDO's prepared statement support is particularly valuable for preventing SQL injection attacks, one of the most critical security concerns for database-driven web applications.

PHP runs across various operating systems (Linux, Windows, macOS, Unix variants) and web servers (Apache, Nginx, Microsoft IIS), providing deployment flexibility. The language benefits from decades of widespread use that have produced comprehensive documentation, extensive libraries and frameworks (Laravel, Symfony, CodeIgniter), large developer communities, and abundant hosting options ranging from inexpensive shared hosting to enterprise cloud infrastructure. As open-source software, PHP incurs no licensing costs, making it economically accessible for academic projects and small business applications.

Modern PHP (versions 7.x and 8.x) has addressed earlier criticisms regarding performance and language consistency. PHP 8 introduced significant performance improvements through the JIT (Just-In-Time) compiler, union types, named arguments, attributes, match expressions, and nullsafe operator, bringing the language's capabilities closer to contemporary programming standards while maintaining backward compatibility.

MySQL is an open-source relational database management system (RDBMS) originally developed by MySQL AB, now owned by Oracle Corporation. As a central component of the LAMP stack (Linux, Apache, MySQL, PHP/Python/Perl), MySQL is one of the most widely deployed database systems globally, powering applications ranging from small personal

projects to massive enterprise systems handling millions of transactions per second.

MySQL's relational data model organizes data into tables with defined schemas, supporting complex queries across multiple related tables through JOIN operations. This relational structure is ideally suited for real estate systems where entities such as properties, clients, agents, appointments, and messages have defined relationships that must be maintained consistently. MySQL's ACID compliance (Atomicity, Consistency, Isolation, Durability) for transactions using the InnoDB storage engine ensures that database operations complete reliably even in the face of system failures or concurrent access.

Security features in MySQL include SSL/TLS encryption for data in transit, user authentication with password policies, fine-grained access control through privilege management, and audit logging capabilities. The system supports replication for data redundancy and read scaling, backup utilities for disaster recovery, and import/export tools for data migration. For the proposed real estate system, MySQL provides robust data integrity, reliable transaction processing, and sufficient performance for anticipated user loads.

HTML (HyperText Markup Language) is the standard markup language for creating web pages and applications. The current version, HTML5, developed through cooperation between the World Wide Web Consortium (W3C) and the Web Hypertext Application Technology Working Group (WHATWG), introduces semantic elements (header, nav, article, section, footer, aside), native multimedia support (video and audio elements without plugins), improved form controls (date pickers, email validation,

range sliders), canvas for dynamic graphics, and enhanced accessibility features.

For real estate systems, HTML provides the structural foundation for all user interface components: property listing pages with organized sections for images, descriptions, specifications, and contact forms; inquiry forms with appropriate input types and validation attributes; administrative dashboards with tabular data displays; and navigation structures that help users move efficiently between system areas. Semantic HTML5 elements improve both accessibility for users with disabilities and search engine optimization by clearly indicating content structure and importance.

CSS (Cascading Style Sheets) controls the visual presentation of HTML documents, separating content structure from visual design. CSS3, the current standard, enables responsive layouts through Flexbox and Grid systems, sophisticated animations and transitions, custom fonts, shadows, gradients, and transforms. Media queries allow styles to adapt based on device characteristics, enabling responsive design that works across desktop, tablet, and mobile screens.

In real estate web design, CSS ensures that property images are displayed attractively with appropriate sizing, aspect ratios, and hover effects; that forms are user-friendly with clear labels, focus indicators, and error styling; that navigation menus are intuitive and accessible; and that the overall aesthetic projects professionalism, trustworthiness, and brand identity. The visual design of a real estate website significantly influences user trust and engagement, making CSS expertise essential for system success.

JavaScript is a versatile, high-level programming language that enables dynamic, interactive web experiences. Originally developed for client-side

browser scripting, JavaScript has evolved into a full-stack language capable of server-side execution (Node.js), mobile application development, and even desktop software creation. For browser-based applications, JavaScript manipulates the Document Object Model (DOM), handles user events, validates form input, makes asynchronous server requests, and creates interactive visual effects.

Modern JavaScript (ECMAScript 6 and later) supports advanced features including arrow functions, template literals, destructuring, modules, promises for asynchronous operations, and the `async/await` syntax for cleaner asynchronous code. These features enable sophisticated client-side functionality without requiring heavy frontend frameworks.

For real estate systems, JavaScript enables critical interactive features: dynamic property search filters that update results without full page reloads, improving user experience and reducing server load; client-side form validation that provides immediate feedback before submission, reducing errors and server processing; interactive image galleries with thumbnail navigation, zoom capabilities, and slideshow functionality; real-time chat interfaces for client-agent communication; AJAX requests that load property data, submit inquiries, and update listings seamlessly; and interactive maps that display property locations with clickable markers and information popups.

2.3.9 Agile Software Development Methodology

Agile software development methodology represents a paradigm shift from traditional, plan-driven development approaches toward adaptive, repeating, and human-centered methods. First formally articulated in the Agile Manifesto (Beck et al., 2001), the methodology emerged from

recognition that rigid, sequential development models often failed to deliver satisfactory software in rapidly changing business environments.

The Agile Manifesto establishes four foundational values: individuals and interactions over processes and tools; working software over comprehensive documentation; customer collaboration over contract negotiation; and responding to change over following a plan. These values do not reject processes, documentation, contracts, or planning, but rather prioritize the former elements when conflicts arise.

Twelve principles elaborate these values, including: satisfying the customer through early and continuous delivery of valuable software; welcoming changing requirements even late in development; delivering working software frequently (weeks rather than months); maintaining daily collaboration between business stakeholders and developers; building projects around motivated individuals with proper support and trust; face-to-face conversation as the most efficient communication method; working software as the primary measure of progress; sustainable development maintaining constant pace indefinitely; continuous attention to technical excellence and good design; simplicity maximizing work not done; self-organizing teams producing best architectures and designs; and regular reflection on team effectiveness with corresponding adjustment.

Scrum, the most widely adopted Agile framework, organizes work into time-boxed iterations called sprints, typically lasting 1-4 weeks. Scrum defines three core roles: the Product Owner, responsible for maximizing product value by managing the product backlog and prioritizing features based on business value; the Scrum Master, responsible for facilitating the Scrum process, removing impediments, and coaching the team; and the

Development Team, a cross-functional group responsible for delivering potentially releasable product increments at the end of each sprint.

Scrum ceremonies provide structured opportunities for planning, synchronization, review, and improvement: Sprint Planning establishes the sprint goal and selects backlog items for the sprint; Daily Stand-ups (15-minute time-boxed meetings) synchronize team members on progress, plans, and impediments; Sprint Review demonstrates completed work to stakeholders and gathers feedback; and Sprint Retrospective reflects on the sprint process and identifies improvements for subsequent sprints.

Scrum artifacts include: the Product Backlog, an ordered list of all desired product features, functions, requirements, and fixes; the Sprint Backlog, the subset of product backlog items selected for the current sprint plus the plan for delivering them; and the Increment, the sum of all completed product backlog items at the end of a sprint, representing a potentially releasable product version.

For capstone projects such as the proposed real estate system, a simplified Agile approach offers several advantages over traditional waterfall methodologies:

repeating delivery allows for early demonstration of core functionalities such as property browsing, user registration, and basic inquiry submission, providing tangible progress that can be evaluated and refined based on advisor feedback.

Requirement flexibility accommodates the learning curve inherent in academic projects, where understanding of technical possibilities and user needs evolves throughout development. Agile's welcoming attitude toward

changing requirements prevents the project from being locked into early, potentially flawed specifications.

Risk reduction through frequent delivery ensures that technical challenges are identified and addressed early rather than discovered during final integration phases. If a particular feature proves technically infeasible within project constraints, Agile allows for scope adjustment while preserving core functionality.

Stakeholder engagement through regular reviews keeps academic advisors, potential users, and team members aligned on project direction and progress, preventing misalignment that often derails waterfall projects.

Quality focus through continuous testing integrates quality assurance throughout development rather than compressing it into a final testing phase, resulting in more robust, better-tested systems.

For the proposed real estate system, Agile development will proceed through repeating cycles: Sprint 1 focusing on database design, basic HTML structure, and user authentication; Sprint 2 implementing property listing creation, image uploads, and basic browsing; Sprint 3 developing search and filter functionality; Sprint 4 creating the inquiry and chat system; Sprint 5 building the reservation management module; Sprint 6 developing the administrative dashboard; and subsequent sprints refining features, improving security, enhancing responsive design, and conducting comprehensive testing.

2.3.10 User Experience Design and Usability Principles

User Experience (UX) design theory encompasses the comprehensive set of practices, processes, and principles used to create products that provide meaningful, relevant, and satisfying experiences to users. UX extends beyond

visual design to encompass the entire user journey, including discovery, learning, usage, troubleshooting, and ongoing relationship maintenance. In web-based real estate systems, UX design determines whether clients can efficiently find suitable properties, confidently submit inquiries, and successfully complete reservations without frustration, confusion, or abandonment.

User-centered design (UCD), the foundational philosophy of effective UX, emphasizes understanding the needs, goals, limitations, and contexts of target users through systematic research before and during design. UCD methods include user interviews, surveys, persona development, task analysis, usability testing, and analytics review. For the proposed real estate system, UCD requires understanding how local buyers in Pavia search for properties, what information OFWs need to make remote decisions, how agents currently manage client interactions, and what administrative tasks consume excessive time.

Information architecture (IA) theory addresses the organization, structuring, and labeling of content to support findability and usability. For real estate platforms, IA encompasses how properties are categorized (by type, location, price range, features), how search and filter functions are structured, how navigation menus organize system areas, and how individual property pages present information hierarchically. Effective IA aligns with users' mental models, matching how they think about real estate rather than imposing alien organizational schemes.

Usability, a core UX attribute, refers to the effectiveness, efficiency, and satisfaction with which users can achieve specific goals in a given context. The International Organization for Standardization (ISO 9241-11) defines

usability as the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use. For real estate systems, effectiveness means users successfully find properties matching their criteria; efficiency means they do so quickly without unnecessary steps; and satisfaction means they feel confident, comfortable, and pleased with the experience.

Accessibility ensures that systems can be used by people with diverse abilities, including those with visual, auditory, motor, or cognitive impairments. Web Content Accessibility Guidelines (WCAG) 2.1 provides internationally recognized standards for web accessibility, organized around four principles: content must be perceivable (available to senses through text alternatives, captions, adaptable layouts, and distinguishable elements); operable (interface components must be navigable and usable with various input methods including keyboards and assistive technologies); understandable (information and interface operation must be clear, predictable, and assist users in avoiding and correcting mistakes); and robust (content must work reliably with current and future assistive technologies).

Findability, a specialized UX concern for information-rich systems, ensures that users can locate specific properties or information efficiently. Findability techniques include prominent search functionality with autocomplete suggestions, faceted filtering that allows progressive refinement of results, clear category navigation, breadcrumb trails showing current location within site hierarchy, and related content recommendations that surface relevant properties users might otherwise miss.

User Interface (UI) design principles address the visual and interactive elements through which users engage with the system. Jakob Nielsen's (1994)

ten usability heuristics, while developed decades ago, remain remarkably relevant for contemporary web design:

Visibility of system status keeps users informed about what is happening through appropriate feedback within reasonable time. Examples include loading indicators when property images are fetching, confirmation messages after inquiry submission, and progress indicators during multi-step reservation processes.

Match between system and real world ensures that the system speaks users' language with familiar concepts rather than system-oriented technical terms. Real estate systems should use standard property terminology (bedrooms, bathrooms, floor area, lot area) rather than database field names or internal codes.

User control and freedom provides clearly marked exits from unwanted states, supporting undo and redo functionality. Users who accidentally submit incomplete inquiries should be able to edit or withdraw them; clients who reserve a property but change their minds need clear cancellation procedures.

Consistency and standards maintains consistent terminology, layout, and interaction patterns throughout the system. If property cards on search results pages display price below the image, detail pages should follow the same convention. Navigation menus should remain in fixed locations across all pages.

Error prevention designs interfaces that prevent problems from occurring, through careful constraint design, confirmation dialogs for destructive actions, and input validation that catches errors before submission.

Recognition rather than recall minimizes memory load by making objects, actions, and options visible. Users should not need to remember property IDs or agent names from previous pages; information should be readily available where needed.

Flexibility and efficiency of use accommodates both inexperienced and experienced users through accelerators, shortcuts, and customization options. While novice users may navigate through step-by-step wizards, experienced agents might benefit from keyboard shortcuts or bulk operations.

Aesthetic and minimalist design avoids unnecessary elements that distract from primary tasks. Property pages should highlight essential information and images without cluttering the interface with peripheral content or excessive advertisements.

Help users recognize, diagnose, and recover from errors through clear, plain-language error messages that precisely indicate problems and suggest constructive solutions. Rather than generic "Error 404" messages, the system should explain that a property listing may have been removed and suggest browsing similar properties.

Help and documentation provides accessible, searchable, concise assistance that guides users through tasks without overwhelming them with excessive information. Context-sensitive help, tooltips, and brief explanatory text can reduce the need for separate documentation.

For real estate systems specifically, UI design must prioritize property imagery, as visual content significantly influences buyer interest and engagement. Research consistently demonstrates that listings with professional photographs receive substantially more views and inquiries than

those with poor-quality or absent images. The interface must therefore showcase property images prominently, support high-resolution displays, enable easy navigation through photo galleries, and potentially integrate virtual tour or video viewing capabilities.

Mobile-first design theory advocates for designing the mobile version of websites before creating desktop versions, ensuring that core functionality works effectively on smaller screens with touch interfaces and potentially slower network connections. Given the widespread use of smartphones in the Philippines for internet access, mobile-first design is essential for reaching the broadest possible audience, including OFWs accessing the platform from mobile devices abroad and local users who may not own desktop computers.

2.3.11 Database Design Theory and Normalization

Database design theory provides systematic frameworks for structuring data to minimize redundancy, ensure integrity, support efficient querying, and accommodate future growth. The database design process typically involves three stages: conceptual design using entity-relationship modeling to identify entities and their relationships independent of specific database technologies; logical design translating the conceptual model into a schema appropriate for the chosen database type (relational, document, graph); and physical design optimizing storage, indexing, and access paths for performance.

Entity-Relationship (ER) modeling, developed by Peter Chen (1976), identifies three core components: entities representing objects of interest (things about which data is stored), attributes describing properties of entities, and relationships representing associations between entities. For the proposed real estate system, core entities and their attributes include:

Users/Clients entity stores individual property seekers with attributes such as unique identifier, full name, email address, phone number, password hash, registration date, and account status. This entity supports authentication, personalization, and communication functions.

Administrators/Agents entity represents system operators with attributes including unique identifier, username, email, password hash, role designation (administrator or agent), account creation date, and activity status. Role designation enables RBAC implementation by distinguishing between full administrators and agents with limited permissions.

Properties entity contains real estate listings with attributes such as unique identifier, title, description, price, location address, geographic coordinates, property type, number of bedrooms, number of bathrooms, floor area, lot area, availability status, listing date, and associated agent identifier. The availability status attribute is particularly critical for preventing inquiries about already-reserved or sold properties.

Property Images entity stores visual assets with attributes including unique identifier, property identifier (foreign key linking to the Properties entity), image file path or URL, upload timestamp, display order, and caption text. Separating images into a distinct entity supports multiple images per property while normalizing the database structure.

Appointments/Reservations entity records client requests with attributes such as unique identifier, client identifier (foreign key), property identifier (foreign key), requested date, requested time, appointment status (pending, confirmed, completed, cancelled), submission timestamp, and any special notes or requirements.

Messages/Chat Logs entity stores communication records with attributes including unique identifier, sender identifier, recipient identifier, property identifier (optional, for context), message content, timestamp, and read status. This entity enables both client-to-agent inquiries and system-generated automated messages.

Contact Messages entity captures general inquiries from website visitors who may not be registered users, storing attributes such as unique identifier, sender name, sender email, sender phone, message content, submission timestamp, and read/response status.

Relationships between these entities define how data connects across the system:

Properties to Property Images: One-to-Many relationship, as each property can have multiple associated images, but each image belongs to exactly one property. This relationship is implemented through the property identifier foreign key in the Property Images entity.

Properties to Appointments: One-to-Many relationship, as each property can be the subject of multiple appointment requests over time, but each appointment references exactly one property.

Users to Appointments: One-to-Many relationship, as each client can submit multiple appointments, but each appointment is submitted by one client.

Users to Messages: One-to-Many relationship for both senders and recipients, as each user can send and receive multiple messages.

Properties to Messages: One-to-Many relationship, as multiple messages may reference the same property for context, though messages can also exist without property references for general inquiries.

Normalization theory organizes relational databases to reduce data redundancy and prevent update, insertion, and deletion anomalies. The normal forms provide progressive rules for table structure:

First Normal Form (1NF) requires that tables have a primary key, contain only atomic (indivisible) values in each column, and have no repeating groups. For example, a property listing should not store multiple image filenames in a single column separated by commas; instead, each image should have its own row in a separate Property Images table.

Second Normal Form (2NF) builds on 1NF by requiring that all non-key attributes depend on the entire primary key, not just part of it. This primarily applies to tables with composite keys. In the proposed system, most tables use simple surrogate keys (auto-incrementing integers), making 2NF automatically satisfied if 1NF is achieved.

Third Normal Form (3NF) requires that all non-key attributes depend only on the primary key, not on other non-key attributes (eliminating transitive dependencies). For example, if an agent's email address were stored in both the Administrators/Agents table and the Properties table (assuming properties reference their listing agent), this would violate 3NF because the agent's email depends on the agent identifier, not directly on the property identifier. Proper 3NF design stores agent emails only in the Administrators/Agents table, with Properties referencing agents through foreign keys.

While normalization reduces redundancy, practical database design sometimes intentionally denormalizes specific tables to improve query performance, particularly for read-heavy applications like property search where joining multiple normalized tables for every query might introduce unacceptable latency. The proposed system will primarily follow 3NF principles while considering strategic denormalization if performance testing identifies bottlenecks.

2.3.12 Geographic Information Systems and Location-Based Services

Geographic Information Systems (GIS) theory addresses the comprehensive set of technologies, processes, and methods for capturing, storing, analyzing, managing, and presenting spatial or geographic data. In real estate, GIS enables property mapping, neighborhood analysis, proximity calculations, and location-based search functionality that significantly enhances property discovery and evaluation.

The integration of web mapping services such as Google Maps, OpenStreetMap, or Mapbox into real estate platforms allows developers to leverage sophisticated geographic infrastructure without building proprietary mapping systems. These services provide APIs that enable embedding interactive maps, geocoding addresses to geographic coordinates, reverse geocoding coordinates to addresses, calculating distances and routes, and displaying satellite or street view imagery.

For the proposed real estate system, GIS integration supports several valuable features: map-based property search displaying available properties as interactive markers on zoomable, pannable maps; location autocomplete suggesting addresses and neighborhood names as users type location queries; proximity filtering finding properties within specified distances of landmarks,

schools, or business districts; and static map images showing property locations in listing previews.

Location-based services (LBS) theory examines how applications can use geographic positioning to deliver contextually relevant information and services. While primarily associated with mobile applications using GPS positioning, LBS principles apply to web-based real estate systems through user-specified location preferences and map-based interactions. Key LBS capabilities for real estate include geofencing (defining geographic boundaries for targeted notifications or search results), location history analysis (understanding which areas users search most frequently to personalize recommendations), and proximity alerts (notifying users when new listings appear in their areas of interest).

For the Pavia, Iloilo City context, location-based features must account for local geographic conventions including barangay boundaries, proximity to key landmarks (plazas, churches, markets, schools), transportation access, and neighborhood characteristics that influence property values. While full GIS implementation may exceed capstone project scope, understanding GIS principles ensures that property location data is structured effectively for future enhancements and that any map integrations follow established usability patterns.

2.4 Synthesis of the Reviewed Literature and Studies

This synthesis section integrates findings from foreign studies, local studies, and related literature to establish convergent themes, identify critical research gaps, articulate the theoretical framework guiding the proposed system, and justify the study's specific contributions to knowledge and practice. The synthesis is organized thematically to build a coherent

argumentative structure that moves from empirical validation through gap analysis to integrated justification.

2.4.1 Convergence of Findings on Digital Transformation in Real Estate

The reviewed literature demonstrates remarkable consistency across international and local sources regarding the ongoing transformation of real estate marketing and management from traditional, manual methods to digital, automated, web-based systems. This convergence provides strong empirical and theoretical validation for the proposed system's fundamental premise: that digital transformation is essential for real estate competitiveness in contemporary markets.

Foreign studies uniformly confirm the benefits of web-based real estate platforms. Al Ghawail (2026) demonstrated that PHP and MySQL-based systems successfully replace traditional methods while improving communication and data management. The online platform research (IRJMETs, 2025) confirmed that web-based systems significantly reduce time, paperwork, and communication delays. The cloud-based Salesforce system (2025) showed that automated workflows improve lead management efficiency. Digital marketing research (European Journal, 2025; Birdeye, 2026; NetSuite, 2025) consistently found that digital strategies outperform traditional methods in reach, engagement, and conversion. AI and automation studies (2026; Whippy, 2025; Nylas, 2024) provided quantitative evidence of dramatic improvements in response times, conversion rates, and operational efficiency.

Local studies similarly establish the digital imperative. The Philippine Statistics Authority (2021) documented nationwide digital adoption trends. The Department of Trade and Industry (2022) formally mandated SME digital

transformation. Market analysis (Ken Research, 2025) revealed a rapidly expanding digital real estate ecosystem. The Tokkatok case study (Codic, 2025) exposed the fragmentation and informality of current Philippine property marketing channels. Government initiatives (PNA, 2025) explicitly support digital platforms for OFW investors. OFW investment research (Unicapital, 2024; Reelist8, 2023; Bria Homes, 2024) consistently identified digital accessibility as a critical need for overseas investors.

The convergence of foreign and local findings across diverse methodologies, contexts, and sources establishes that digital transformation in real estate is not a niche trend or luxury option but a global and national imperative. Businesses that fail to adopt digital systems face documented disadvantages in customer reach, operational efficiency, and competitive positioning. This convergence is particularly significant because it spans both developed and developing markets, suggesting that the benefits of real estate digitalization are not conditional on high-income economies but are achievable in contexts similar to Pavia, Iloilo City.

2.4.2 Validation of Proposed System Features

The reviewed literature provides robust validation for each major feature planned in the proposed web-based real estate management system:

Property browsing and listing management is validated by Al Ghawail (2026), the online platform study (2025), the advanced PMS research (2025), and the Tokkatok case study (2025), all of which successfully implemented and tested similar features. These studies confirm that users can effectively navigate property catalogs, view detailed information, and make informed decisions through web interfaces.

Online reservation and appointment management is supported by reservation system research (Jurnal Galaksi, 2025; IJEAIS, 2023) demonstrating that digital booking reduces administrative burden, prevents double-booking, and improves user convenience. While these studies focused on hospitality applications, their findings regarding scheduling integrity, confirmation processes, and reminder notifications directly apply to real estate contexts.

Automated communication and chat functionality receives particularly strong validation from AI-CRM research (2026), which documented conversion rate improvements from 20% to 40% and query handling improvements from 40% to 85% after chatbot implementation. Whippy (2025) and Nylas (2024) further confirmed that automated follow-ups, real-time alerts, and integrated messaging significantly improve client engagement and agent productivity.

Administrative dashboards for system management are validated by multiple studies including the cloud-based Salesforce research (2025) and the advanced PMS study (2025), which both implemented comprehensive dashboards for monitoring metrics, managing listings, and tracking client activities. ISO/IEC 25010 evaluation of the PMS dashboard achieved a 4.53 security rating and 4.62 usability rating, demonstrating that well-designed dashboards meet high quality standards.

Security measures including authentication, input validation, and defensive programming are extensively validated by the advanced PMS research (2025), which implemented bcrypt hashing, MFA, OAuth 2.0, RBAC, and ISO 25010 compliance. OWASP principles and security

frameworks reviewed in the related literature provide additional theoretical foundations for the proposed system's security architecture.

The cumulative validation across all feature categories suggests that the proposed system is not experimental in its feature selection but rather assembles proven, empirically supported functionalities into an integrated package tailored to a specific local context.

2.4.3 Identification of Critical Research Gaps

Despite extensive literature on real estate digital platforms, several significant gaps justify and position the current study:

Geographic specificity gap: Existing Philippine real estate platforms (Lamudi, Property24, Carousell) serve national or regional markets with generic interfaces that lack deep localization for specific municipalities. These platforms treat Pavia, Iloilo City as merely one of thousands of locations, providing no community-specific context, neighborhood-level insights, or locally relevant features. The current study addresses this gap by focusing specifically on properties in Pavia, Iloilo City, ensuring that content, navigation, and features reflect local geographic conventions, community characteristics, and market conditions.

Small agency integration gap: Major platforms target large developers and established agencies with substantial marketing budgets and technical resources. Small, local real estate businesses lack access to integrated, affordable systems that combine property management, digital marketing, automated communication, and reservation management in customizable solutions. The current study addresses this gap by designing a system specifically for a local real estate business with limited technical resources,

using cost-effective open-source technologies (PHP, MySQL) and straightforward deployment options.

OFW-specific functionality gap: While general platforms serve various user segments, limited research and development focus specifically on the unique needs of OFW property investors. OFWs require timezone-aware communication, comprehensive property documentation for remote decision-making, simplified reservation processes that accommodate international schedules, and interfaces optimized for mobile access from abroad. The current study explicitly identifies OFWs as primary beneficiaries and incorporates features designed to serve their specific circumstances.

Academic-practical integration gap: Many existing studies either focus on theoretical marketing strategies without accompanying system implementation, or present commercial system development without rigorous academic grounding. The current study bridges this gap by applying established information systems theories (TAM, UTAUT, DeLone and McLean's Success Model) to guide the development of a practical, functional system within an academic capstone framework. This integration produces both a working software solution and a theoretically informed scholarly contribution.

Technology stack validation gap: While numerous studies demonstrate the viability of various technology stacks (MERN, Python Flask, Salesforce, Java Spring Boot), limited recent research validates the continued effectiveness of PHP and MySQL for real estate applications in the Philippine context. Al Ghawail (2026) provides valuable validation, but additional research in different geographic and market contexts strengthens the evidence base for this technology choice.

These gaps are not merely abstract academic deficiencies but represent real market failures that affect actual buyers, sellers, and agents in Pavia, Iloilo City. By addressing these gaps, the current study contributes both to scholarly knowledge and to practical market improvement.

2.4.4 Integrated Theoretical Framework

The proposed study synthesizes multiple theoretical frameworks into an integrated model that guides system design, implementation, and evaluation:

From Information Systems theory, the study adopts the Socio-Technical Systems perspective to ensure the system balances technical capabilities with social and organizational factors. The system must fit within existing real estate business workflows, respect established client-agent relationships, and accommodate the learning curves of users with varying digital literacy levels. The Technology Acceptance Model and Unified Theory of Acceptance and Use of Technology guide interface design and feature prioritization to maximize perceived usefulness, ease of use, performance expectancy, and facilitating conditions. DeLone and McLean's IS Success Model provides the evaluation framework, ensuring assessment across system quality, information quality, service quality, use, user satisfaction, and net benefits dimensions.

From Digital Marketing theory, the 7Ps Marketing Mix Model structures the system's marketing features, ensuring comprehensive coverage of product presentation (property listings), pricing display, digital placement (web platform), promotional tools, people (agent profiles and communication), process (inquiry and reservation workflows), and physical evidence (visual design and imagery quality). SEO and content marketing

principles inform the structure of property listings and website content to maximize organic visibility for location-specific searches.

From Web Development theory, three-tier architecture guides the system's technical organization, separating presentation, business logic, and data layers for maintainability and scalability. Responsive web design principles ensure accessibility across the diverse devices used by Philippine consumers and OFWs abroad. Security frameworks (OWASP, RBAC, defensive programming) protect system integrity and user data against common web application threats.

From CRM and Communication theory, the CRM value chain model structures client interaction workflows from initial inquiry through post-reservation follow-up. Automated communication principles guide the design of chat functionality and inquiry response systems, ensuring immediate acknowledgment and systematic follow-up. Service quality dimensions (responsiveness, reliability, assurance, empathy, tangibles) inform the design of client communication interfaces to maintain professional standards even in automated interactions.

From Software Engineering theory, Agile development methodology provides the project management framework, enabling repeating delivery, continuous feedback incorporation, and adaptability to evolving requirements. Database normalization principles ensure data integrity and efficiency, while defensive programming techniques protect against security vulnerabilities and system failures.

This integrated framework does not merely list theories but actively combines them into a coherent design philosophy. For example, when designing the chat functionality, the system draws simultaneously from TAM

(ensuring perceived usefulness and ease of use), SERVQUAL (ensuring responsiveness and assurance), security frameworks (ensuring input validation and output encoding), and Agile methodology (delivering chat functionality incrementally for early testing).

2.4.5 Technological and Methodological Justification

The choice of PHP, MySQL, HTML, CSS, and JavaScript for system development is thoroughly justified by the reviewed literature. Al Ghawail (2026) successfully implemented a real estate system using this exact technology stack, demonstrating its viability for property management applications in a comparable context. The LAMP stack remains one of the most widely deployed web development platforms globally, with extensive documentation, community support, and hosting availability that reduce development risks and deployment costs.

PHP's native database integration capabilities, combined with MySQL's relational data model and ACID compliance, provide robust foundations for managing property listings, user accounts, reservations, and message logs. The combination has been refined over decades of widespread use, resulting in mature, stable, well-understood technologies with abundant learning resources suitable for academic projects.

HTML5, CSS3, and JavaScript enable modern, responsive user interfaces without requiring specialized frontend frameworks that might increase development complexity beyond capstone project scope. These foundational web technologies are universally supported, ensuring compatibility across all modern browsers and devices without requiring plugin installations or specialized runtime environments.

The Agile development methodology, supported by software engineering literature and validated in multiple studies, provides an appropriate project management framework for repeating development within academic timelines. The methodology's emphasis on working software, continuous feedback, and adaptability aligns with the exploratory and learning-oriented nature of capstone projects while ensuring regular deliverables and demonstrable progress.

The technological choices also reflect practical constraints and opportunities specific to the deployment context. PHP and MySQL hosting is widely available and inexpensive in the Philippines, making ongoing system operation feasible for a small local agency. The technologies require no proprietary licensing fees, aligning with the resource constraints of small business users. The large pool of PHP developers in the Philippines ensures that future maintenance and enhancement can be sourced locally if needed.

2.4.6 Synthesis of Local Market Needs and Opportunities

The local studies collectively establish clear market needs and opportunities that the proposed system directly addresses:

Fragmented market structure: The Philippine real estate market outside major metropolitan areas lacks structured, trustworthy digital platforms for property transactions. Informal channels such as Facebook groups dominate, creating verification challenges, inconsistent listing quality, and unreliable communication. The proposed system provides a dedicated, professional platform with structured listings, verified information, and organized inquiry management.

OFW market importance and underserved needs: Multiple studies confirm that OFWs represent a critical, high-value market segment for Philippine real estate, yet they face significant challenges in remote property research and investment. The system's web-based accessibility, comprehensive property information, automated communication, and mobile-responsive design directly address documented OFW pain points including time zone differences, inability to visit properties physically, and need for immediate, reliable information.

Government and industry alignment: Government backing for digital real estate platforms (PNA, 2025) and DTI encouragement for SME digital transformation (2022) create favorable policy conditions for system adoption. The proposed system aligns with national digitalization trends, government initiatives supporting OFW investment, and industry movement toward online property marketing.

Competitive differentiation: While national platforms serve broad markets, localized systems can offer deeper community context, more personalized service, and stronger local trust relationships. The Pavia, Iloilo City focus enables the system to capture market share among buyers specifically interested in this area, who may prefer a specialized local platform over generic national alternatives.

The synthesis of market needs also reveals a temporal opportunity: the post-pandemic acceleration of digital adoption, combined with sustained OFW investment flows and government digitalization support, creates a window where the introduction of a localized digital platform is likely to encounter receptive users and favorable market conditions.

2.4.7 Summary of Justification and Study Positioning

The comprehensive synthesis of reviewed literature and studies provides multi-faceted justification for the proposed web-based real estate management system:

Empirical justification derives from multiple foreign and local studies demonstrating that similar systems have successfully improved real estate operations, client satisfaction, and business performance across diverse contexts. The consistency of positive outcomes suggests high probability of success for the proposed system in the Pavia, Iloilo City context.

Theoretical justification is established through the integration of established information systems, digital marketing, CRM, and web development theories that provide robust frameworks for system design, implementation, and evaluation. These theories ensure the system is grounded in proven principles rather than ad hoc development decisions.

Technological justification is validated by existing real estate system implementations using the chosen technology stack and supported by extensive development resources. The PHP/MySQL/HTML/CSS/JavaScript combination offers proven capability, development efficiency, and deployment flexibility suitable for academic and small business contexts.

Market justification emerges from Philippine market research, government initiatives, documented OFW investment patterns, and identified gaps in existing digital platforms. The specific focus on Pavia, Iloilo City addresses an underserved local market with distinct characteristics and needs.

Gap justification is established through the identification of geographic specificity gaps, small agency integration gaps, OFW functionality gaps, academic-practical integration gaps, and technology validation gaps that the

current study explicitly addresses. These gaps ensure the study contributes meaningfully to existing knowledge and practice rather than merely replicating prior work in identical contexts.

In conclusion, the reviewed literature and studies collectively establish that the development of a web-based real estate management system with integrated digital marketing and automated communication features is theoretically sound, empirically supported, technologically feasible, market-justified, and academically warranted. The system addresses documented inefficiencies in traditional real estate operations while serving the specific needs of the Pavia, Iloilo City market and its significant OFW client base. The convergence of international best practices, local market insights, and robust theoretical frameworks provides a solid foundation upon which the subsequent methodology and system development chapters will build. The following chapter will detail the research methodology through which this theoretically grounded and empirically informed system will be developed, tested, and evaluated against established quality criteria.

CHAPTER 3

METHODOLOGY

This chapter explains the methods, tools, and steps that the researchers used to build the web-based real estate management system. It shows how the study was planned, how the system was designed, what technologies were used, and how the researchers made sure the system works properly. All the

information in this chapter is written in simple words so that readers can easily understand the process even if they do not have a strong background in information technology.

The main goal of this chapter is to give a clear picture of how the researchers turned the problems identified in Chapter 1 into a working digital solution. It covers the research design, the development method, the system structure, the software and hardware used, how data was collected, how the system was tested, and how the researchers measured if the system was successful. Each part of this chapter is explained in detail to make sure that the development process is easy to follow and understand.

3.1 Research Design

The research design is the overall plan that guides how the study was conducted. It acts like a roadmap that shows the researchers what to do, when to do it, and how to do it. For this study, the researchers used a combination of descriptive research and development research. These two approaches were chosen because they work well together when the goal is to understand a real-world problem and then create a practical solution for it.

Descriptive research is a type of study that aims to describe the current situation, condition, or problem in a detailed way. It does not try to change anything or prove cause-and-effect relationships. Instead, it focuses on collecting facts and information about what is happening right now. In this study, the researchers used descriptive research to understand the real problems that real estate businesses in Pavia, Iloilo City are facing. They looked at how these businesses handle property listings, how they communicate with clients, and what difficulties they experience in their daily operations.

The researchers chose descriptive research because it allowed them to gather real and accurate information about the existing problems before trying to solve them. By understanding the problem deeply, the researchers could design a system that truly meets the needs of the users. For example, through descriptive research, the researchers found out that many real estate agents still use manual methods to record client inquiries. They also learned that property listings are often outdated and that communication with clients is slow. Without this descriptive phase, the researchers might have built a system that does not address the real needs of the business.

Development research, on the other hand, is a type of study that focuses on creating a new product, system, or solution to solve a specific problem. It is also called Research and Development or R&D. In this study, the researchers used development research to build the web-based real estate management system. This approach involves designing, building, testing, and improving a product until it becomes useful and effective.

The combination of descriptive and development research is very effective for capstone projects because it follows a logical flow: first understand the problem, then build the solution. The researchers did not just jump into coding and designing. Instead, they took time to study the situation, interview potential users, review existing systems, and analyze the data they collected. Only after this understanding phase did they begin the actual development of the system.

The research design also includes the overall strategy for collecting and analyzing information. The researchers used both primary and secondary data sources. Primary data refers to information that is collected directly from the source. In this study, primary data came from interviews with real estate

business owners, agents, and potential clients. Secondary data refers to information that already exists, such as books, articles, and studies from other researchers. The researchers reviewed related literature and studies to support their understanding of the problem and to learn from the experiences of others.

The researchers also decided to follow a structured and organized approach to avoid confusion and wasted effort. They created a timeline, set clear milestones, and divided the work among team members. This organized approach is important in any research project because it helps the team stay focused and finish the work on time.

Another important part of the research design is the setting or location of the study. This study focused on real estate properties located in Pavia, Iloilo City. The researchers chose this location because it is a growing area with active real estate development. It is also a place where the researchers can easily access real estate businesses for interviews and testing. By limiting the study to one specific area, the researchers were able to focus their efforts and create a system that is relevant to that location.

The target users of the system were also clearly defined in the research design. The system is intended for three main groups of users: clients who want to buy or rent properties, real estate agents who manage listings and communicate with clients, and administrators who oversee the entire system. By identifying these users early in the research design, the researchers were able to include features that are useful for each group.

In summary, the research design of this study is a combination of descriptive and development research. It follows a clear and logical process: understand the problem first, then develop the solution. The researchers collected both primary and secondary data, focused on a specific location, and

identified the target users. This design ensures that the final system is based on real needs and real data, making it more likely to be useful and effective in solving the identified problems.

3.2 System Development Methodology

The system development methodology is the step-by-step process that the researchers followed to build the web-based real estate management system. It is like a recipe that tells the researchers what ingredients to use, what steps to take, and in what order. Choosing the right methodology is very important because it affects how fast the system is built, how well it works, and how easy it is to fix problems along the way.

For this study, the researchers chose to use the Agile Development Model. Agile is a popular method for building software because it is flexible, adaptive, and focused on delivering working features quickly. Unlike older methods that require all planning to be done before any coding starts, Agile allows the team to build the system in small parts, test each part, get feedback, and then improve it. This makes Agile ideal for student projects like capstones because it allows for continuous learning and adjustment.

The Agile Development Model is based on several important ideas. First, it values people and communication more than strict processes and tools. This means that the research team worked closely together, talked often, and made decisions as a group. Second, it values working software more than detailed documentation. While documentation is still important, the main goal is to create a system that actually works and can be used. Third, it values customer collaboration more than fixed contracts. The researchers worked with their client, asked for feedback, and made changes based on what the client needed. Fourth, it values responding to change more than following a

rigid plan. If the researchers discovered a better way to do something, they were free to change their approach.

The Agile method works in cycles called iterations or sprints. Each sprint is a short period of time, usually a few weeks, during which the team plans, designs, builds, and tests a small part of the system. At the end of each sprint, the team has a working piece of software that can be shown to the client. The client gives feedback, and the team uses that feedback to plan the next sprint. This cycle continues until the entire system is complete.

The researchers chose Agile for several reasons. First, real estate business needs can change quickly. A client might ask for a new feature or realize that an old feature is not needed. Agile makes it easy to handle these changes without starting over. Second, Agile reduces risk. By building and testing small parts of the system at a time, the researchers could find and fix problems early instead of discovering them at the end. Third, Agile improves quality. Because each sprint includes testing, the final system is more reliable and has fewer bugs. Fourth, Agile keeps the client involved. The real estate business owner and agents were able to see progress regularly and suggest improvements.

The Agile Development Model used in this study consisted of five main phases. Each phase is explained in detail below.

3.2.1 Phase 1: Requirements Analysis

The first phase of the Agile Development Model is Requirements Analysis. This phase is all about understanding what the system needs to do. The researchers spent time talking to the client, observing how the real estate

business currently operates, and writing down all the things that the new system should be able to do.

During this phase, the researchers conducted interviews with the real estate business owner and several agents. They asked questions like: How do you currently list your properties? How do clients contact you? What problems do you face every day? What features would make your work easier? The researchers also looked at how the business handles appointments, how they update property information, and how they communicate with potential buyers.

The researchers divided the requirements into two types: functional requirements and non-functional requirements. Functional requirements are the specific features and functions that the system must have. For this study, the functional requirements included:

Property browsing: Users should be able to view a list of available properties with images, descriptions, prices, and locations. Property search and filter: Users should be able to search for properties by location, price range, or property type. Online reservation: Users should be able to select a property and make a reservation or appointment to view it. Automated chat: Users should be able to send messages and receive quick responses about property details. Admin dashboard: Administrators should be able to add, edit, and delete property listings. Inquiry management: Administrators should be able to view and respond to client messages and reservations. User authentication: The system should have login and registration features to protect user accounts. Database management: The system should store all property data, user data, and transaction records securely.

Non-functional requirements are the qualities that the system should have, rather than specific features. These include: Usability: The system should be easy to use, even for people who are not tech-savvy. Security: The system should protect user data and prevent unauthorized access. Performance: The system should load quickly and handle multiple users at the same time. Reliability: The system should work correctly most of the time without crashing. Compatibility: The system should work on different devices such as computers, tablets, and smartphones.

The researchers documented all these requirements in a requirements specification document. This document served as a guide throughout the development process. Whenever the team was unsure about what to build, they could look back at the requirements document to remind themselves of the client's needs.

Requirements analysis is a very important phase because mistakes made here can cause big problems later. If the researchers did not understand what the client needed, they might build the wrong system. That is why they spent extra time on this phase, asking many questions and confirming their understanding with the client before moving forward.

3.2.2 Phase 2: System Design

The second phase of the Agile Development Model is System Design. In this phase, the researchers planned how the system would look and how it would work behind the scenes. This phase is like drawing the blueprints for a house before construction begins. Without good design, the system might be confusing to use or difficult to build.

The system design phase was divided into two main parts: front-end design and back-end design. The front-end is the part of the system that users see and interact with. It includes the colors, buttons, menus, images, and overall layout of the website. The back-end is the part of the system that users do not see. It includes the server, the database, and the logic that processes user actions.

For the front-end design, the researchers created wireframes and mockups. A wireframe is a simple sketch that shows where different elements will be placed on a web page. It does not include colors or images; it just shows the structure. A mockup is a more detailed design that includes colors, fonts, and sample images. The researchers used these tools to visualize the user interface before writing any code.

The researchers designed several key pages for the system:

Home Page: This is the first page that visitors see. It includes a welcome message, featured properties, and navigation links. Property Listings Page: This page shows all available properties in a grid or list format. Each property card includes an image, title, price, and brief description. Property Details Page: When a user clicks on a property, they see this page. It includes large images, full descriptions, location details, and a reservation button. Reservation Page: This page contains a form where users can enter their name, contact information, preferred date, and time for viewing the property. Chat Page: This page allows users to send messages and receive automated or manual responses. Login and Registration Pages: These pages allow users to create accounts and log in to the system. Admin Dashboard: This is a special area for administrators. It includes menus for managing properties, viewing reservations, reading messages, and generating reports.

For each page, the researchers considered user experience principles. They made sure that the navigation was simple and intuitive. They used consistent colors and fonts throughout the site. They placed important buttons in easy-to-find locations. They also made sure that the design was responsive, meaning that the pages would look good on both large computer screens and small smartphone screens.

For the back-end design, the researchers planned the system architecture, the database structure, and the flow of data. They decided to use a client-server architecture, which is explained in detail in Section 3.3. They also designed the database tables, which is explained in detail in Section 3.8.

The researchers created several design documents during this phase. These included:

System Architecture Diagram: This shows how the different parts of the system connect to each other. Database Schema: This shows all the tables in the database and how they relate to each other. Data Flow Diagrams: These show how information moves through the system from the user to the database and back. Entity Relationship Diagram: This shows the relationships between different types of data, such as users, properties, and reservations. Use Case Diagram: This shows the different actions that users and administrators can perform in the system.

All these design documents were reviewed by the team and the client before moving to the next phase. This review process helped catch errors early and ensured that everyone agreed on the plan.

3.2.3 Phase 3: Development

The third phase of the Agile Development Model is Development. This is the phase where the researchers actually built the system by writing code, creating the database, and connecting all the parts together. In Agile, development is done in small pieces called iterations. Each iteration focuses on building one or a few features, testing them, and then adding them to the growing system.

The researchers planned their development work into several iterations. Each iteration lasted about two to three weeks. At the start of each iteration, the team held a planning meeting where they decided which features to build. They chose features based on priority, with the most important features being built first. This approach is called prioritization, and it ensures that even if the project runs out of time, the most critical parts of the system are already working.

Iteration 1 focused on setting up the project environment and building the basic structure. The researchers installed the necessary software, configured the development server, and created the main folders and files for the project. They also built the database and created the basic login and registration system. This foundation was important because all other features depend on it.

Iteration 2 focused on property management features for the admin side. The researchers built the admin dashboard and created the pages where administrators can add new properties, upload images, edit property details, and delete listings. They also built the database queries needed to store and retrieve property information.

Iteration 3 focused on the client-side property browsing features. The researchers built the home page, the property listings page, and the property

details page. They made sure that property information was pulled correctly from the database and displayed in an attractive way. They also added search and filter features so that users could find properties based on their preferences.

Iteration 4 focused on the reservation and appointment system. The researchers built the reservation form, created the database table for appointments, and developed the logic for recording and displaying reservations. They also built the admin pages for viewing and managing appointments.

Iteration 5 focused on the communication features. The researchers built the chat interface where users can send messages. They also built the contact form and the admin message management page. Automated responses were programmed for common questions about property availability and pricing.

Iteration 6 focused on security and polishing. The researchers added input validation to prevent users from entering harmful data. They improved the login system with password hashing. They tested all features together and fixed any bugs they found. They also improved the visual design based on feedback from the client.

During each iteration, the researchers followed good coding practices. They wrote clean and organized code with comments explaining what each part does. They used meaningful names for variables and functions so that the code is easy to understand. They also kept backups of their work to prevent data loss.

Communication was very important during the development phase. The team met regularly to discuss progress, share problems, and help each other. When a team member was stuck on a difficult programming task, the others would help find a solution. This teamwork made the development process faster and more enjoyable.

3.2.4 Phase 4: Testing

The fourth phase of the Agile Development Model is Testing. Testing is the process of checking if the system works correctly and if it meets the requirements. In Agile, testing is not done only at the end. Instead, it is done continuously throughout the development process. Every time a new feature is built, it is tested before being added to the main system.

The researchers used several types of testing to make sure their system was reliable and effective. These testing methods are explained in detail in Section 3.6, but a brief overview is provided here.

Unit Testing was done by the developers themselves. In unit testing, individual parts of the code are tested separately. For example, if the researchers built a function that calculates the total price of a property reservation, they would test that function with different numbers to make sure it always gives the correct answer. Unit testing helps find small errors in the code before they become big problems.

System Testing was done to check if all the parts of the system work together correctly. Even if each individual feature works well on its own, problems can appear when they are combined. System testing involves going through the entire system from start to finish, performing all possible actions, and checking if everything flows smoothly.

User Acceptance Testing was done with the help of the client and some volunteer users. In this type of testing, real users try the system and give feedback. The researchers asked the users to perform specific tasks, such as searching for a property, making a reservation, and sending a message. The researchers watched how the users interacted with the system and asked them questions about their experience. This feedback was very valuable because it showed the researchers how actual users feel about the system.

During testing, the researchers kept a record of all the problems they found. This record is called a bug list or issue tracker. For each problem, they wrote down what went wrong, where it happened, and how serious it was. They then fixed the problems one by one, starting with the most serious ones. After fixing a problem, they tested again to make sure the fix worked and did not cause new problems.

Testing is a critical phase because it directly affects the quality of the final system. A system that has not been tested properly might have hidden errors that cause frustration for users or even data loss. The researchers took testing very seriously and spent a significant amount of time on it.

3.2.5 Phase 5: Deployment (Prototype)

The fifth phase of the Agile Development Model is Deployment. Deployment means making the system available for use. In this study, the researchers deployed a prototype of the system. A prototype is an early version of the system that is used for demonstration and testing purposes. It may not have all the features of the final product, but it shows how the system will work.

The researchers deployed the prototype on a local server and also made it accessible through a web hosting service. This allowed the client and test users to access the system from their own devices. The deployment process involved several steps.

First, the researchers prepared the server environment. They made sure that the server had all the necessary software installed, including the web server software, the PHP programming language, and the MySQL database system. They also configured the security settings to protect the system from unauthorized access.

Second, the researchers transferred all the system files to the server. This included the HTML, CSS, JavaScript, and PHP files, as well as the images and other media files. They made sure that all files were placed in the correct folders and that the file permissions were set correctly.

Third, the researchers set up the database on the server. They created the database, created all the necessary tables, and imported any sample data that was needed for testing. They also tested the connection between the system files and the database to make sure data could be read and written correctly.

Fourth, the researchers performed final checks before announcing that the prototype was ready. They tested the login system, checked all the main pages, and verified that the system loaded properly on different browsers and devices.

After deployment, the researchers collected feedback from users. They asked questions like: Is the system easy to use? Do the pages load quickly?

Are there any features missing? Is there anything confusing? This feedback was used to plan improvements for the next iteration of development.

The Agile approach means that deployment is not the final step. Even after the prototype is live, the researchers continue to improve the system based on user feedback. This cycle of building, testing, deploying, and improving continues until the system meets all the requirements and satisfies the users.

In summary, the Agile Development Model provided a flexible and effective framework for building the web-based real estate management system. It allowed the researchers to understand requirements, design carefully, build in small steps, test continuously, and deploy a working prototype. The iterative nature of Agile ensured that the system evolved based on real feedback and real needs, making it more likely to succeed in solving the problems identified in Chapter 1.

3.3 System Architecture

System architecture refers to the overall structure of the system. It shows how the different parts of the system are organized and how they communicate with each other. Think of system architecture like the floor plan of a building. It shows where each room is located, how big it is, and how it connects to other rooms. Without a good architecture, a system can become messy, slow, and difficult to maintain.

The web-based real estate management system uses a client-server architecture. This is one of the most common and reliable ways to build web applications. In a client-server architecture, there are two main sides: the client side and the server side. The client side is what the user sees and interacts

with. The server side is where data is stored and processed. These two sides communicate with each other through the internet.

3.3.1 The Client Side (Front End)

The client side, also called the front end, is the part of the system that runs on the user's device. When a person opens the real estate website on their computer or smartphone, the front end is what appears on their screen. The front end includes all the visual elements such as text, images, buttons, forms, and menus.

The front end of this system was built using three main technologies: HTML, CSS, and JavaScript. HTML stands for HyperText Markup Language. It is the basic building block of web pages. HTML is used to define the structure of a page, such as where the headings go, where the paragraphs are, and where images are placed. Every web page on the internet uses HTML.

CSS stands for Cascading Style Sheets. CSS is used to make web pages look attractive. It controls colors, fonts, spacing, sizes, and layouts. Without CSS, web pages would look plain and boring, like plain text documents. With CSS, the researchers were able to create a professional and visually appealing design for the real estate system.

JavaScript is a programming language that adds interactivity to web pages. It allows the page to respond to user actions. For example, when a user clicks a button to view more property images, JavaScript makes that action happen. JavaScript can also validate form inputs, show popup messages, and create animations. In this system, JavaScript was used to make the user experience smooth and interactive.

The front end communicates with the back end by sending requests. For example, when a user searches for properties in Pavia, the front end sends a request to the server asking for properties that match the search. The server processes the request, looks in the database, and sends back the results. The front end then displays those results to the user.

The researchers made sure that the front end was responsive. A responsive design means that the website automatically adjusts its layout to fit the screen size of the device being used. Whether the user is on a large desktop monitor, a tablet, or a small smartphone, the website should still look good and be easy to use. This was achieved by using CSS techniques such as media queries and flexible grid layouts.

3.3.2 The Server Side (Back End)

The server side, also called the back end, is the part of the system that runs on the web server. The user does not see the back end, but it is responsible for all the important work that happens behind the scenes. The back end processes user requests, runs business logic, and communicates with the database.

The back end of this system was built using PHP. PHP stands for PHP: Hypertext Preprocessor. It is a popular programming language for web development. PHP is especially good for creating dynamic web pages, which are pages that change based on user input or database content. For example, when a user views a property details page, PHP retrieves the property information from the database and inserts it into the HTML page before sending it to the user's browser.

PHP was chosen for this project because it is widely used, well-documented, and works very well with MySQL databases. It is also supported by most web hosting services, which makes deployment easier. The researchers had previous experience with PHP, which allowed them to develop the system more efficiently.

The back end handles many important tasks in the system. It checks if users have entered correct login information. It validates form data to make sure users are not sending harmful or incorrect information. It processes reservation requests and stores them in the database. It handles file uploads when administrators add property images. It also manages the automated chat responses by checking the user's message and returning an appropriate reply.

Security is a major concern on the server side. The back end must protect the database and user information from attacks. The researchers implemented several security measures, which are discussed in detail in Section 3.11. These include input validation, password hashing, and protection against SQL injection attacks.

3.3.3 The Database

The database is the storage area of the system. It is where all the information is kept in an organized way. The database stores user accounts, property listings, images, reservation records, chat messages, and contact form submissions. Without a database, the system would not be able to remember anything from one visit to the next.

The database for this system was built using MySQL. MySQL is a relational database management system. It organizes data into tables, and each table contains rows and columns. For example, the properties table has

columns for id, title, price, location, description, and image. Each row in the table represents one property.

MySQL was chosen because it is free, reliable, and works very well with PHP. It is also easy to learn and has a large community of users who can provide help and resources. The researchers used a tool called phpMyAdmin to manage the database. phpMyAdmin provides a visual interface for creating tables, adding data, and running queries.

The database design is very important because a poorly designed database can cause problems such as duplicate data, slow performance, and data loss. The researchers carefully planned the database structure to make sure it was efficient and scalable. They used primary keys to uniquely identify each record, foreign keys to link related tables, and proper data types to save storage space.

The relationship between the front end, back end, and database can be summarized as follows: The user interacts with the front end. The front end sends requests to the back end. The back end processes the requests and talks to the database. The database returns the requested data. The back end formats the data and sends it back to the front end. The front end displays it to the user. This cycle happens every time a user clicks a button, fills out a form, or loads a page.

This three-layer architecture is strong, flexible, and easy to maintain. If the researchers want to change how the website looks in the future, they only need to update the front end. If they want to change how data is processed, they only need to update the back end. If they need to store more types of information, they only need to update the database. This separation of concerns makes the system easier to develop and improve over time.

3.4 Tools and Technologies Used

To build the web-based real estate management system, the researchers used several tools and technologies. Each tool was chosen for a specific purpose, and together they form the complete development environment. This section explains each tool in detail, including what it does and why it was selected for this project.

3.4.1 HTML (HyperText Markup Language)

HTML is the standard language for creating web pages. It provides the basic structure and content of a website. Every web page on the internet is built using HTML. HTML uses tags to define different elements such as headings, paragraphs, images, links, tables, and forms.

In this project, HTML was used to create all the pages of the real estate system. The home page, property listings page, property details page, reservation page, chat page, login page, registration page, and admin dashboard were all built with HTML. HTML tags were used to organize the content in a logical way so that users can easily understand and navigate the information.

The researchers used HTML5, which is the latest version of HTML. HTML5 includes new features that make web development easier and more powerful. For example, HTML5 has new form input types such as email, date, and number, which help validate user input automatically. It also supports video and audio elements, though these were not heavily used in this project.

HTML is essential because it is the foundation of every web page. Without HTML, there would be no structure or content. Even though HTML

alone cannot create beautiful or interactive websites, it is the starting point upon which CSS and JavaScript are added.

3.4.2 CSS (Cascading Style Sheets)

CSS is the language used to style and layout web pages. While HTML provides the content, CSS provides the appearance. CSS controls colors, fonts, sizes, spacing, borders, backgrounds, and the overall layout of the page.

In this project, CSS was used to create an attractive and professional design for the real estate system. The researchers chose a color scheme that looks clean and trustworthy, which is important for a real estate business. They used consistent fonts that are easy to read. They created layouts that organize information clearly, such as grid layouts for property listings and centered layouts for login forms.

The researchers used CSS3, which is the latest version of CSS. CSS3 includes powerful features such as animations, transitions, gradients, and flexible box layouts. These features were used to create smooth hover effects on property cards and responsive layouts that adapt to different screen sizes.

A special technique called media queries was used to make the website responsive. Media queries allow the CSS to change based on the width of the screen. For example, on a large screen, property listings might be displayed in three columns. On a small smartphone screen, the same listings might be displayed in one column for easier reading. This adaptability is very important because many users access websites from mobile devices.

CSS was also used to create the admin dashboard layout. The dashboard includes a sidebar menu, a top navigation bar, and a main content area. CSS

ensured that these elements stayed in place and looked organized regardless of the amount of content being displayed.

3.4.3 JavaScript

JavaScript is a programming language that adds behavior and interactivity to web pages. While HTML and CSS create static pages, JavaScript makes them dynamic and responsive to user actions.

In this project, JavaScript was used for many purposes. It was used to validate form inputs before they are sent to the server. For example, if a user tries to submit a reservation form without entering their name, JavaScript can show an error message immediately without needing to reload the page. This saves time and reduces server load.

JavaScript was also used to create interactive elements such as image sliders on property details pages. When a property has multiple images, JavaScript allows users to click through the images or swipe on mobile devices. This creates a better viewing experience than simply showing one static image.

Another important use of JavaScript was in the search and filter features. When users type in the search box or select filter options, JavaScript can update the displayed results instantly without reloading the entire page. This technique is called AJAX (Asynchronous JavaScript and XML), and it makes the website feel faster and more modern.

JavaScript was also used for the chat feature. When a user sends a message, JavaScript sends it to the server in the background and displays the response immediately. This creates a smooth chat experience similar to popular messaging applications.

The researchers used plain JavaScript without heavy frameworks, which kept the system lightweight and easy to understand. They also made sure that the JavaScript code was well-organized and commented for future maintenance.

3.4.4 PHP (PHP: Hypertext Preprocessor)

PHP is a server-side programming language designed specifically for web development. It runs on the web server and generates dynamic content that is sent to the user's browser.

In this project, PHP was the main back-end language. It was used to handle all server-side tasks, including user authentication, database queries, form processing, file uploads, and session management. Every time a user interacts with the system in a way that requires data storage or retrieval, PHP handles that request.

PHP was chosen for several reasons. First, it is free and open-source, which means there are no licensing costs. This is important for a student project with limited budget. Second, it has excellent support for MySQL databases, which was the chosen database system. Third, it has a large community and extensive documentation, making it easy to find solutions to problems. Fourth, it is supported by almost all web hosting providers, making deployment simple.

The researchers wrote PHP scripts to perform specific tasks. For example, there are scripts for logging in users, registering new accounts, adding properties, editing properties, deleting properties, making reservations, retrieving chat messages, and updating appointment statuses. Each script was designed to be secure and efficient.

PHP also handles the inclusion of common page elements. For example, the navigation menu and footer appear on almost every page. Instead of copying the same HTML code to every file, the researchers created separate PHP files for the menu and footer, and then included them in each page. This makes maintenance easier because changes only need to be made in one place.

3.4.5 MySQL

MySQL is a database management system that stores and organizes data. It is one of the most popular database systems in the world and is commonly used with PHP in web development.

In this project, MySQL was used to create and manage the database. The database contains several tables, each designed to store a specific type of information. The main tables include:

Users table: Stores client account information such as names, emails, phone numbers, and passwords. Admin users table: Stores administrator and agent account information. Properties table: Stores property listings including titles, descriptions, prices, locations, images, and availability status. Property images table: Stores additional images for each property, linked to the main properties table. Appointments table: Stores reservation requests including client information, property details, preferred dates and times, and status. Chat logs table: Stores messages exchanged between users and the system. Contact messages table: Stores messages sent through the contact form.

MySQL uses a language called SQL (Structured Query Language) to interact with the database. The researchers wrote SQL queries to insert new data, update existing data, delete old data, and retrieve information. For

example, when a user searches for properties in a specific location, the system runs a SQL query that looks for matching records in the properties table.

MySQL was chosen because it is reliable, fast, and free. It also integrates seamlessly with PHP through built-in functions. The researchers used prepared statements in their SQL queries to prevent security vulnerabilities such as SQL injection attacks.

3.4.6 Visual Studio Code

Visual Studio Code, often called VS Code, is a code editor used by developers to write and manage their code. It is free, lightweight, and supports many programming languages through extensions.

In this project, Visual Studio Code was the main development tool. The researchers used it to write HTML, CSS, JavaScript, and PHP code. VS Code provides helpful features such as syntax highlighting, which colors different parts of the code to make it easier to read. It also provides code completion, which suggests possible code as the developer types, saving time and reducing errors.

VS Code has a built-in file explorer that makes it easy to navigate between different project files. The researchers could quickly switch between the front-end files, back-end scripts, and configuration files. VS Code also has an integrated terminal, which allowed the researchers to run commands and manage the local server without leaving the editor.

Extensions were installed to enhance productivity. For example, extensions for PHP provided better code suggestions and error detection. Extensions for HTML and CSS helped with formatting and previewing

changes. A live server extension allowed the researchers to see their web pages update automatically as they made changes.

VS Code was chosen because it is modern, free, and widely used by developers around the world. Its user-friendly interface and powerful features made the development process faster and more organized.

3.4.7 XAMPP

XAMPP is a software package that provides a complete local server environment for web development. The name XAMPP stands for Cross-Platform, Apache, MariaDB, PHP, and Perl. It includes everything needed to run a PHP and MySQL application on a local computer.

In this project, XAMPP was used during the development and testing phases. It allowed the researchers to build and test the system on their own computers before deploying it to a live web server. XAMPP includes Apache, which is the web server software that handles HTTP requests. It also includes MySQL (through MariaDB) for the database, and PHP for processing scripts.

The researchers installed XAMPP on their development computers. They placed their project files in the htdocs folder, which is the root directory for web files in XAMPP. They started the Apache and MySQL services from the XAMPP control panel, and then they could access their website by typing localhost in their web browser.

XAMPP was chosen because it is free, easy to install, and provides all the necessary components in one package. It eliminated the need to install and configure each component separately, which saved time and reduced technical difficulties.

3.4.8 phpMyAdmin

phpMyAdmin is a web-based tool for managing MySQL databases. It provides a visual interface that makes it easy to create databases, create tables, add data, run queries, and manage users.

In this project, phpMyAdmin was used extensively during development. The researchers used it to create the database schema, define table structures, set primary keys, and establish relationships between tables. They also used it to insert sample data for testing purposes and to verify that their PHP scripts were storing data correctly.

phpMyAdmin was especially useful for debugging. When a database query did not work as expected, the researchers could use phpMyAdmin to test the query directly and see the results. They could also browse the tables to check if data was being saved correctly.

phpMyAdmin comes included with XAMPP, so no additional installation was needed. Its user-friendly interface made database management accessible even for team members who were less experienced with SQL commands.

3.4.9 Web Browser (Google Chrome and Mozilla Firefox)

Web browsers were essential tools for testing and viewing the system. The researchers used Google Chrome and Mozilla Firefox to open the website, test features, and check the appearance of pages.

Different browsers can sometimes display the same website slightly differently. Therefore, the researchers tested the system on both Chrome and Firefox to ensure consistency. They also used the browser's developer tools, which allow developers to inspect HTML elements, view CSS styles, debug JavaScript, and monitor network requests.

The developer tools were very helpful for fixing layout problems, identifying JavaScript errors, and measuring page load times. The responsive design mode in Chrome's developer tools allowed the researchers to see how the website would look on different screen sizes without needing multiple physical devices.

3.4.10 Git and GitHub

Git is a version control system that tracks changes to code over time. GitHub is a web-based platform that hosts Git repositories and provides collaboration features.

In this project, the researchers used Git to manage their source code. Every time a team member made significant changes, they committed those changes with a descriptive message explaining what was done. This created a history of all changes, which made it easy to see who did what and when.

GitHub was used to store the project repository online. This provided a backup of the code and allowed team members to access the latest version from any computer. GitHub also made it easy to merge changes from different team members and resolve conflicts when two people edited the same file.

Version control is very important in team projects because it prevents loss of work and helps coordinate efforts. The researchers could experiment with new features on separate branches without affecting the main codebase, and then merge the successful changes when ready.

3.5 Data Gathering Techniques

Data gathering is the process of collecting information that helps the researchers understand the problem and design the right solution. Without accurate and relevant data, the researchers might build a system that does not

solve the real needs of the users. This section explains the techniques used to gather data for this study.

3.5.1 Review of Existing Real Estate Platforms

The first data gathering technique was the review of existing real estate platforms. The researchers studied popular real estate websites and applications both locally and internationally. They looked at how these platforms display property listings, how they handle user searches, how they process reservations, and what communication features they offer.

Some of the platforms reviewed included international sites like Zillow and Realtor.com, as well as local Philippine real estate websites. The researchers paid attention to the user interface design, the types of information shown for each property, the filtering and sorting options, and the overall user experience.

This review provided several benefits. First, it showed the researchers what features are considered standard in the industry. For example, most real estate platforms include high-quality images, detailed descriptions, and map locations. Second, it revealed what features make a platform stand out. Third, it helped the researchers identify common mistakes or annoying features that they should avoid in their own system.

The researchers documented their findings from each platform review. They took screenshots, listed features, and noted both positive and negative aspects. This documentation was used during the requirements analysis and design phases to guide decision-making.

3.5.2 Interviews with the Client

The second and most important data gathering technique was conducting interviews with the client. The client in this study is the real estate business owner or manager who will use the system. Interviews allowed the researchers to ask direct questions and get detailed answers about the business operations, challenges, and needs.

The researchers prepared an interview guide with a list of questions organized by topic. The questions covered areas such as:

Current business processes: How do you currently list properties? How do clients find you? How do you handle inquiries? Challenges and problems: What difficulties do you face every day? Where do you lose potential clients? What tasks take too much time? Desired features: What would you like the system to do? What information should be included for each property? How should reservations work? User expectations: Who will use the system? What level of computer skills do they have? What devices will they use?

The interviews were conducted in person at the real estate office in Pavia, Iloilo City. The researchers recorded the answers (with permission) and took detailed notes. They asked follow-up questions when something was unclear. The interviews were conversational and friendly, which helped the client feel comfortable sharing honest information.

The information gathered from interviews was extremely valuable because it came directly from the people who would use the system. For example, the client explained that they often receive inquiries from OFWs who are abroad and cannot visit properties in person. This insight led the researchers to prioritize high-quality images and detailed descriptions. The client also mentioned that they sometimes forget to follow up with potential buyers, which led to the development of the inquiry tracking feature.

After the interviews, the researchers analyzed the responses. They looked for common themes, identified the most frequently mentioned problems, and ranked the requested features by importance. This analysis was used to create the requirements specification document that guided the entire development process.

3.5.3 Observation

In addition to interviews, the researchers also used observation as a data gathering technique. Observation means watching how the real estate business operates in its natural setting. The researchers visited the office and observed how agents handle phone calls, how they write down client information, how they update property boards or lists, and how they manage their daily tasks.

Observation provided information that interviews alone could not capture. Sometimes people say they do things one way, but observation shows that they actually do things differently. Observation also revealed small details that the client might not think to mention, such as how often the phone rings, how many clients walk in without appointments, or how agents search through paper files to find property information.

The researchers took notes during observation sessions and later discussed what they saw. These observations helped them understand the real workflow of the business and design a system that fits naturally into that workflow rather than forcing the business to change completely.

3.5.4 Document Analysis

The researchers also analyzed existing documents used by the real estate business. These included property brochures, listing sheets, appointment logs, and any existing spreadsheets or databases. Document

analysis showed what information the business currently collects and how it is organized.

For example, looking at a property listing sheet revealed what details are typically recorded for each property: address, lot area, floor area, number of bedrooms, number of bathrooms, price, and special features. This information was used to design the database fields for the properties table. Looking at appointment logs showed what information is needed when booking a viewing: client name, contact number, preferred date and time, and property of interest.

Document analysis ensured that the digital system would capture all the information that the business already considers important. It also helped identify information that was missing or inconsistently recorded, which the new system could improve.

3.6 Testing Procedures

Testing is one of the most important steps in building a software system. It is the process of checking if the system works correctly, if it meets the requirements, and if it is ready for real users. Testing helps find and fix problems before the system is deployed. A system that has not been tested properly can have errors that frustrate users, lose data, or create security risks.

The researchers used three main types of testing for the web-based real estate management system: Unit Testing, System Testing, and User Acceptance Testing. Each type of testing serves a different purpose and was conducted at different stages of the development process.

3.6.1 Unit Testing

Unit testing is the process of testing individual parts of the code to make sure each part works correctly on its own. A unit is the smallest testable part of a program, such as a single function, method, or procedure. Unit testing is usually done by the developers themselves as they write the code.

In this project, the researchers performed unit testing throughout the development phase. Every time they wrote a new function or feature, they tested it immediately before moving on to the next task. This approach is sometimes called test-driven development or continuous testing.

For example, when the researchers built the login function, they tested it with different combinations of usernames and passwords. They tested what happens when the correct credentials are entered, what happens when the wrong password is entered, what happens when an empty form is submitted, and what happens when special characters are used. They checked if the function correctly validates the input, queries the database, starts a user session, and redirects the user to the right page.

Another example is the property search function. The researchers tested it with various search terms: exact matches, partial matches, empty searches, searches with no results, and searches with many results. They verified that the function returns the correct properties, sorts them properly, and handles errors gracefully.

Unit testing was also performed on database operations. The researchers tested insert operations to make sure new records are added correctly. They tested update operations to confirm that changes are saved. They tested delete operations to ensure that records are removed without affecting other data. They tested select operations to verify that the right data is retrieved.

The researchers kept a log of all unit tests performed. For each test, they recorded what was being tested, what input was used, what the expected result was, what the actual result was, and whether the test passed or failed. Failed tests were investigated immediately, and the code was fixed before proceeding.

Unit testing is important because it catches errors early when they are still small and easy to fix. If a coding error is not discovered until the end of the project, it might be buried under many layers of other code, making it much harder to find and fix. By testing each unit as it is built, the researchers maintained a higher quality codebase throughout the project.

3.6.2 System Testing

System testing is the process of testing the entire system as a whole. While unit testing checks individual parts, system testing checks if all the parts work together correctly. It verifies that the complete system meets the requirements and performs all intended functions properly.

System testing was conducted after several iterations of development were completed and the major features were built. The researchers performed system testing by following test cases that covered all the main functions of the system.

A test case is a detailed description of a specific test scenario. It includes the steps to perform, the expected outcome, and the actual outcome. The researchers created test cases for every major feature of the system.

For the user registration feature, the test case included steps like: open the registration page, fill in the name field, fill in the email field, fill in the password field, click the register button, and check if the account is created

successfully. The expected outcome was that a new user record appears in the database and the user is redirected to the login page.

For the property reservation feature, the test case included: log in as a user, browse properties, select a property, click reserve, fill in the reservation form, submit the form, and check if the reservation is recorded. The expected outcome was that the appointment appears in the admin dashboard and the database.

For the admin features, test cases covered adding a property, uploading images, editing property details, deleting a property, viewing reservations, and responding to messages. Each test case was performed step by step, and the results were recorded.

System testing also included testing for error conditions. The researchers tested what happens when invalid data is entered, when required fields are left blank, when users try to access pages without logging in, and when the system receives unexpected requests. The system should handle these situations gracefully by showing appropriate error messages instead of crashing.

Performance was also checked during system testing. The researchers measured how long pages take to load, how quickly search results appear, and how the system behaves when multiple users are active at the same time. While this system was not expected to handle thousands of simultaneous users, basic performance testing ensured that normal usage would be smooth.

System testing was repeated multiple times. After bugs were fixed, the researchers ran the tests again to make sure the fixes worked and did not break

anything else. This repetition is called regression testing, and it is important because sometimes fixing one problem accidentally creates another problem.

3.6.3 User Acceptance Testing

User Acceptance Testing, often called UAT, is the final phase of testing. In UAT, real users test the system to determine if it meets their needs and if they are satisfied with it. UAT is important because developers might think a system works well, but the actual users might find it confusing or missing important features.

For this project, the researchers invited the client, several real estate agents, and a few potential clients to participate in User Acceptance Testing. The testing sessions were conducted at the real estate office and at the school computer laboratory. Each participant was given a set of tasks to complete using the system.

The tasks were designed to cover all the main features from the user's perspective. For regular users, the tasks included:

Create a new account. Log in to the system. Browse available properties. Use the search feature to find a specific type of property. View the details of a property. Make a reservation for a property viewing. Send a message through the chat feature. Log out.

For administrators and agents, the tasks included:

Log in to the admin dashboard. Add a new property listing with images. Edit an existing property. View all reservations. Respond to a client message. Update the status of an appointment. Generate or view reports.

The researchers observed the participants as they performed these tasks. They watched where users clicked, where they hesitated, where they made

mistakes, and what questions they asked. They did not help the users during the tasks because the goal was to see if the system was intuitive enough to use without assistance.

After completing the tasks, each participant filled out a feedback form. The form asked questions about the overall experience, the ease of use, the visual design, the usefulness of features, and any problems encountered. It also included space for open comments where participants could suggest improvements or mention anything they liked or disliked.

The feedback collected during UAT was very valuable. Some participants suggested adding more filter options for property searches. Others requested clearer confirmation messages after submitting forms. Some mentioned that certain buttons were hard to find. The real estate agents suggested improvements to the admin dashboard layout to make their daily tasks faster.

The researchers analyzed all the feedback and prioritized the suggested changes. They fixed the most critical issues immediately and noted less urgent improvements for future development. After making changes, they invited some participants to test again and confirm that the problems were solved.

UAT gave the researchers confidence that the system was ready for real use. It also helped build trust with the client because the client could see that their input was valued and incorporated into the final product.

3.7 Evaluation Criteria

Evaluation criteria are the standards or measures used to judge whether the system is successful. They answer the question: How do we know if the system is good? The researchers established clear evaluation criteria before

testing began so that they could objectively measure the system's performance.

The system was evaluated based on five main criteria: Functionality, Usability, Accuracy of Real Estate Properties, Security, and Efficiency. Each criterion is explained in detail below.

3.7.1 Functionality

Functionality refers to whether the system does what it is supposed to do. It measures if all the required features are present and if they work correctly. A system with good functionality meets all the requirements specified during the requirements analysis phase.

To evaluate functionality, the researchers checked each feature against the requirements list. They asked questions such as:

Can users successfully register and create accounts? Can users log in and log out securely? Can users browse and search for properties? Can users view detailed property information including images? Can users make reservations for property viewings? Can users send messages through the chat or contact form? Can administrators add, edit, and delete property listings? Can administrators view and manage reservations? Can administrators respond to client messages? Does the automated chat provide appropriate responses?

For each question, the researchers tested the feature and recorded whether it worked as expected. They also checked if there were any missing features or incomplete functions. The functionality criterion was considered satisfied if all major features worked correctly and matched the requirements.

Functionality is the most basic and important criterion because a system that does not function properly cannot serve its purpose no matter how beautiful or fast it is.

3.7.2 Usability

Usability refers to how easy the system is to use. A system with good usability can be used effectively by people with different levels of technical skill. Usability is important because even a fully functional system will fail if users find it too confusing or frustrating.

To evaluate usability, the researchers considered several factors:

Learnability: How quickly can a new user learn to use the system? Can they complete basic tasks without training? Efficiency: Once learned, can users perform tasks quickly? Memorability: If a user stops using the system for a while, can they remember how to use it when they return? Error prevention and recovery: Does the system help users avoid mistakes? If users make mistakes, can they easily recover? Satisfaction: Do users enjoy using the system? Are they satisfied with the experience?

The researchers measured usability primarily through User Acceptance Testing. They observed how long it took participants to complete tasks, how many errors they made, and whether they needed help. They also analyzed the feedback forms where participants rated the ease of use.

Specific usability checks included:

Is the navigation menu clear and consistent on all pages? Are buttons and links easy to find and click? Is the text readable and appropriately sized? Do forms have clear labels and helpful instructions? Are error messages easy

to understand? Does the system work well on mobile devices? Are page layouts clean and uncluttered?

The researchers aimed for a system that users could understand within a few minutes of first use. They made improvements based on usability feedback, such as adding clearer labels, improving button placement, and simplifying complex processes.

3.7.3 Accuracy of Real Estate Properties

Accuracy of real estate properties refers to whether the property information displayed in the system is correct, complete, and up to date. This criterion is especially important for a real estate system because clients make important decisions based on the information they see. If the information is wrong, clients might lose trust in the business.

To evaluate accuracy, the researchers checked:

Are property details such as price, location, and description correct? Do property images match the actual properties? Is the availability status accurate? Are sold or reserved properties marked correctly? Are there any duplicate listings? Is property information updated promptly when changes occur? Are location details specific enough for clients to understand where the property is?

The researchers worked with the client to verify that the sample data entered into the system was accurate. They also designed the admin dashboard to make it easy for administrators to update information quickly, which helps maintain accuracy over time.

Accuracy also depends on how the system handles data input. The researchers added validation rules to prevent administrators from entering

impossible values, such as negative prices or invalid dates. They also included image upload requirements to ensure that property listings always have visual representations.

3.7.4 Security

Security refers to how well the system protects data and prevents unauthorized access. A secure system keeps user information private, prevents hackers from stealing data, and ensures that only authorized users can perform sensitive actions.

To evaluate security, the researchers implemented and tested several security measures:

Authentication: Does the login system correctly verify user identities? Are passwords stored securely using hashing? Authorization: Can regular users access admin pages? Can one user access another user's private data? Input validation: Does the system reject harmful input such as SQL injection attempts or script tags? Session management: Are user sessions properly managed? Do sessions expire after a period of inactivity? Data protection: Is sensitive information such as passwords encrypted? Is the database protected from direct access? Error handling: Does the system hide technical error details from users? Do error messages reveal too much information?

The researchers tested security by attempting common attacks on their own system. They tried SQL injection by entering special characters in form fields. They tried to access admin pages without logging in. They checked if passwords were visible in the database. They verified that error messages were user-friendly and did not expose database structure or file paths.

Security is a continuous concern. While the researchers implemented best practices for a student project, they also acknowledged that a live production system would need additional security measures such as SSL certificates, regular backups, and server-level protections.

3.7.5 Efficiency

Efficiency refers to how well the system uses resources such as time, memory, and processing power. An efficient system responds quickly to user actions and does not waste server resources.

To evaluate efficiency, the researchers measured:

Page load time: How long does it take for pages to fully load? The researchers aimed for load times under three seconds for most pages. Search response time: How quickly do search results appear when users type queries? Database query performance: Are database queries optimized? Do they return results quickly even as the database grows? Resource usage: Does the system use reasonable amounts of server memory and CPU? Scalability: Can the system handle an increasing number of users and properties without significant slowdown?

The researchers optimized efficiency by:

Writing efficient SQL queries that only retrieve necessary data. Using indexes on database tables to speed up searches. Optimizing images so they load quickly without losing quality. Minimizing the use of heavy scripts that slow down page rendering. Using pagination for long lists so that only a few items are loaded at a time.

Efficiency testing was done by monitoring page load times using browser developer tools. The researchers tested the system with various

amounts of data to simulate real-world usage. They fixed any bottlenecks that caused slow performance.

3.8 Database Design and Implementation

The database is the heart of the web-based real estate management system. It stores all the information that the system needs to function. A well-designed database makes the system fast, reliable, and easy to maintain. A poorly designed database can cause data duplication, slow performance, and errors.

3.8.1 Database Planning

Before creating the database tables, the researchers carefully planned what information needed to be stored. They reviewed the requirements from Phase 1 and identified all the data entities in the system. A data entity is any person, place, thing, or concept about which information is stored. In this system, the main entities are users, properties, appointments, messages, and administrators.

For each entity, the researchers listed the attributes or details that needed to be stored. For example, the property entity needed attributes such as title, description, price, location, number of bedrooms, number of bathrooms, and images. The user entity needed attributes such as full name, email address, phone number, and password.

The researchers also identified the relationships between entities. For example, one property can have many appointments. One user can make many appointments. One property can have many images. These relationships helped determine how the tables should be connected using primary keys and foreign keys.

3.8.2 Database Tables

The database was implemented using MySQL and contains the following main tables:

Users Table (customers). This table stores information about clients who register on the website. The fields include: **id:** A unique number that identifies each user. This is the primary key. **fullname:** The complete name of the user. **email:** The email address of the user, used for login and communication. **phone:** The contact number of the user. **password:** The hashed password for secure login.

Admin Users Table (admin_users). This table stores information about administrators and agents who manage the system. The fields include: **id:** A unique identifier and primary key. **username:** The login name of the administrator. **gmail:** The email address used for admin login. **password:** The hashed password for security.

Properties Table (properties). This table stores all the property listings. The fields include: **id:** A unique identifier for each property. Primary key. **title:** The name or headline of the property listing. **display_image:** The main image shown in property previews. **price:** The selling or rental price of the property. **location:** The address or area where the property is located. **description:** A detailed explanation of the property features. **bedrooms:** The number of bedrooms. **bathrooms:** The number of bathrooms. **property_page:** Additional categorization or page assignment. **availability_status:** Indicates if the property is available, sold, or reserved.

Property Images Table (property_images). This table stores additional images for each property. It supports a one-to-many relationship with the

properties table. The fields include: id: Unique identifier. property_id: A foreign key linking to the properties table. image_path: The file path or URL of the additional image.

Appointments Table (appointments). This table stores reservation requests from clients. The fields include: id: Unique identifier. fullname: The name of the client making the reservation. email: The email address of the client. phone: The contact number of the client. property_id: A foreign key linking to the properties table. preferred_date: The date requested for property viewing. preferred_time: The time requested for property viewing. status: The current status of the appointment (pending, confirmed, completed, cancelled). notes: Additional comments from the client or admin.

Chat Logs Table (chat_logs). This table stores messages exchanged through the automated chat system. The fields include: id: Unique identifier. user_message: The message sent by the user. bot_response: The automated response generated by the system. timestamp: The date and time when the message was sent. user_id: Optional link to identify the user if logged in.

Contact Messages Table (contact_messages). This table stores messages sent through the contact form. The fields include: id: Unique identifier. name: The name of the sender. email: The email address of the sender. phone: The phone number of the sender. message: The content of the message. date_sent: When the message was submitted. is_read: Whether the admin has viewed the message.

3.8.3 Database Relationships

The tables in the database are connected through relationships that ensure data integrity and prevent duplication.

One-to-Many Relationships: One property can have many images. The `property_images` table has a foreign key (`property_id`) that references the `id` in the `properties` table. One property can have many appointments. The `appointments` table has a foreign key (`property_id`) that references the `properties` table. One user can make many appointments. The `appointments` table can also include a `user_id` field referencing the `customers` table.

These relationships are enforced using foreign key constraints in MySQL. Foreign key constraints ensure that a record cannot be deleted if other records depend on it. For example, if a property has appointments, the system will prevent accidental deletion of that property unless the appointments are handled first.

3.8.4 Database Normalization

Normalization is the process of organizing database tables to reduce redundancy and improve integrity. The researchers normalized the database to at least the third normal form (3NF).

First Normal Form (1NF) was achieved by ensuring that each table has a primary key and that each column contains only atomic (indivisible) values. For example, instead of storing multiple phone numbers in one field, the database uses one phone field per user.

Second Normal Form (2NF) was achieved by removing partial dependencies. All non-key attributes depend fully on the primary key. For example, property details depend only on the property id, not on any other column.

Third Normal Form (3NF) was achieved by removing transitive dependencies. Non-key attributes do not depend on other non-key attributes.

For example, user details are stored in the users table rather than being duplicated in the appointments table. The appointments table only stores the user_id, and the full user information is retrieved through the relationship.

Normalization helps prevent data anomalies such as update anomalies, insert anomalies, and delete anomalies. It also saves storage space by avoiding unnecessary duplication.

3.9 System Features and Functionalities

This section provides a detailed description of all the features and functions included in the web-based real estate management system. Each feature is explained in terms of what it does, who uses it, and why it is important.

3.9.1 User Registration and Authentication

The user registration feature allows new clients to create accounts on the system. When a visitor wants to make a reservation or use the chat feature, they are encouraged to register. The registration form asks for basic information: full name, email address, phone number, and password.

The system validates the registration form to ensure that all required fields are filled and that the email address is in the correct format. It also checks if the email address is already registered to prevent duplicate accounts. Passwords are hashed using a secure algorithm before being stored in the database. Hashing transforms the password into a random-looking string that cannot be easily reversed, which protects user accounts even if the database is compromised.

The authentication feature allows registered users to log in using their email and password. The system checks the credentials against the database,

starts a user session if they match, and redirects the user to their intended page. If the credentials do not match, the system shows an error message without revealing which part was wrong, which adds a layer of security.

Session management keeps users logged in as they navigate between pages. Sessions expire after a period of inactivity to protect accounts on shared computers. Users can also manually log out to end their session immediately.

3.9.2 Property Browsing and Search

The property browsing feature is the main attraction for visitors. It displays all available properties in an organized and visually appealing format. Each property is shown as a card containing the main image, title, location, price, and a brief description. Users can click on any property card to see more details.

The search feature allows users to find specific properties based on their preferences. Users can search by keyword, location, price range, number of bedrooms, or number of bathrooms. The system processes the search query, looks through the properties table, and returns matching results.

The search results are displayed dynamically without requiring a full page reload, thanks to JavaScript and AJAX. This makes the search experience fast and smooth. If no properties match the search criteria, the system displays a friendly message suggesting that the user try different keywords or browse all listings.

Sorting options allow users to arrange results by price (low to high or high to low), by date added (newest first), or alphabetically by title. These options help users find the most relevant properties quickly.

3.9.3 Property Details Viewing

The property details page provides comprehensive information about a specific property. When a user clicks on a property from the listings page, they are taken to this detailed view.

The page includes: A large image gallery showing all photos of the property. Users can click through images or view them in a larger format. The full property title and price prominently displayed. A complete description including lot area, floor area, features, and nearby amenities. Location information, sometimes including a map or landmarks for reference. Property specifications such as number of bedrooms, bathrooms, and parking spaces. Availability status clearly indicated (Available, Reserved, or Sold). A reservation button that allows interested users to book a viewing appointment. A contact or chat option for immediate inquiries.

The property details page is designed to answer as many client questions as possible without requiring them to contact the agent. This is especially important for OFW clients who cannot visit the property in person and rely entirely on digital information for their initial assessment.

3.9.4 Online Reservation System

The online reservation system is one of the most important features of the platform. It allows clients to request appointments to view properties without making phone calls or visiting the office.

When a user clicks the reserve button on a property details page, they are taken to the reservation form. The form asks for: Full name. Email address. Phone number. Preferred date for viewing. Preferred time for viewing. Any special requests or notes.

The system validates all inputs to ensure completeness and correctness. It checks that the selected date is not in the past and that the time is within reasonable business hours. Once submitted, the reservation is saved in the appointments table with a status of "Pending."

The user receives a confirmation message on the screen indicating that their request has been received. Meanwhile, the administrator sees the new appointment in the admin dashboard and can confirm, reschedule, or cancel it. The status updates are reflected in the system, and if the client is logged in, they can view their appointment history.

This feature solves the major problem of manual appointment tracking. Instead of writing appointments in notebooks or relying on memory, agents have a centralized digital record of all viewing requests. The system also prevents double-booking by showing which time slots are already taken.

3.9.5 Automated Communication and Chat

The automated communication feature improves the speed and consistency of responses to client inquiries. It includes two main components: the contact form and the chat system.

The contact form is a simple form on the website where visitors can send messages without needing an account. It collects name, email, phone, and message. Submissions are stored in the `contact_messages` table and displayed in the admin dashboard for response.

The chat system provides instant interaction. When a user opens the chat interface, they can type questions about properties, availability, or procedures. The system uses predefined responses for common questions. For example, if a user asks "What are your office hours?" the system can

automatically reply with the business hours. If a user asks about a specific property's availability, the system can check the database and respond instantly.

For questions that the automated system cannot answer, the message is flagged for a human agent to respond. The chat logs are saved so that agents can review past conversations and provide consistent follow-up.

This feature significantly reduces the workload of agents by handling repetitive questions automatically. It also ensures that clients receive immediate responses at any time of day, even outside business hours. This 24/7 availability improves client satisfaction and prevents potential buyers from losing interest while waiting for a response.

3.9.6 Administrative Dashboard

The administrative dashboard is the control center of the system. Only authorized administrators and agents can access it after logging in. The dashboard provides all the tools needed to manage the real estate business digitally.

The main sections of the dashboard include:

Property Management: Add new properties with detailed information and image uploads. Edit existing properties to update prices, descriptions, or availability. Delete properties that are no longer available. View all properties in a list or grid format with quick status indicators.

Appointment Management: View all incoming reservations in chronological order. Filter appointments by status (pending, confirmed, completed, cancelled). Update appointment status and add internal notes. Contact clients directly regarding their appointments.

Message Management: View all messages from the contact form and chat system. Mark messages as read or unread. Reply to messages or assign them to specific agents. Archive old messages for record-keeping.

User Management: View registered client accounts. Manage admin and agent accounts. Control access permissions.

Reports and Analytics: View summary statistics such as total properties, total appointments, and total messages. See which properties are generating the most interest. Monitor response times and appointment conversion rates.

The dashboard is designed to be intuitive and efficient. Common tasks are accessible within one or two clicks. The interface uses clear labels, color coding for statuses, and responsive tables that work on both desktop and mobile devices.

3.9.7 Content Management

The content management features allow administrators to control the information displayed on the public-facing website. This includes not only property listings but also static content such as the business description, contact information, and any announcements.

Administrators can update the home page content to feature specific properties or promotions. They can manage the about page to reflect current business information. They can also update footer information such as address, phone numbers, and social media links.

While the system does not include a full blog management feature as originally considered, the content management capabilities ensure that the

website remains current and relevant without requiring technical knowledge to update.

3.10 User Interface Design Approach

The user interface, often called UI, is the visual and interactive part of the system that users see and touch. A good user interface makes the system enjoyable and easy to use. A bad user interface can make even the most powerful system frustrating and confusing.

The researchers followed several important principles when designing the user interface for the real estate system.

3.10.1 Simplicity

Simplicity means keeping the design clean and avoiding unnecessary complexity. The researchers aimed to present information clearly without overwhelming the user. Each page has a clear purpose and a clear next step. The home page introduces the business and highlights featured properties. The listings page focuses on helping users find properties. The details page focuses on providing comprehensive information about one property.

Unnecessary elements were removed. The researchers avoided cluttering pages with too many advertisements, pop-ups, or distracting animations. Every element on the page serves a purpose.

3.10.2 Consistency

Consistency means using the same design patterns throughout the system. When users learn how to do something on one page, they should be able to do the same thing on another page without relearning.

The researchers maintained consistency by: Using the same color scheme across all pages. Using the same fonts and text sizes for similar types of content. Placing the navigation menu in the same location on every page. Using the same button styles for primary actions. Using the same form layout patterns for all input fields. Maintaining the same footer information across all pages.

Consistency reduces the mental effort required to use the system. Users feel more confident and comfortable when they know what to expect.

3.10.3 Feedback

Feedback means telling users what is happening in response to their actions. When users click a button, submit a form, or perform any action, the system should confirm that the action was received and indicate what will happen next.

The researchers included feedback mechanisms such as: Success messages after form submissions. Error messages when something goes wrong, with suggestions for how to fix it. Loading indicators when the system is processing a request. Confirmation dialogs before destructive actions like deleting a property. Visual changes when hovering over clickable elements. Status indicators showing if a property is available, reserved, or sold.

Good feedback prevents users from feeling confused or uncertain. It assures them that the system is working and that their actions have effects.

3.10.4 Accessibility

Accessibility means designing the system so that it can be used by as many people as possible, including those with disabilities or limitations.

While the researchers did not pursue formal accessibility certification, they incorporated basic accessibility practices.

These practices included: Using sufficient color contrast so that text is readable against backgrounds. Using descriptive labels for form fields. Providing alternative text for images (where technically implemented). Ensuring that the system can be navigated using a keyboard. Avoiding design elements that might cause seizures or physical discomfort. Using clear and simple language that is easy to understand.

Accessibility is important because real estate clients come from all backgrounds and may have varying levels of technical ability, vision quality, or device capabilities.

3.10.5 Responsive Design

Responsive design means that the website adapts to the screen size of the device being used. The researchers recognized that many users would access the system from smartphones, as mobile internet usage is very high in the Philippines.

The responsive design approach used flexible layouts, flexible images, and media queries. On large screens, property listings might display in multiple columns with large images. On small screens, the same listings stack vertically in a single column with appropriately sized images. The navigation menu collapses into a hamburger icon on mobile devices to save space. Forms and buttons resize to remain tappable on touch screens.

The researchers tested the responsive design by resizing their browser windows and using the device simulation tools in Chrome and Firefox. They also tested on actual smartphones and tablets to ensure real-world usability.

3.11 Security Measures and Data Protection

Security is a critical aspect of any web-based system that handles personal information and business data. The researchers implemented multiple layers of security to protect the system from common threats and to ensure that user data remains confidential.

3.11.1 Input Validation

Input validation is the process of checking data entered by users to ensure it is correct, complete, and safe. The researchers validated all form inputs on both the client side and the server side.

Client-side validation uses JavaScript to check inputs before the form is submitted. It provides immediate feedback to users and reduces unnecessary server requests. For example, it checks if an email field contains the @ symbol, if required fields are filled, and if phone numbers contain only numbers.

Server-side validation uses PHP to check inputs again after they reach the server. This is essential because client-side validation can be bypassed by malicious users. Server-side validation checks for:

- Required fields: Ensuring that important information is not missing.
- Data type: Ensuring that numbers are actually numbers and dates are valid dates.
- Length limits: Preventing excessively long inputs that might cause errors.
- Format patterns: Using regular expressions to validate email addresses and phone numbers.
- Harmful content: Stripping or escaping special characters that might be used in attacks.

3.11.2 Password Hashing

Password hashing is a security technique that transforms passwords into unreadable strings before storing them in the database. The researchers used a strong hashing algorithm to protect user passwords.

When a user creates an account or changes their password, the system hashes the password immediately. The hash, not the original password, is stored in the database. When the user logs in, the system hashes the password they entered and compares it to the stored hash. If they match, the login is successful.

Hashing is important because even if an attacker gains access to the database, they cannot easily determine the actual passwords. The researchers also encouraged users to choose strong passwords by requiring a minimum length.

3.11.3 SQL Injection Prevention

SQL injection is a common attack where malicious users insert harmful SQL code into form inputs to manipulate the database. The researchers prevented SQL injection by using prepared statements in all database queries.

Prepared statements separate the SQL command from the data. The database knows which parts are commands and which parts are data, so even if a user enters SQL code as input, it is treated as plain text rather than executable commands. This technique effectively eliminates the risk of SQL injection attacks.

3.11.4 Cross-Site Scripting (XSS) Prevention

Cross-site scripting is an attack where malicious scripts are injected into web pages viewed by other users. The researchers prevented XSS by escaping all user-generated content before displaying it on web pages.

When users submit messages, comments, or any text that will be shown to others, the system converts special HTML characters into their safe equivalents. For example, the less-than symbol < becomes < and the greater-than symbol > becomes >. This ensures that any script tags entered by users are displayed as plain text rather than being executed by the browser.

3.11.5 Session Security

Session security ensures that user sessions cannot be hijacked or impersonated. The researchers implemented several session security measures:

Session regeneration: When a user logs in, the system generates a new session ID to prevent session fixation attacks. Session timeout: Sessions expire automatically after a period of inactivity, requiring the user to log in again. Secure session storage: Session data is stored on the server, not in the user's browser, and session IDs are transmitted securely. Logout functionality: Users can manually destroy their session by clicking the logout button.

3.11.6 Access Control

Access control ensures that users can only access the parts of the system they are authorized to use. The researchers implemented role-based access control with two main roles: regular users and administrators.

Regular users can browse properties, make reservations, send messages, and manage their own accounts. They cannot access the admin dashboard, edit properties, view other users' reservations, or delete records.

Administrators can access all management features. The system checks the user's role before allowing access to restricted pages. If a non-admin user

tries to access an admin page by typing the URL directly, they are redirected to the home page or login page.

3.11.7 Data Backup

While the prototype did not include automated backup features, the researchers manually backed up the database and project files regularly during development. Backups were stored in multiple locations to prevent loss due to hardware failure or accidental deletion. For a production deployment, the researchers recommended implementing automated daily backups.

3.12 Project Timeline and Development Schedule

A project timeline is a schedule that shows when each phase of the project will be completed. It helps the team stay organized, meet deadlines, and allocate time properly. The researchers created a timeline at the beginning of the project and updated it as needed.

The development of the web-based real estate management system was completed over the course of one semester, approximately four to five months. The timeline was divided into the following major periods:

Weeks 1 to 2: Research and Planning. During this period, the researchers finalized their proposal, conducted interviews with the client, reviewed existing platforms, and gathered all necessary data. They wrote the first chapters of the documentation and created the requirements specification.

Weeks 3 to 4: System Design. During this period, the researchers created wireframes, mockups, and design documents. They designed the database schema, drew system diagrams, and planned the system architecture. They presented the design to their adviser and the client for approval.

Weeks 5 to 10: Development (Iterations 1 to 4). During this period, the researchers built the core features of the system. The first iteration established the project foundation and user authentication. The second iteration built the property management features. The third iteration built the client browsing and search features. The fourth iteration built the reservation system. Each iteration included coding, unit testing, and team review.

Weeks 11 to 12: Development (Iterations 5 to 6). During this period, the researchers built the communication features and the admin dashboard. They also implemented security measures and optimized performance. Iteration 5 focused on chat and contact forms. Iteration 6 focused on polishing, security, and responsive design improvements.

Weeks 13 to 14: System Testing and Bug Fixing. During this period, the researchers conducted comprehensive system testing. They tested all features together, fixed bugs, and performed regression testing. They also optimized database queries and page load times.

Weeks 15 to 16: User Acceptance Testing and Documentation. During this period, the researchers invited users to test the system. They collected feedback, made final improvements, and completed the system documentation. They prepared the deployment environment and created user manuals.

Weeks 17 to 18: Deployment and Final Presentation. During this period, the researchers deployed the prototype, conducted final demonstrations for the client and panel, and prepared for the oral defense. They also compiled all project materials and ensured that the documentation was complete.

The researchers used a Gantt chart to visualize the timeline. A Gantt chart is a bar chart that shows the start and end dates of each task. It makes it easy to see which tasks overlap and which tasks depend on others. The team reviewed the Gantt chart during weekly meetings to track progress and adjust the schedule if any delays occurred.

Time management was crucial for the success of the project. The researchers avoided procrastination by setting mini-deadlines for each iteration. They also built buffer time into the schedule to handle unexpected problems or additional requests from the client.

3.13 Supporting Diagrams and Visual Representations

Diagrams and visual representations are important tools for understanding how a system works. They provide a clear and concise way to show complex information that would be difficult to explain with words alone. This section describes the main diagrams used in this study.

3.13.1 Data Flow Diagram (DFD) Level 0

The Data Flow Diagram Level 0, also called the Context Diagram, shows the system as a single process and illustrates how external entities interact with it. The external entities in this system are the Client (user) and the Administrator.

The Context Diagram shows that clients send inputs such as registration information, login credentials, search queries, reservation requests, and messages. The system processes these inputs and returns outputs such as property listings, property details, confirmation messages, and chat responses.

Administrators send inputs such as login credentials, property data, and status updates. The system processes these and returns outputs such as admin dashboard views, property management confirmations, and appointment lists.

The Level 0 DFD provides a high-level overview that is useful for understanding the system's boundaries and its interaction with the outside world.

3.13.2 Data Flow Diagram (DFD) Level 1

The Data Flow Diagram Level 1 breaks down the single process from Level 0 into major sub-processes. It shows more detail about how data moves within the system.

The main processes in the Level 1 DFD include: User Authentication Process: Handles registration, login, and session management. Property Browsing Process: Handles search queries and returns property information. Reservation Process: Handles appointment requests and updates the database. Communication Process: Handles messages and chat interactions. Admin Management Process: Handles property CRUD operations, appointment management, and message handling.

Data stores shown in the DFD include the Users Database, Properties Database, Appointments Database, and Messages Database. The diagram shows how data flows from external entities to processes, from processes to data stores, and back to external entities.

The Level 1 DFD is useful for understanding the main functional components of the system and how they relate to each other.

3.13.3 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram shows the logical structure of the database. It identifies the entities (tables), their attributes (fields), and the relationships between them.

The main entities in the ERD are: Customer: Represents clients who register on the website. Admin User: Represents administrators and agents. Property: Represents real estate listings. Property Image: Represents additional photos linked to properties. Appointment: Represents reservation requests. Chat Log: Represents automated chat conversations. Contact Message: Represents messages from the contact form.

The relationships shown in the ERD include: One Customer can make Many Appointments. One Property can have Many Property Images. One Property can have Many Appointments. One Admin User can manage Many Properties and Many Appointments.

The ERD uses crow's foot notation to show the cardinality of relationships. Lines connect related entities, and symbols at the ends of lines indicate whether the relationship is one-to-one, one-to-many, or many-to-many.

The ERD is essential for database design because it ensures that all necessary data is captured and that relationships are properly structured.

3.13.4 Use Case Diagram

The Use Case Diagram shows the different ways that users interact with the system. It identifies the actors (types of users) and the use cases (actions they can perform).

The actors in this system are: User (Client): A person who visits the website to browse properties and make reservations. Administrator: A person who manages the system, properties, and client interactions.

The use cases for the User actor include: Browse Properties: View the list of available properties. View Property Details: See comprehensive information about a specific property. Make Reservation: Submit a request to view a property. Send Message: Communicate with agents through chat or contact form. Register Account: Create a user account. Log In: Access personalized features.

The use cases for the Administrator actor include: Log In: Access the admin dashboard. Manage Properties: Add, edit, and delete property listings. View Reservations: See all appointment requests. Manage Messages: Read and respond to client inquiries. Manage Users: Oversee registered accounts. View Reports: See system statistics and analytics.

The Use Case Diagram uses ovals to represent use cases, stick figures to represent actors, and lines to show which actors participate in which use cases. This diagram is helpful for verifying that all required features are included and that each user type has appropriate access.

3.14 Summary of Methodology

This chapter has presented a comprehensive explanation of the methods and approaches used to develop the web-based real estate management system. The research design combined descriptive research to understand the problem and development research to create the solution. The Agile Development Model guided the entire process through its five phases:

Requirements Analysis, System Design, Development, Testing, and Deployment.

The system architecture follows a proven client-server model with clear separation between the front end, back end, and database. The tools and technologies were carefully selected for their suitability, availability, and the team's familiarity with them. HTML, CSS, and JavaScript power the user interface. PHP handles server-side processing. MySQL stores all data securely. Visual Studio Code, XAMPP, and phpMyAdmin support the development workflow.

Data was gathered through multiple techniques including platform reviews, client interviews, observation, and document analysis. These techniques ensured that the system was built on real needs rather than assumptions.

Testing was thorough and multi-layered, including unit testing, system testing, and user acceptance testing. Evaluation criteria covered functionality, usability, accuracy, security, and efficiency, ensuring a well-rounded assessment of the system's quality.

The database was carefully designed with proper normalization and relationships. System features were developed to address the specific problems identified in Chapter 1, including property browsing, online reservations, automated communication, and administrative management.

The user interface was designed with simplicity, consistency, feedback, accessibility, and responsiveness in mind. Security was addressed through input validation, password hashing, SQL injection prevention, XSS prevention, session security, and access control.

The project followed a realistic timeline with clear milestones and iterative development cycles. Supporting diagrams provided visual clarity for the system's structure and behavior.

All of these methodological choices work together to ensure that the final system is functional, user-friendly, secure, and effective in addressing the challenges of digital marketing and client communication for real estate properties in Pavia, Iloilo City. The systematic approach documented in this chapter provides a replicable framework and a solid foundation for the results and discussions presented in the following chapters.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.